

BOOK 5

WORK, ENERGY & POWER

5 Work, energy and power

An understanding of the forms of energy and energy transfers from Cambridge IGCSE/O Level Physics or equivalent is assumed.

5.1 Energy conservation

Candidates should be able to:

- 1 understand the concept of work, and recall and use *work done* = *force* × *displacement* in the direction of the force
- 2 recall and apply the principle of conservation of energy
- 3 recall and understand that the efficiency of a system is the ratio of useful energy output from the system to the total energy input
- 4 use the concept of efficiency to solve problems
- 5 define power as work done per unit time
- 6 solve problems using $P = W/t$
- 7 derive $P = Fv$ and use it to solve problems

5.2 Gravitational potential energy and kinetic energy

Candidates should be able to:

- 1 derive, using $W = Fs$, the formula $\Delta E_p = mg\Delta h$ for gravitational potential energy changes in a uniform gravitational field
- 2 recall and use the formula $\Delta E_p = mg\Delta h$ for gravitational potential energy changes in a uniform gravitational field
- 3 derive, using the equations of motion, the formula for kinetic energy $E_k = \frac{1}{2}mv^2$
- 4 recall and use $E_k = \frac{1}{2}mv^2$

1 9702/1/M/J/02/Q5

The power loss P in a resistor is calculated using the formula $P = V^2/R$.

The uncertainty in the potential difference V is 3% and the uncertainty in the resistance R is 2%.
What is the uncertainty in P ?

- A 4% B 7% C 8% D 11%

2 9702/1/M/J/02/Q9

Two similar spheres, each of mass m and travelling with speed v , are moving towards each other.

The spheres have a head-on elastic collision.

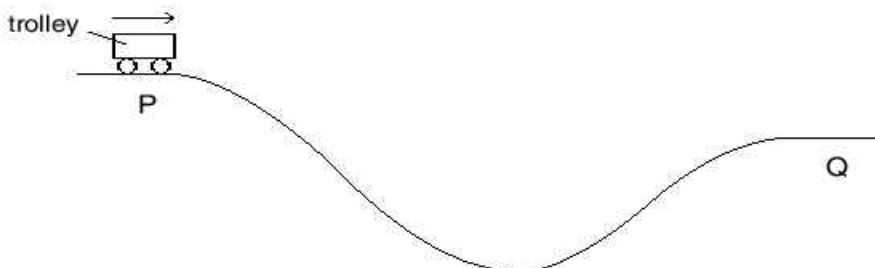
Which statement is correct?

- A The spheres stick together on impact.
B The total kinetic energy after impact is mv^2 .
C The total kinetic energy before impact is zero.
D The total momentum before impact is $2mv$.



3 9702/11/O/N/11/Q18

A trolley runs from P to Q along a track. At Q its potential energy is 50 kJ less than at P.



At P, the kinetic energy of the trolley is 5 kJ. Between P and Q the work the trolley does against friction is 10 kJ.

What is the kinetic energy of the trolley at Q?

- A 35 kJ B 45 kJ C 55 kJ D 65 kJ

4 9702/1/M/J/02/Q19

To travel at a constant speed, a car engine provides 24 kW of useful power. The driving force on the car is 600 N.

At what speed does it travel?

- A 2.5 m s^{-1} B 4.0 m s^{-1} C 25 m s^{-1} D 40 m s^{-1}

5 9702/1/O/N/02/Q16

Which of the following is an expression for power?

- A energy $\times$ time B force $\times$ displacement C force $\times$ velocity D mass $\times$ velocity

6 9702/1/O/N/02/Q18

An area of land is an average of 2.0 m below sea level. To prevent flooding, pumps are used to lift rainwater up to sea level.

What is the minimum pump output power required to deal with 1.3×10^9 kg of rain per day?

- A 15 kW B 30 kW C 150 kW D 300 kW

7 9702/1/O/N/02/Q19

A twig from a tree drops from a 200 m high cliff on to a beach below. During its fall, 40% of the twig's energy is converted into thermal energy.

What is the speed with which the twig hits the beach?

- A 35 m s^{-1} B 40 m s^{-1} C 49 m s^{-1} D 63 m s^{-1}

8 9702/01/M/J/03/Q16

Which of the following expressions **defines** power?

- A force x distance moved in the direction of the force C work done $\div$ time taken
B force x velocity D work done x time taken

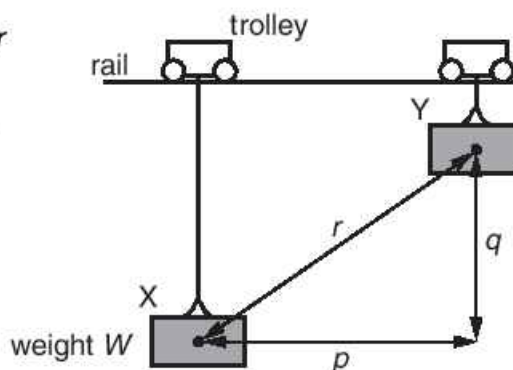
9 9702/01/M/J/03/Q17

A weight W hangs from a trolley that runs along a rail. The trolley moves horizontally through a distance p and simultaneously raises the weight through a height q .

As a result, the weight moves through a distance r from X to Y. It starts and finishes at rest.

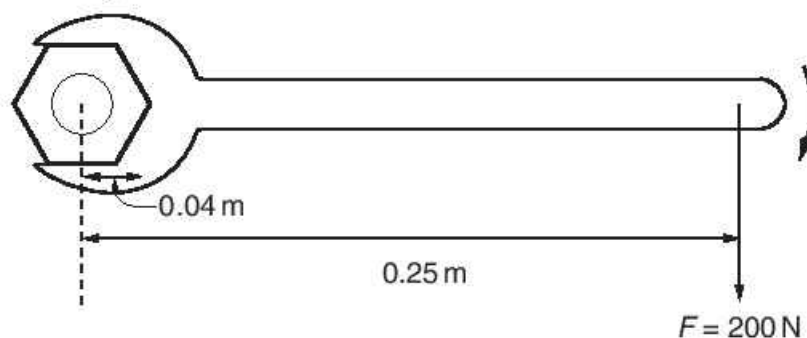
How much work is done on the weight during this process?

- A Wp B $W(p + q)$
C Wq D Wr



10 9702/01/O/N/03/Q13

A spanner is used to tighten a nut as shown.



A force F is applied at right-angles to the spanner at a distance of 0.25 m from the centre of the nut. When the nut is fully tightened, the applied force is 200 N.

What is the resistive torque, in an anticlockwise direction, preventing further tightening?

- A 8 N m B 25 N m C 50 N m D 800 N m

11 9702/01/O/N/03/Q18

A boat moving at constant speed v through still water experiences a total frictional drag F . What is the power developed by the boat?

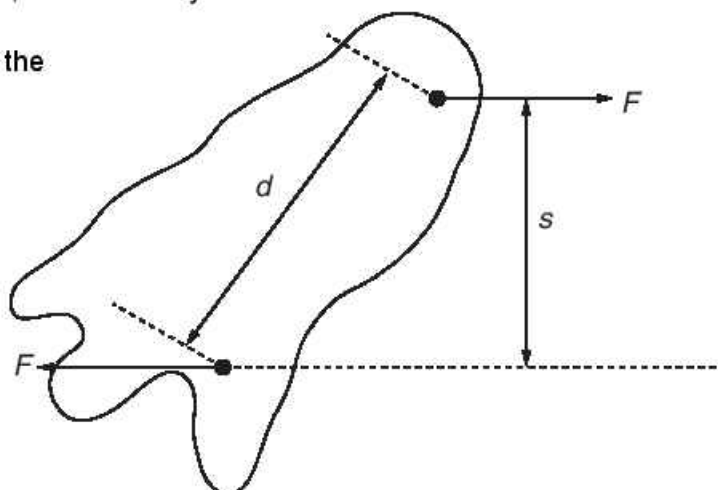
- A $\frac{1}{2}Fv$ B Fv C $\frac{1}{2}Fv^2$ D Fv^2

12 9702/01/O/N/03/Q14

Two parallel forces, each of magnitude F , act on a body as shown.

What is the magnitude of the torque on the body produced by these forces?

- A Fd
- B Fs
- C $2Fd$
- D $2Fs$

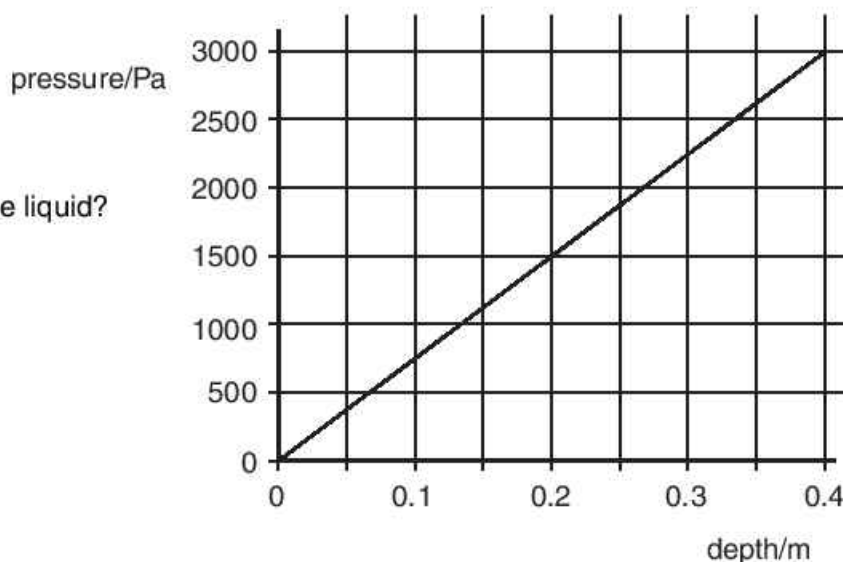


13 9702/01/O/N/03/Q19

The graph shows how the pressure exerted by a liquid varies with depth below the surface.

What is the density of the liquid?

- A 600 kg m^{-3}
- B 760 kg m^{-3}
- C 5900 kg m^{-3}
- D 7500 kg m^{-3}



14 9702/01/O/N/03/Q20

In an experiment to demonstrate Brownian motion, smoke particles in a container are illuminated by a strong light source and observed through a microscope.

The particles are seen as small specks of light that are in motion.

What causes this motion?

- A collisions between the smoke particles and air molecules
- B collisions between the smoke particles and the walls of the container
- C convection currents within the air as it is warmed by the light source
- D kinetic energy gained by the smoke particles on absorption of light

15 9702/01/M/J/04/Q15

What is the expression used to **define** power?

- A $\frac{\text{energy output}}{\text{energy input}}$
- B energy x time taken
- C force x velocity
- D $\frac{\text{work done}}{\text{time taken}}$

16 9702/12/M/J/13/Q15

A ball is thrown vertically upwards.

Neglecting air resistance, which statement is correct?

- A The kinetic energy of the ball is greatest at the greatest height attained.
- B By the principle of conservation of energy, the total energy of the ball is constant throughout its motion.
- C By the principle of conservation of momentum, the momentum of the ball is constant throughout its motion.
- D The potential energy of the ball increases uniformly with time during the ascent.

17 9702/01/M/J/04/Q17*

Car X is travelling at half the speed of car Y. Car X has twice the mass of car Y.

Which statement is correct?

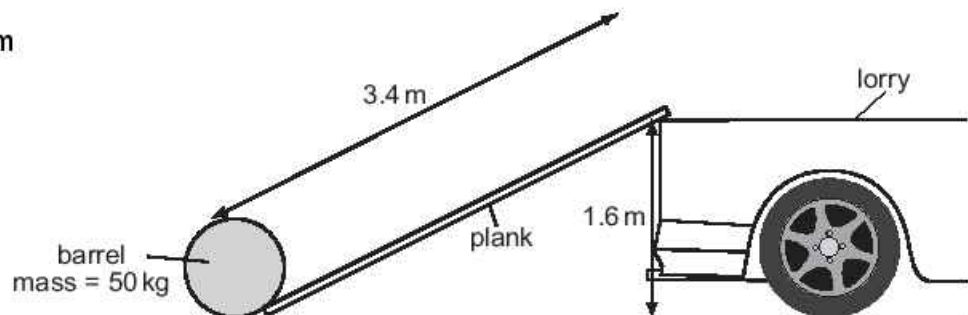
- A Car X has half the kinetic energy of car Y.
- B Car X has one quarter of the kinetic energy of car Y.
- C Car X has twice the kinetic energy of car Y.
- D The two cars have the same kinetic energy.

18 9702/01/M/J/04/Q18

A barrel of mass 50 kg is loaded onto the back of a lorry 1.6 m high by pushing it up a smooth plank 3.4 m long.

What is the minimum work done?

- A 80 J
- B 170 J
- C 780 J
- D 1700 J

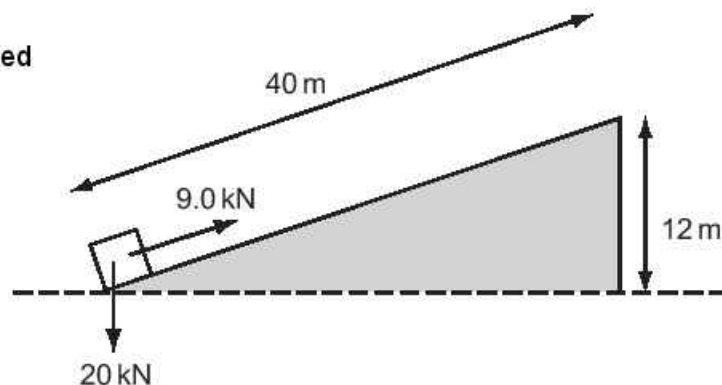


19 9702/01/O/N/04/Q18

A constant force of 9.0 kN, parallel to an inclined plane, moves a body of weight 20 kN through a distance of 40 m along the plane at constant speed. The body gains 12 m in height, as shown.

How much of the work done is dissipated as heat?

- A 120 kJ
- B 240 kJ
- C 360 kJ
- D 600 kJ



20 **9702/01/O/N/04/Q15**

The kinetic energy of a particle is increased by a factor of 4.
By what factor does its speed increase?

- A 2 B 4 C 8 D 16

21 **9702/01/O/N/04/Q16**

A horizontal force of 90 N is used to push a box across a horizontal floor. The frictional force on the box is 50 N.

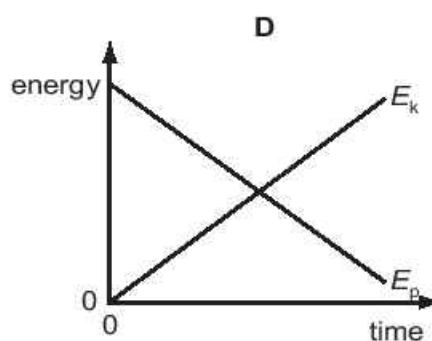
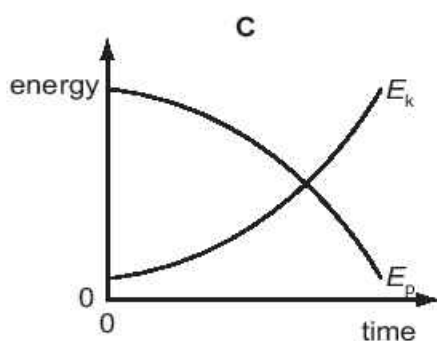
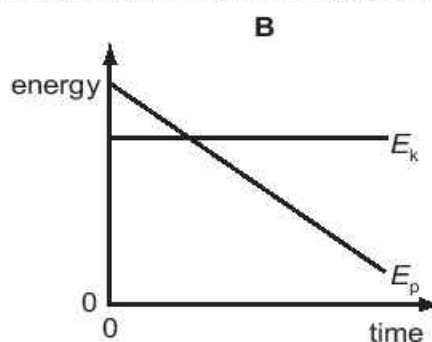
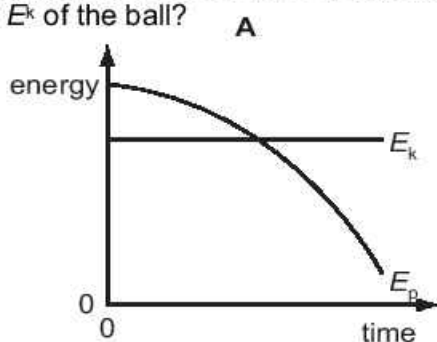
What is the gain in kinetic energy of the box when it is moved through a distance of 6.0 m?

- A 240 J B 300 J C 540 J D 840 J

22 **9702/01/M/J/05/Q15**

A steel ball is falling at constant speed in oil.

Which graph shows the variation with time of the gravitational potential energy E_p and the kinetic energy E_k of the ball?

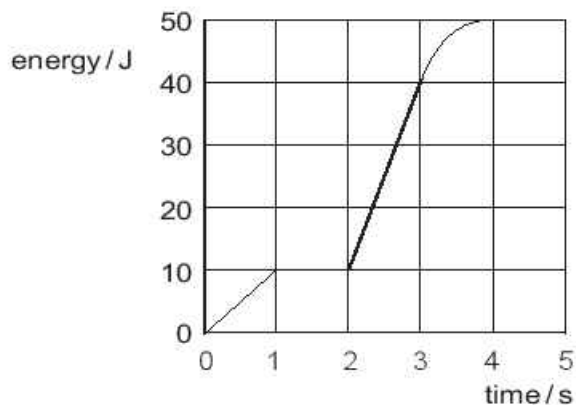


23 **9702/01/M/J/05/Q16***

An electrical generator is started at time zero. The total electrical energy generated during the first 5 seconds is shown in the graph.

What is the maximum electrical power generated at any instant during these first 5 seconds?

- A 10 W B 13 W
C 30 W D 50 W



24 9702/01/M/J/05/Q17

A concrete cube of side 0.50 m and uniform density $2.0 \times 10^3 \text{ kg m}^{-3}$ is lifted 3.0 m vertically by a crane.

What is the change in potential energy of the cube?

- A** 0.75 kJ **B** 7.4 kJ **C** 29 kJ **D** 470 kJ

25 9702/01/O/N/05/Q14

A car with a total mass of 1400 kg is travelling at 30 m s^{-1} .

What is the kinetic energy of the car?

- A** 21 kJ **B** 42 kJ **C** 630 kJ **D** 1260 kJ

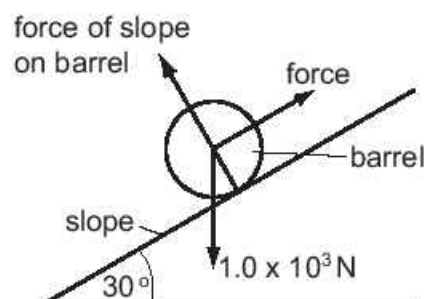
26 9702/01/O/N/05/Q16

The diagram shows a barrel of weight $1.0 \times 10^3 \text{ N}$ on a frictionless slope inclined at 30° to the horizontal.

A force is applied to the barrel to move it up the slope at constant speed. The force is parallel to the slope.

What is the work done in moving the barrel a distance of 5.0 m up the slope?

- A** $1.0 \times 10^4 \text{ J}$ **B** $2.5 \times 10^3 \text{ J}$
C $4.3 \times 10^3 \text{ J}$ **D** $5.0 \times 10^3 \text{ J}$



27 9702/01/M/J/06/Q16

What is the internal energy of an object?

- A** It is the energy associated with the object's movement through space.
B It is the energy associated with the random movement of the molecules in the object.
C It is the energy due to the attractions between the molecules within the object.
D It is the sum of all the microscopic potential and kinetic energies of the molecules.

28 9702/01/M/J/06/Q17

A motorist travelling at 10 m s^{-1} can bring his car to rest in a braking distance of 10 m.

In what distance could he bring the car to rest from a speed of 30 m s^{-1} using the same braking force?

- A** 17 m **B** 30 m **C** 52 m **D** 90 m

29 9702/01/O/N/06/Q16

To get to his office from the entrance of the building, a man has to walk up six flights of stairs. The height of each flight is 2.5 m and the man has a mass of 80 kg.

What is the approximate gain in the man's gravitational potential energy during the climb?

- A** 1200 J **B** 2000 J **C** 4800 J **D** 12 000 J

30 9702/01/O/N/06/Q17

In many old-style filament lamps, as much as 93 J of energy is emitted as thermal energy for every 7 J of energy emitted as light.

What is the efficiency of the lamp, as the percentage of electrical energy converted to light energy?

- A** 7 % **B** 8 % **C** 92 % **D** 93 %

31 9702/01/M/J/07/Q12

The diagram shows the masses and velocities of two trolleys about to collide.

After the impact they move off together.

What is the total kinetic energy of the trolleys after the collision?

- A 1.3 J B 12 J
C 18 J D 19 J

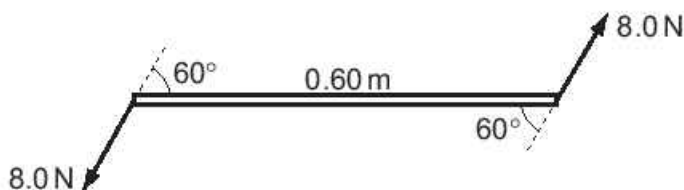


32 9702/01/M/J/07/Q13

Two 8.0 N forces act at each end of a beam of length 0.60 m. The forces are parallel and act in opposite directions. The angle between the forces and the beam is 60° .

What is the torque of the couple exerted on the beam?

- A 2.4 N m B 4.2 N m
C 4.8 N m D 9.6 N m



33 9702/01/M/J/07/Q14

Which expression **defines** power?

- A force $\times$ distance moved in the direction of the force
B force $\times$ velocity
C work done $\div$ time taken
D work done $\times$ time taken

34 9702/01/O/N/07/Q15

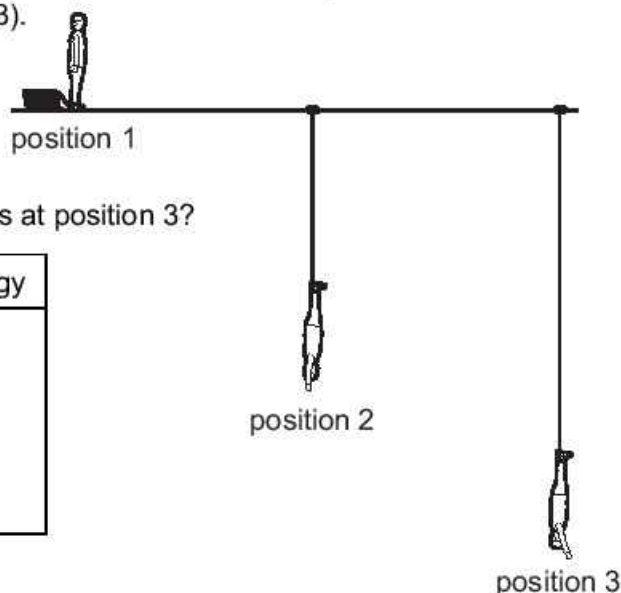
A car of mass 1000 kg first travels forwards at 25 m s^{-1} and then backwards at 5 m s^{-1} .

What is the change in the kinetic energy of the car?

- A 200 kJ B 300 kJ C 325 kJ D 450 kJ

35 9702/01/O/N/07/Q16

When bungee jumping, a student starts with maximum gravitational potential energy (position 1), then falls freely until the rope fully unwinds (position 2), after which the rope starts to stretch until the lowest point of the jump is reached (position 3).



What are the kinetic and elastic potential energies at position 3?

	kinetic energy	elastic potential energy
A	maximum	maximum
B	maximum	minimum
C	minimum	maximum
D	minimum	minimum

36 9702/01/M/J/08/Q17

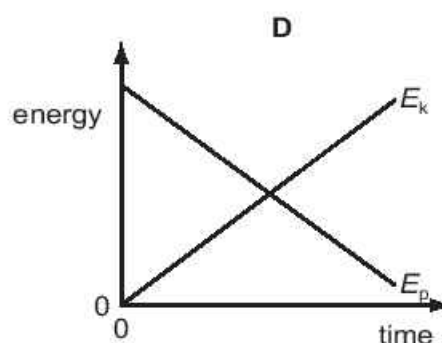
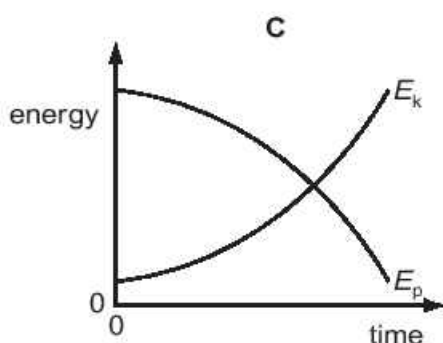
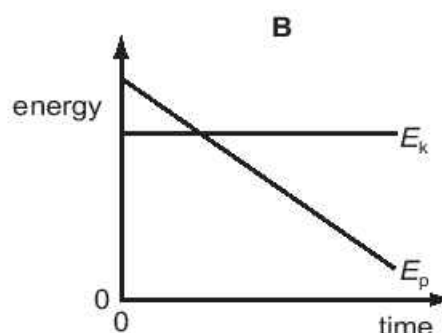
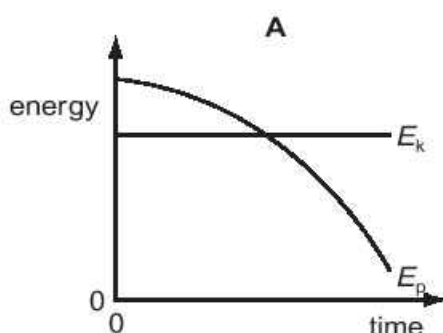
Which quantities are conserved in an inelastic collision?

	kinetic energy	total energy	linear momentum
A	conserved	not conserved	conserved
B	conserved	not conserved	not conserved
C	not conserved	conserved	conserved
D	not conserved	conserved	not conserved

37 9702/01/M/J/08/Q18

A steel ball is falling at constant speed in oil.

Which graph shows the variation with time of the gravitational potential energy E_p and the kinetic energy E_k of the ball?



38 9702/01/M/J/08/Q19

The total energy input E_{in} in a process is partly transferred to useful energy output U , and partly to energy that is wasted W .

What is the efficiency of the process?

- A $\frac{U}{W} \times 100\%$ B $\frac{W}{E_{in}} \times 100\%$ C $\frac{U}{E_{in}} \times 100\%$ D $\frac{U+W}{E_{in}} \times 100\%$

39 9702/01/M/J/09/Q14

The forward motion of a motor-boat is opposed by forces F which vary with the boat's speed v in accordance with the relation $F = kv^2$, where k is a constant.

The effective power of the propellers required to maintain the speed v is P .

Which expression relates k , P and v ?

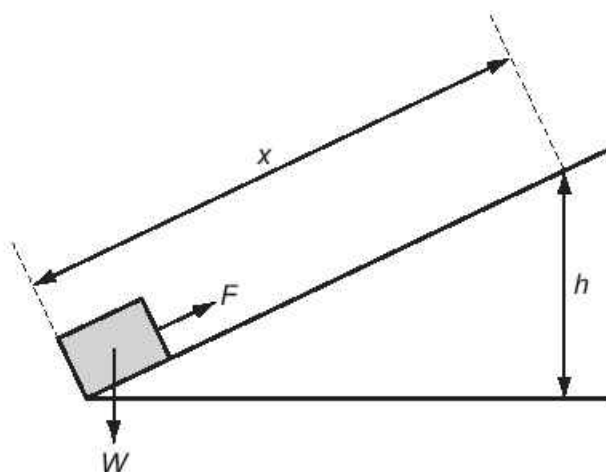
- A $k = \frac{P}{v}$ B $k = \frac{P}{v^2}$ C $k = \frac{P}{v^3}$ D $k = \frac{P}{v^4}$

40 9702/01/O/N/08/Q15

A block of weight W is pulled up a rough slope by a force F .
When the block has moved a distance x along the slope, it has risen height h .

Which expressions give the amount of work done on the block and the amount of gravitational potential energy gained by the block?

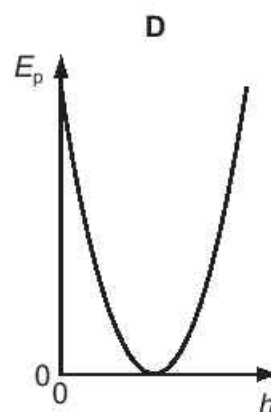
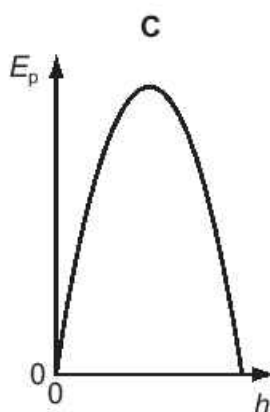
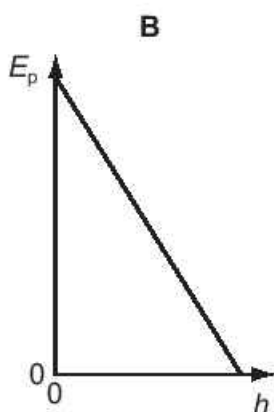
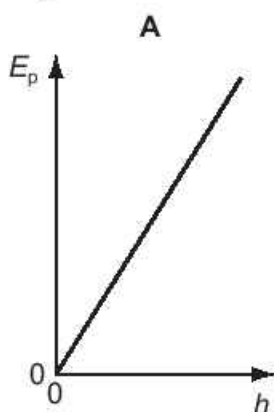
	work done	gravitational potential energy
A	Fx	Wh
B	Fh	Wx
C	Wx	Fh
D	Wh	Fx



41 9702/01/O/N/08/Q16

An object is thrown into the air.

Which graph shows how the potential energy E_p of the object varies with height h above the ground?

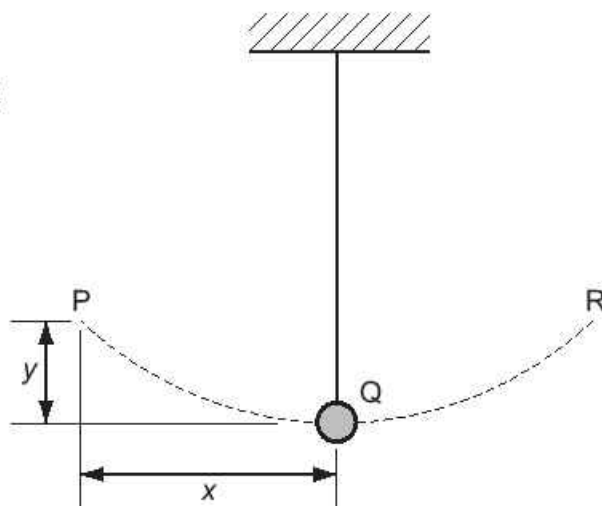


42 9702/01/O/N/08/Q17

A pendulum bob oscillates between P and R.

Assuming the gravitational potential energy lost in moving from P to Q is converted into kinetic energy, what is the speed of the bob at Q?

- A $\sqrt{2gx}$
- B $2gx$
- C $\sqrt{2gy}$
- D $2gy$



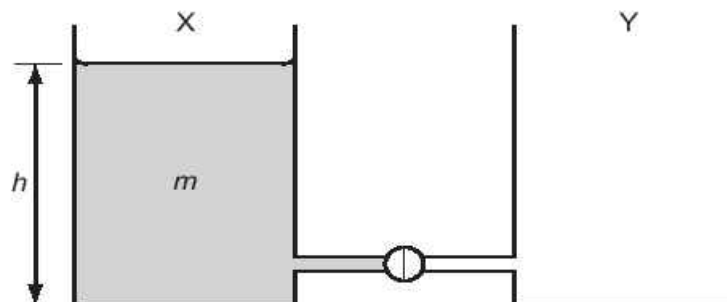
43 9702/01/O/N/08/Q18

Which operation involves the greatest mean power?

- A a car moving against a resistive force of 0.4 kN at a constant speed of 20 m s^{-1}
- B a crane lifting a weight of 3 kN at a speed of 2 m s^{-1}
- C a crane lifting a weight of 5 kN at a speed of 1 m s^{-1}
- D a weight being pulled across a horizontal surface at a speed of 6 m s^{-1} against a frictional force of 1.5 kN

44 9702/01/M/J/09/Q15

The diagram shows two identical vessels X and Y connected by a short pipe with a tap.



Initially, X is filled with water of mass m to a depth h , and Y is empty.

When the tap is opened, water flows from X to Y until the depths of water in both vessels are equal.

How much potential energy is lost by the water during this process? (g = acceleration of free fall)

- A 0
- B $\frac{mgh}{4}$
- C $\frac{mgh}{2}$
- D mgh

45 9702/12/O/N/09/Q13

A projectile is launched at 45° to the horizontal with initial kinetic energy E .

Assuming air resistance to be negligible, what will be the kinetic energy of the projectile when it reaches its highest point?

- A $0.50E$
- B $0.71E$
- C $0.87E$
- D E

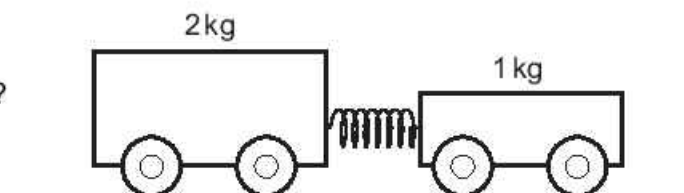
46 9702/12/O/N/09/Q14

Two trolleys are placed together on a horizontal runway with a compressed spring between them.

When they are released, the 2 kg trolley moves to the left at 2 m s^{-1} .

How much energy was stored in the spring?

- A 4 J
- B 6 J
- C 8 J
- D 12 J



47 9702/12/M/J/10/Q16

A force of 1000 N is needed to lift the hook of a crane at a steady velocity. The crane is then used to lift a load of mass 1000 kg at a velocity of 0.50 m s^{-1} .

How much of the power developed by the motor of the crane is used in lifting the hook and the load? Assume that the acceleration of free fall g is equal to 10 m s^{-2} .

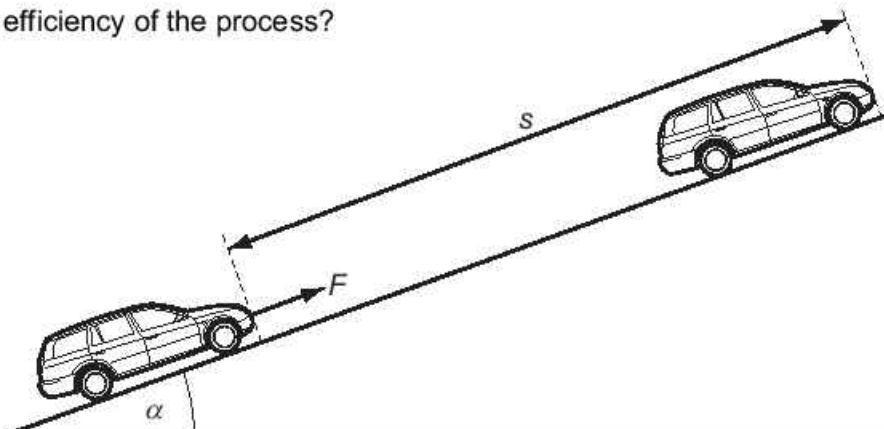
- A 5.0 kW
- B 5.5 kW
- C 20 kW
- D 22 kW

48 9702/12/M/J/10/Q15

A constant force F , acting on a car of mass m , moves the car up the slope through a distance s at constant velocity v . The angle of the slope to the horizontal is α .

Which expression gives the efficiency of the process?

- A $\frac{mgs \sin \alpha}{Fv}$
- B $\frac{mv}{Fs}$
- C $\frac{mv^2}{2Fs}$
- D $\frac{mg \sin \alpha}{F}$



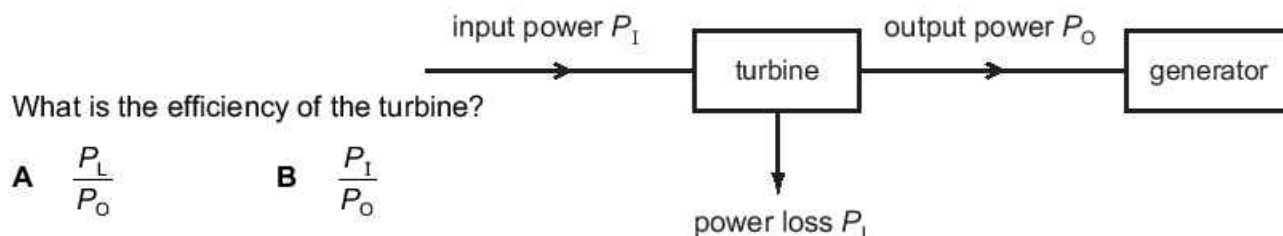
49 9702/11/O/N/10/Q16

What is the internal energy of a system?

- A the amount of heat supplied to the system
- B the energy of the atoms of the system
- C the total kinetic energy of the system
- D the total potential energy of the system

50 9702/11/O/N/10/Q17

A steam turbine is used to drive a generator. The input power to the turbine is P_I and the output power P_O . The power loss in the turbine is P_L , as shown below.



What is the efficiency of the turbine?

- A $\frac{P_L}{P_O}$
- B $\frac{P_I}{P_O}$
- C $\frac{P_L}{P_I}$
- D $\frac{P_O}{P_I}$

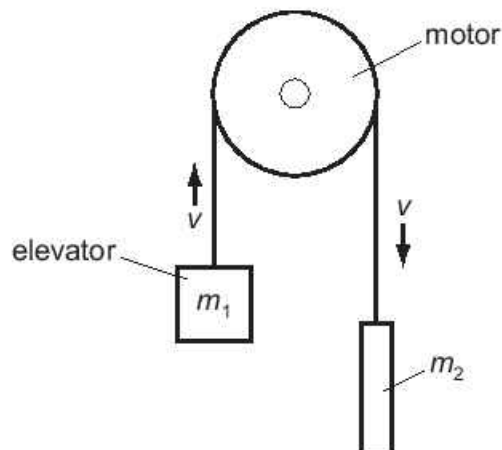
51 9702/11/O/N/10/Q18

The diagram shows a lift system in which the elevator (mass m_1) is partly counterbalanced by a heavy weight (mass m_2).

At what rate does the motor provide energy to the system when the elevator is rising at a steady speed v ?

(g = acceleration of free fall)

- A $\frac{1}{2} m_1 v^2$
- B $\frac{1}{2} (m_1 - m_2) v^2$
- C $m_1 g v$
- D $(m_1 - m_2) g v$

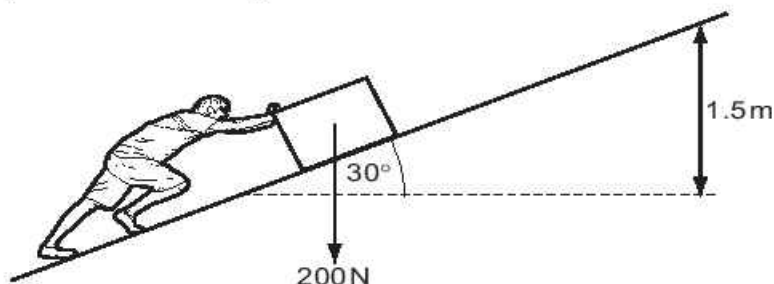


52 9702/12/O/N/10/Q14

A box of weight 200 N is pushed so that it moves at a steady speed along a ramp, through a height of 1.5 m. The ramp makes an angle of 30° with the ground. The frictional force on the box is 150 N while the box is moving.

What is the work done by the person?

- A 150 J
- B 300 J
- C 450 J
- D 750 J



53 9702/12/O/N/10/Q15

A raindrop of mass m is falling vertically through the air with a steady speed v . The raindrop experiences a retarding force kv due to the air, where k is a constant. The acceleration of free fall is g .

Which expression gives the kinetic energy of the raindrop?

- A $\frac{mg}{k}$
- B $\frac{mg^2}{2k^2}$
- C $\frac{m^3g^2}{k^2}$
- D $\frac{m^3g^2}{2k^2}$

54 9702/12/O/N/10/Q16

The kinetic energy of a vehicle of mass 1000 kg is 4.5×10^5 J. It is stopped by applying a constant braking force of 6000 N.

What is its stopping distance?

- A 37 m
- B 75 m
- C 150 m
- D 300 m

55 9702/12/O/N/10/Q17

What are units of work, energy and power?

	work	energy	power
A	J	Nm	J
B	J s^{-1}	J	J s^{-1}
C	Nm	Nm	W
D	Nm	W	W

56 9702/11/M/J/11/Q14

A steel sphere is dropped vertically onto a horizontal metal plate. The sphere hits the plate with a speed u , leaves it at a speed v , and rebounds vertically to half of its original height.

Which expression gives the value of $\frac{v}{u}$?

- A $\frac{1}{2^2}$
- B $\frac{1}{2}$
- C $\frac{1}{\sqrt{2}}$
- D $1 - \frac{1}{\sqrt{2}}$

57 9702/13/M/J/17/Q17

A man has a mass of 80 kg. He ties himself to one end of a rope which passes over a single fixed pulley. He pulls on the other end of the rope to lift himself up at an average speed of 50 cm s^{-1} .

What is the average useful power at which he is working?

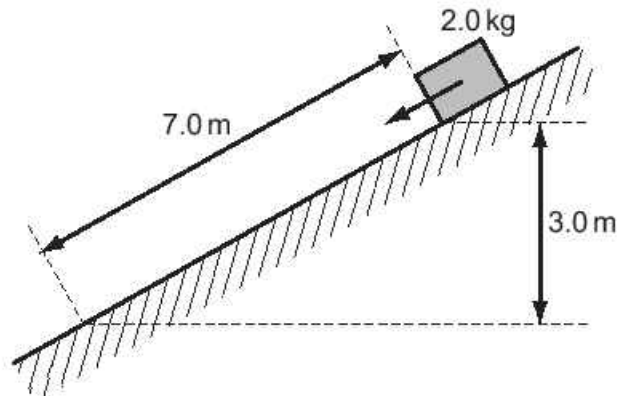
- A 40 W
- B 0.39 kW
- C 4.0 kW
- D 39 kW

58 9702/11/M/J/11/Q15

A block of mass 2.0 kg is released from rest on a slope. It travels 7.0 m down the slope and falls a vertical distance of 3.0 m . The block experiences a frictional force parallel to the slope of 5.0 N .

What is the speed of the block after falling this distance?

- A 4.9 ms^{-1} B 6.6 ms^{-1}
C 8.6 ms^{-1} D 10.1 ms^{-1}



59 9702/11/M/J/11/Q17

A body travelling with a speed of 10 ms^{-1} has kinetic energy 1500 J .

If the speed of the body is increased to 40 ms^{-1} , what is its new kinetic energy?

- A 4500 J B 6000 J C $24\,000\text{ J}$ D $1\,350\,000\text{ J}$

60 9702/11/M/J/11/Q18

Which process does **not** require energy to be supplied?

- A boiling B evaporation C freezing D melting

61 9702/12/M/J/11/Q18

An electric motor produces 120 W of useful mechanical output power. The efficiency of the motor is 60% .

Which row is correct?

	electrical power input / W	waste heat power output / W
A	72	48
B	192	72
C	200	72
D	200	80

62 9702/12/M/J/11/Q19

A hammer with 10 J of kinetic energy hits a nail and pushes it 5.0 mm into a plank.

Both the hammer and nail come to rest after the collision.

What is the average force that acts on the nail while it moves the 5.0 mm ?

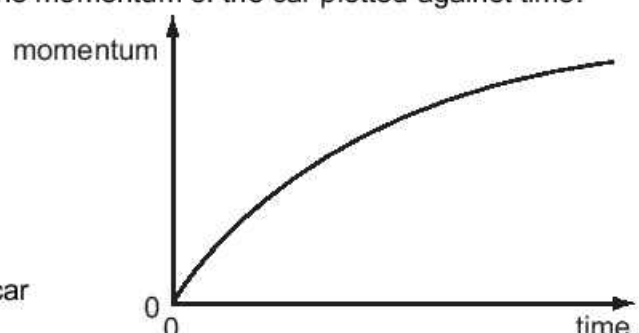
- A 0.050 N B 2.0 N C 50 N D 2000 N

63 9702/11/O/N/11/Q11

A car accelerates from rest. The graph shows the momentum of the car plotted against time.

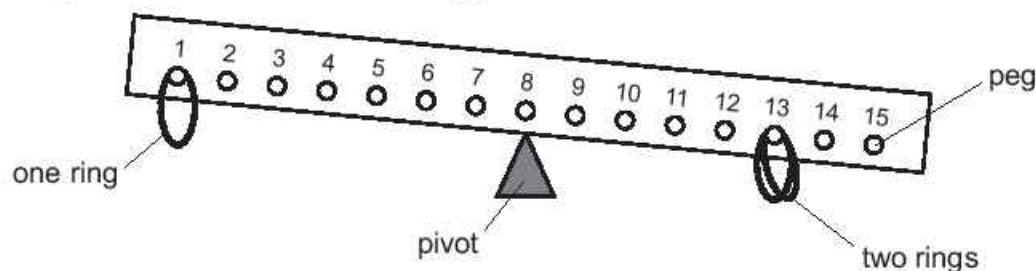
What is the meaning of the gradient of the graph at a particular time?

- A the resultant force on the car
B the velocity of the car
C the kinetic energy of the car
D the rate of change of kinetic energy of the car



64 9702/11/O/N/11/Q15

The diagram shows a child's balancing game.



The wooden rod is uniform and all the rings are of equal mass. Two rings are hung on peg 13 and one on peg 1.

On which hook must a fourth ring be hung in order to balance the rod?

- A 2 B 3 C 5 D 6

65 9702/11/O/N/11/Q16

The diagram shows a particle X, with kinetic energy E_k , about to collide with a stationary particle Y. Both particles have the same mass.

After colliding, X and Y travel onwards together as a single larger particle.

How much kinetic energy is lost in the collision?



- A 0 B $\frac{E_k}{4}$ C $\frac{E_k}{2}$ D $\frac{3E_k}{4}$

66 9702/11/O/N/11/Q17

The first column in the table gives four examples of work being done. The second column gives more detail of the action.

Which row is **not** correct?

	example	detail
A	a girl dives from a diving board into a swimming pool	work is done by the girl against gravity as she falls
B	a man pushes a car along a level road	work is done by the man against friction
C	an electron is accelerated towards a positively-charged plate	work is done on the electron by the electric field of the plate
D	a piston is pushed outwards as a gas expands	work is done on the atmosphere by the gas

67 9702/11/O/N/11/Q19

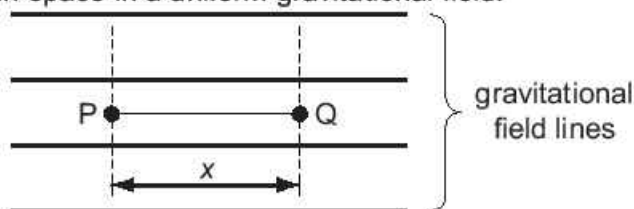
An electric motor is required to produce 120 W of mechanical output power. The efficiency of the motor is 80 %.

Which row is correct?

	electrical power input to motor / W	waste heat output from motor / W
A	120	24
B	120	96
C	150	30
D	150	120

68 9702/11/O/N/11/Q20

A mass m is situated in space in a uniform gravitational field.



When the mass moves through a displacement x , from P to Q, it loses an amount of potential energy E .

Which row correctly specifies the magnitude and the direction of the acceleration due to the gravity in this field?

	magnitude	direction
A	$\frac{E}{mx}$	$\rightarrow$
B	$\frac{E}{mx}$	$\leftarrow$
C	$\frac{E}{x}$	$\rightarrow$
D	$\frac{E}{x}$	$\leftarrow$

69 9702/12/O/N/11/Q15

When a horizontal force F is applied to a frictionless trolley over a distance s , the kinetic energy of the trolley changes from 4 J to 8 J.

If a force of $2F$ is applied to the trolley over a distance of $2s$, what will the original kinetic energy of 4 J become?

- A 16 J B 20 J C 32 J D 64 J

70 9702/12/O/N/11/Q16

The kinetic energy of a vehicle of mass 1000 kg is 4.5×10^5 J. It is braked with a total constant braking force of 6000 N.

What will be its stopping distance?

- A 37 m B 75 m C 150 m D 300 m

71 9702/12/O/N/11/Q17

In many old-style filament lamps, as much as 92 J of energy is emitted as thermal energy for every 8 J of energy emitted as light.

What is the efficiency of the lamp, as the percentage of electrical energy converted to light energy?

- A 8 % B 9 % C 91 % D 92 %

72 9702/12/O/N/11/Q18

What is the unit of power in SI base units?

- A kg ms^{-2} B kg m s^{-3} C $\text{kg m}^2 \text{s}^{-2}$ D $\text{kg m}^2 \text{s}^{-3}$

73 9702/12/M/J/12/Q19

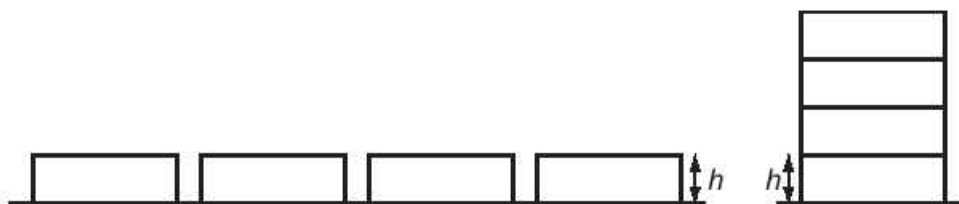
A car engine exerts an average force of 500 N in moving the car 1.0 km in 200 s.

What is the average power developed by the engine?

- A 2.5 W B 2.5 kW C 100 kW D 100 MW

74 9702/12/M/J/12/Q17

Initially, four identical uniform blocks, each of mass m and thickness h , are spread on a table.



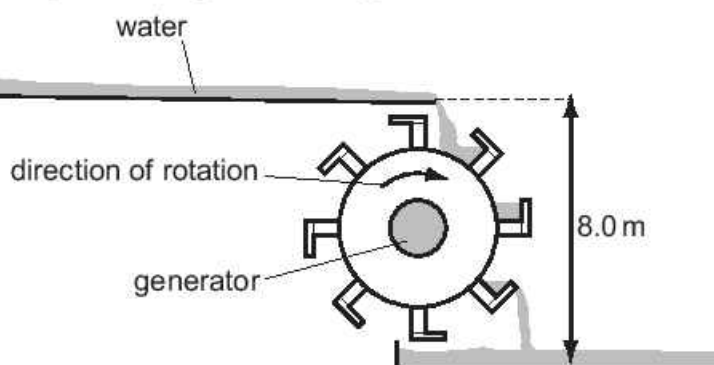
How much work is done on the blocks in stacking them on top of one another?

- A $3mgh$ B $6mgh$ C $8mgh$ D $10mgh$

75 9702/12/M/J/12/Q18

The diagram shows the design of a water wheel which drives a generator to produce electrical energy. The flow rate of the water is 200 kg s^{-1} . The generator supplies a current of 32 A at a voltage of 230 V .

Ignoring any changes in kinetic energy of the water, what is the efficiency of the system?



- A 14 % B 16 %
C 22 % D 47 %

76 9702/13/M/J/12/Q16

The force resisting the motion of a car is taken as being proportional to the square of the car's speed. The magnitude of the force at a speed of 20 m s^{-1} is 800 N .

What effective power is required from the car's engine to maintain a steady speed of 40 m s^{-1} ?

- A 32 kW B 64 kW C 128 kW D 512 kW

77 9702/13/M/J/12/Q17

A concrete cube of side 0.60 m and uniform density $2.0 \times 10^3 \text{ kg m}^{-3}$ is lifted 5.0 m vertically by a crane.

What is the change in potential energy of the cube?

- A 2.2 kJ B 21 kJ C 59 kJ D 450 kJ

78 9702/01/M/J/06/Q16

What is the internal energy of an object?

- A It is the energy associated with the object's movement through space.
B It is the energy associated with the random movement of the molecules in the object.
C It is the energy due to the attractions between the molecules in the object.
D It is the sum of all the microscopic potential and kinetic energies of the molecules in the object.

79 9702/13/M/J/12/Q19

The data below are taken from a test of a petrol engine for a motor car.

power output	150 kW
fuel consumption	20 litres per hour
energy content of fuel	40 MJ per litre

Which expression will evaluate the efficiency of the engine?

- A $\frac{150 \times 10^3}{40 \times 10^6 \times 20 \times 60 \times 60}$ C $\frac{150 \times 10^3 \times 40 \times 10^6 \times 20}{60 \times 60}$
- B $\frac{150 \times 10^3 \times 60 \times 60}{20 \times 40 \times 10^6}$ D $\frac{150 \times 10^3 \times 20}{40 \times 10^3 \times 60 \times 60}$

80 9702/12/O/N/12/Q19

A piston in a gas supply pump has an area of 500 cm^2 and it moves a distance of 30 cm during one stroke.

The pump moves the gas against a fixed pressure of 4000 Pa.

How much work is done by the piston during one stroke?

- A 60 J B $6.0 \times 10^3 \text{ J}$ C $6.0 \times 10^5 \text{ J}$ D $6.0 \times 10^7 \text{ J}$

81 9702/12/O/N/12/Q20

A railway engine accelerates a train of total mass 1200 tonnes (1 tonne = 1000 kg) from rest to a speed of 75 m s^{-1} .

How much useful work must be done on the train to reach this speed?

- A $1.7 \times 10^6 \text{ J}$ B $3.4 \times 10^6 \text{ J}$ C $1.7 \times 10^9 \text{ J}$ D $3.4 \times 10^9 \text{ J}$

82 9702/12/O/N/12/Q21

A crane is being used to lift containers off a ship. One container has a mass of 14 000 kg and is being lifted vertically with a speed of 3.2 m s^{-1} .

The electric motor being used to supply the power to lift the container is using a current of 240 A at a potential difference of 2200 V.

What is the efficiency of the system?

- A 8.1 % B 8.5 % C 48 % D 83 %

83 9702/12/O/N/12/Q22

Trains supply coal to a power station. The table below gives quantities describing the operation of the power station.

	symbol	unit
power station output	P	W
number of trains per day	N	
mass of coal on a train	M	kg
energy from 1 kg of coal	J	J
number of seconds in one day	S	

Which expression gives the efficiency of the power station?

- A $\frac{PS}{NMJ}$ B $\frac{PSN}{MJ}$ C $\frac{NMJ}{PS}$ D $\frac{NM}{PSJ}$

84 9702/13/O/N/12/Q18

A car travelling with speed 28 m s^{-1} leaves a motorway on an exit road. The end of the exit road is 22 m higher than the motorway.

If only the force of gravity is considered, what will be the speed of the car at the end of the exit road?

- A 7.3 m s^{-1} B 19 m s^{-1} C 21 m s^{-1} D 24 m s^{-1}

85 9702/13/O/N/12/Q20

A transformer has the following input and output.

What is the efficiency of the transformer?

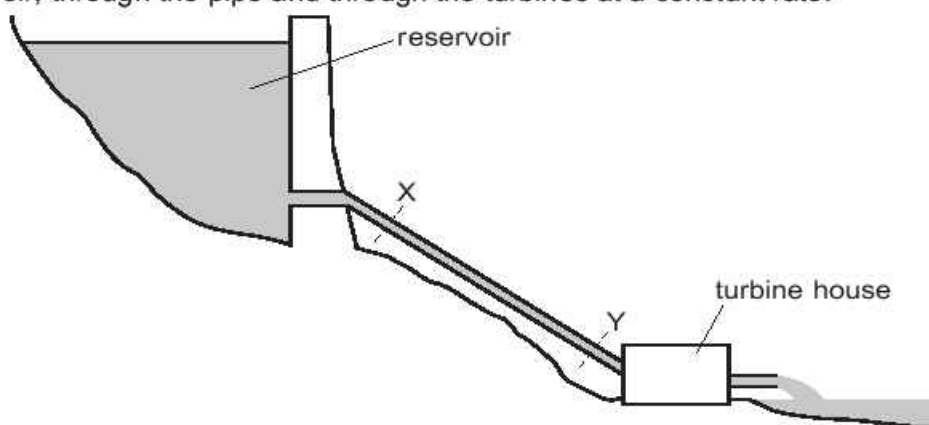
- A 0.94% B 1.0%
C 11% D 94%

	potential difference / V	current / A
input	11 000	28
output	240	1200

86 9702/13/O/N/12/Q21

The diagram shows a hydroelectric power station.

The reservoir is linked to the turbines by a pipe of uniform cross-sectional area. Water flows from the reservoir, through the pipe and through the turbines at a constant rate.



Which statement about the change of energy of the water as it moves from X to Y is correct?

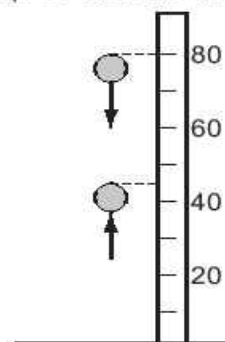
- A It gains both gravitational potential energy and kinetic energy.
B It loses gravitational potential energy and gains elastic potential energy.
C It loses gravitational potential energy and gains kinetic energy.
D It loses both elastic potential energy and gravitational potential energy.

87 9702/11/M/J/13/Q17

A solid rubber ball has a diameter of 8.0 cm . It is released from rest with the top of the ball 80 cm above a horizontal surface. It falls vertically and then bounces back up so that the maximum height reached by the top of the ball is 45 cm , as shown.

If the kinetic energy of the ball is 0.75 J just before it strikes the surface, what is its kinetic energy just after it leaves the surface?

- A 0.36 J B 0.39 J C 0.40 J D 0.42 J



88 9702/11/M/J/13/Q18

A wind turbine has blades that sweep an area of 2000 m^2 . It converts the power available in the wind to electrical power with an efficiency of 50%.

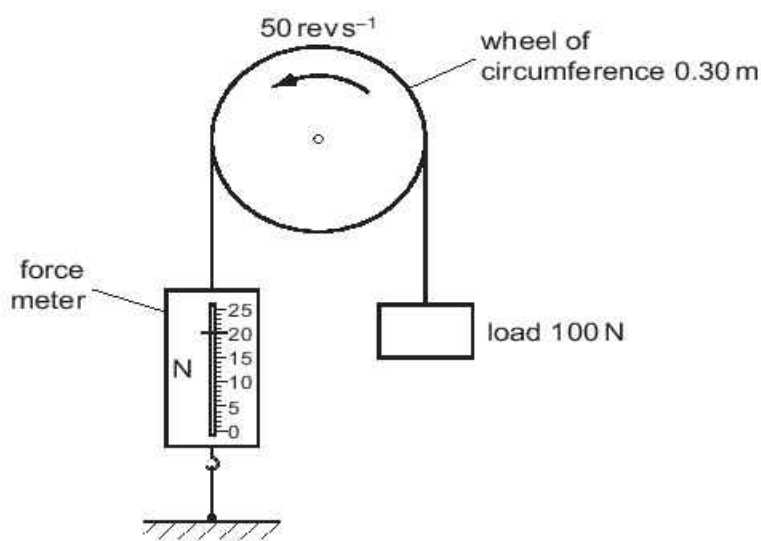
What is the electrical power generated if the wind speed is 10 ms^{-1} ? (The density of air is 1.3 kg m^{-3} .)

- A** 130 kW **B** 650 kW **C** 1300 kW **D** 2600 kW

89 9702/11/M/J/13/Q19

The diagram shows a wheel of circumference 0.30 m. A rope is fastened at one end to a force meter. The rope passes over the wheel and supports a freely hanging load of 100 N. The wheel is driven by an electric motor at a constant rate of 50 revolutions per second.

When the wheel is turning at this rate, the force meter reads 20 N.



What is the output power of the motor?

- A** 0.3 kW **B** 1.2 kW
C 1.8 kW **D** 3.8 kW

90 9702/12/M/J/13/Q15

A ball is thrown vertically upwards.

Neglecting air resistance, which statement is correct?

- A** The kinetic energy of the ball is greatest at the greatest height attained.
B By the principle of conservation of energy, the total energy of the ball is constant throughout its motion.
C By the principle of conservation of momentum, the momentum of the ball is constant throughout its motion.
D The potential energy of the ball increases uniformly with time during its ascent.

91 9702/12/M/J/13/Q16

A bow of mass 400 g shoots an arrow of mass 120 g vertically upwards. The potential energy stored in the bow just before release is 80 J. The system has an efficiency of 28%.

What is the height reached by the arrow when air resistance is neglected?

- A** 4 m **B** 19 m **C** 187 m **D** 243 m

92 9702/11/M/J/14/Q14

What is the average power output of a laser that can deliver 0.20 J of energy in 10 ns?

- A** 2 nW **B** 20 mW **C** 200 kW **D** 20 MW

93 9702/12/M/J/13/Q17

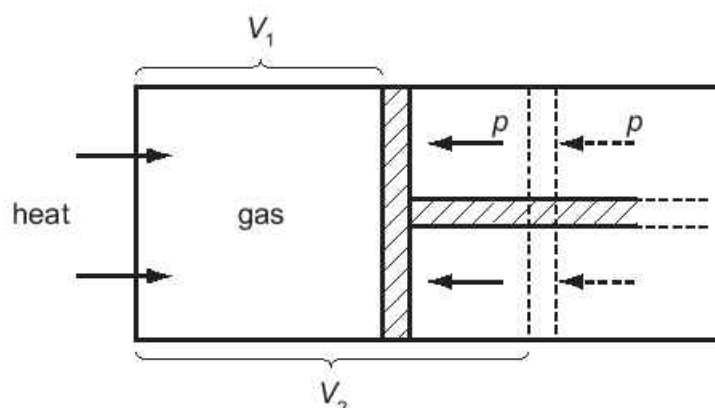
A train on a mountain railway is carrying 200 people of average mass 70 kg up a slope at an angle of 30° to the horizontal and at a speed of 6.0 ms^{-1} . The train itself has a mass of 80 000 kg. The percentage of the power from the engine which is used to raise the passengers and the train is 40 %.

What is the power of the engine?

- A 1.1 MW B 2.8 MW C 6.9 MW D 14 MW

94 9702/12/M/J/13/Q18

A gas is enclosed inside a cylinder which is fitted with a frictionless piston.



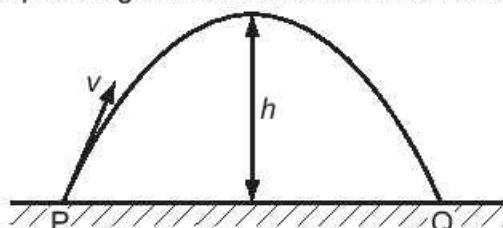
Initially, the gas has a volume V_1 and is in equilibrium with an external pressure p . The gas is then heated slowly so that it expands, pushing the piston back until the volume of the gas has increased to V_2 .

How much work is done by the gas during this expansion?

- A $p(V_2 - V_1)$ B $\frac{1}{2}p(V_2 - V_1)$ C $p(V_2 + V_1)$ D $\frac{1}{2}p(V_2 + V_1)$

95 9702/13/M/J/13/Q14

A ball of mass m is thrown up to height h in air with an initial velocity v , as shown.



Air resistance is considered negligible. The acceleration of free fall is g .

What is the **total** work done by the gravitational force on the ball during its flight from P to Q?

- A zero B $\frac{1}{2}mv^2$ C mgh D $2mgh$

96 9702/13/M/J/13/Q15

A spring of unextended length 40 mm is suspended from a fixed point. A load of 16 N is applied to the free end of the spring. This causes the spring to extend so that its final length is five times its original length. The spring obeys Hooke's Law.

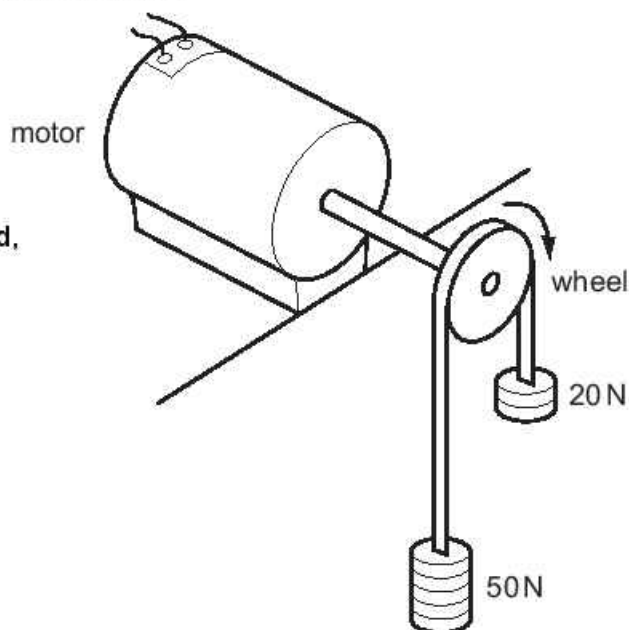
What is the energy stored in the spring due to this extension?

- A 1.3 J B 1.6 J C 2.6 J D 3.2 J

97 9702/13/M/J/13/Q16

The diagram shows an arrangement used to find the output power of an electric motor.

The wheel attached to the motor's axle has a circumference of 0.5 m and the belt which passes over it is stationary when the weights have the values shown.



If the wheel is making 20 revolutions per second, what is the output power of the motor?

- A 300 W B 500 W
C 600 W D 700 W

98 9702/12/O/N/13/Q16

What is the internal energy of a system?

- A the amount of heat supplied to the system C the total kinetic energy of the system
B the random energy of the atoms of the system D the total potential energy of the system

99 9702/12/O/N/13/Q19

A turbine at a hydroelectric power station is situated 30 m below the level of the surface of a large lake. The water passes through the turbine at a rate of 340 m^3 per minute.

The overall efficiency of the turbine and generator system is 90%.

What is the output power of the power station? (The density of water is 1000 kg m^{-3} .)

- A 0.15 MW B 1.5 MW C 1.7 MW D 90 MW

100 9702/13/O/N/13/Q17

The pump of a water pumping system uses 2.0 kW of electrical power when raising water. The pumping system lifts 16 kg of water per second through a vertical height of 7.0 m.

What is the efficiency of the pumping system?

- A 1.8% B 5.6% C 22% D 55%

101 9702/13/O/N/13/Q18

A body travelling with a speed of 20 m s^{-1} has kinetic energy E_k .

If the speed of the body is increased to 80 m s^{-1} , what is its new kinetic energy?

- A $4E_k$ B $8E_k$ C $12E_k$ D $16E_k$

102 9702/11/M/J/14/Q16

The engine of a car exerts a force of 600 N in moving the car 1.0 km in 150 seconds.

What is the average output power of the engine?

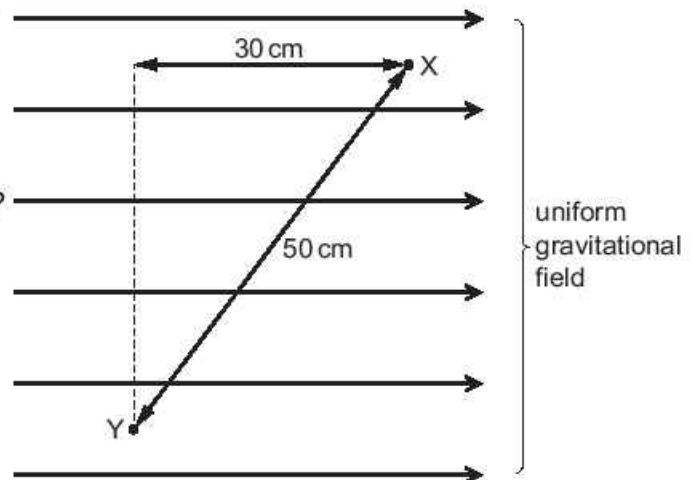
- A 4.0 W B 4.0 kW C 90 kW D 90 MW

103 9702/12/M/J/14/Q14

A mass at point X inside a uniform gravitational field experiences a gravitational force of 0.200 N. It has 1.00 J of gravitational potential energy.

The mass is then moved to point Y.
What is its new gravitational potential energy?

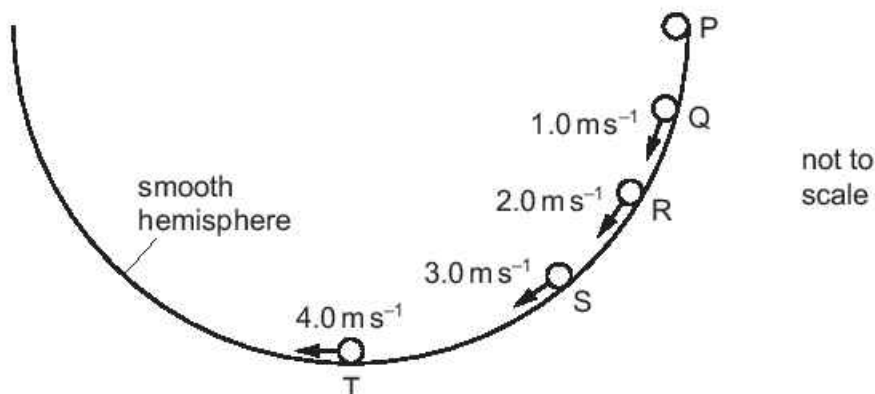
- A 0.90 J B 0.94 J
C 1.06 J D 1.10 J



104 9702/12/M/J/14/Q15

A small mass is placed at point P on the inside surface of a smooth hemisphere. It is then released from rest. When it reaches the lowest point T, its speed is 4.0 m s^{-1} .

The diagram (not to scale) shows the speed of the mass at other points Q, R and S as it slides down. Air resistance is negligible.



The mass loses potential energy E in falling from P to T.

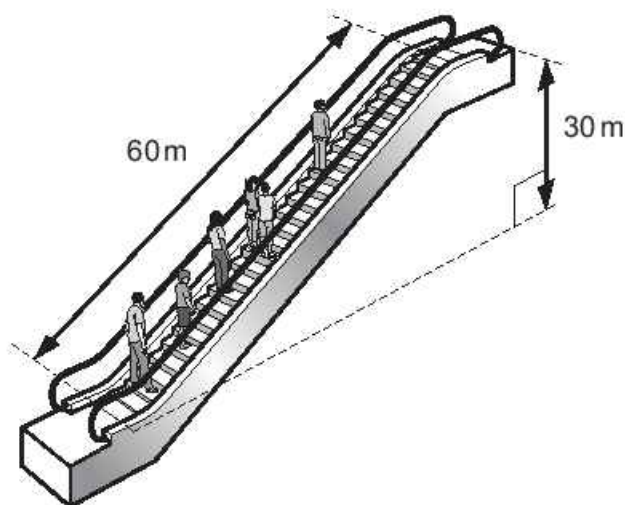
At which point has the mass lost potential energy $\frac{E}{4}$?

- A Q B R C S D none of these

Space for working

105 9702/12/M/J/14/Q16

An escalator is 60 m long and lifts passengers through a vertical height of 30 m, as shown.



To drive the escalator against the forces of friction when there are no passengers requires a power of 2.0 kW.

The escalator is used by passengers of average mass 60 kg and the power to overcome friction remains constant.

How much power is required to drive the escalator when it is carrying 20 passengers and is travelling at 0.75 ms^{-1} ?

- A 4.4 kW B 6.4 kW C 8.8 kW D 10.8 kW

106 9702/13/M/J/14/Q12

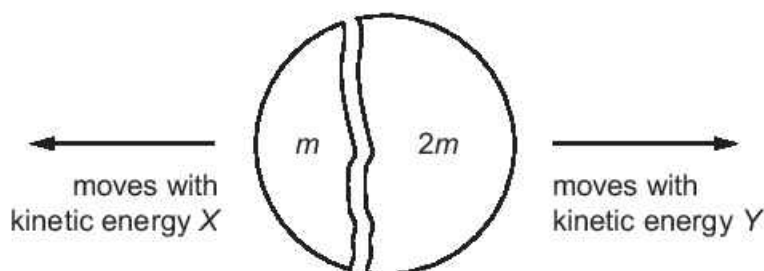
A stationary body explodes into two components of masses m and $2m$.

The components gain kinetic energies X and Y respectively.

What is the value of the ratio $\frac{X}{Y}$?

- A $\frac{1}{4}$ B $\frac{1}{2}$

- C $\frac{2}{1}$ D $\frac{4}{1}$



107 9702/12/O/N/14/Q13

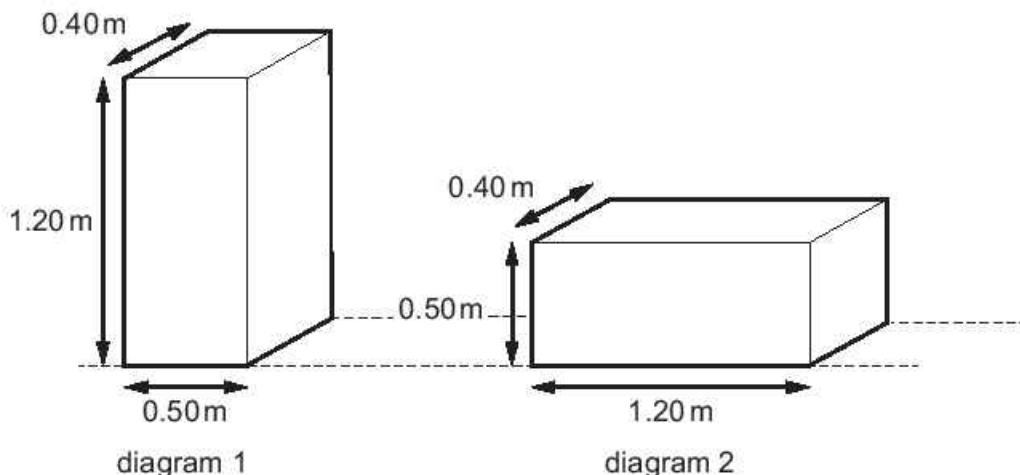
A mass attached to the lower end of a spring bounces up and down.

At which points in the path of the mass do the gravitational potential energy of the mass (GPE), the elastic potential energy in the spring (EPE) and the kinetic energy of the mass (KE) have their highest values?

	GPE	EPE	KE
A	bottom	middle	top
B	bottom	top	middle
C	top	bottom	middle
D	top	bottom	top

108 9702/13/M/J/14/Q14

A uniform solid cuboid of concrete of dimensions $0.50\text{ m} \times 1.20\text{ m} \times 0.40\text{ m}$ and weight 4000 N rests on a flat surface with the 1.20 m edge vertical as shown in diagram 1.

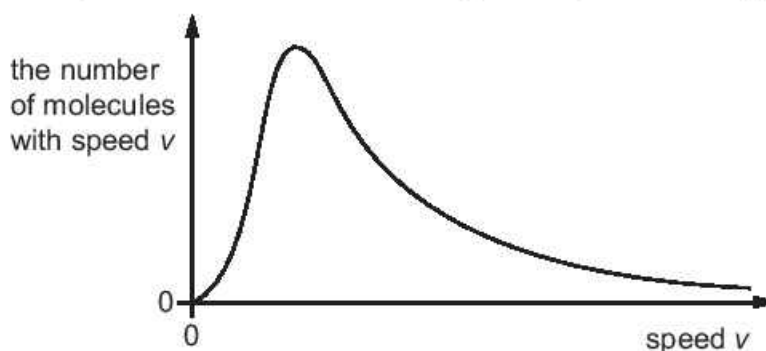


What is the minimum energy required to roll the cuboid through 90° to the position shown in diagram 2 with the 0.50 m edge vertical?

- A 200 J B 400 J C 1400 J D 2600 J

109 9702/13/M/J/14/Q21

The graph shows the distribution of speeds for the molecules of a gas at a particular temperature.



Which statement is correct?

- A All the molecules have the same kinetic energy.
 B The commonest value of speed is also the average speed.
 C The graph shows that the molecules of a gas are widely spaced apart.
 D The peak value of the graph would move to the right if the temperature is increased.

110 9702/11/M/J/15/Q17

Which statement is correct?

- A A ball lands on the ground and bounces. The kinetic energy changes sign, because the ball changes direction.
 B A car drives up a slope at a steady speed. The power generated by the engine equals the potential energy gained per unit time.
 C An electric heater can be 100% efficient.
 D It is impossible for momentum to be conserved in a collision.

111 9702/12/O/N/09/Q13

A projectile is launched at 45° to the horizontal with initial kinetic energy E .

Assuming air resistance to be negligible, what will be the kinetic energy of the projectile when it reaches its highest point?

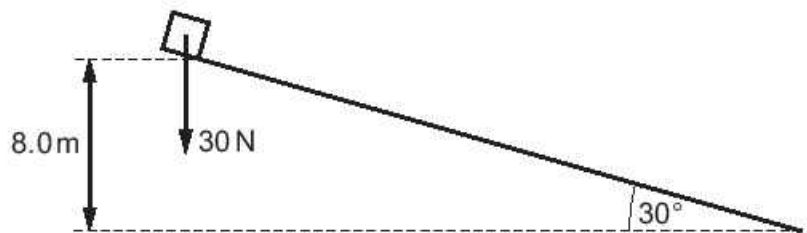
- A $0.50E$ B $0.71E$ C $0.87E$ D E

112 9702/13/O/N/14/Q17

A box of weight 30 N is released from rest on a ramp that is at an angle of 30° to the horizontal. The box slides down the ramp so that it falls through a vertical distance of 8.0 m . A constant frictional force of 10 N acts on the box while it is moving.

What is the kinetic energy of the box after falling through this distance?

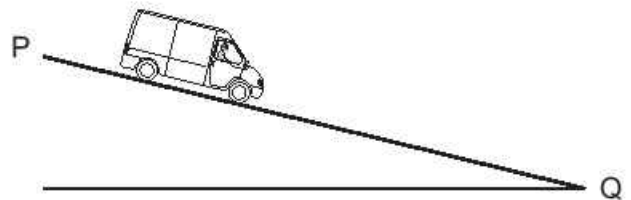
- A 80 J B 160 J
C 240 J D 400 J



113 9702/13/O/N/14/Q19

A van driver adjusts the force on a van's brakes so that the van travels at constant speed down a hill from P to Q.

The magnitude of the change in the van's kinetic energy is ΔE_k . The magnitude of the change in its gravitational potential energy is ΔE_p .



- A $\Delta E_k > \Delta E_p$ B $\Delta E_k = \Delta E_p$
C $\Delta E_p > \Delta E_k > 0$ D $\Delta E_k = 0$

114 9702/11/M/J/15/Q16

A team of nine dogs can pull a sledge with a combined force of 800 N at a speed of 1.5 m s^{-1} for 360 minutes.

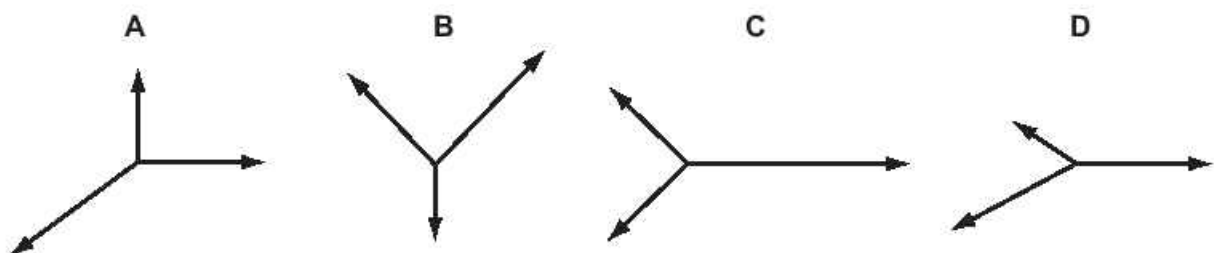
What is the average work done by each dog during this time?

- A $4.8 \times 10^4\text{ J}$ B $4.3 \times 10^5\text{ J}$ C $2.9 \times 10^6\text{ J}$ D $2.6 \times 10^7\text{ J}$

115 9702/1/O/N/02/Q15

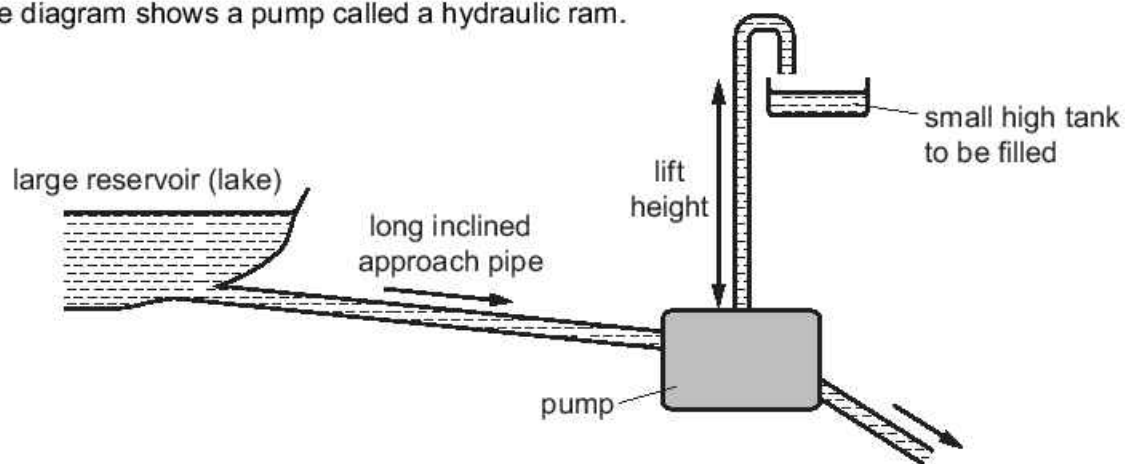
The diagrams represent systems of coplanar forces acting at a point. The lengths of the force vectors represent the magnitudes of the forces.

Which system of forces is in equilibrium?



116 9702/11/M/J/15/Q18

The diagram shows a pump called a hydraulic ram.



In one such pump the long approach pipe holds 500 kg of water. A valve shuts when the speed of this water reaches 2.0 m s^{-1} and the kinetic energy of this water is used to lift a small quantity of water by a height of 15 m.

The efficiency of the pump is 10%.

Which mass of water could be lifted 15 m?

- A** 0.15 kg **B** 0.68 kg **C** 1.5 kg **D** 6.8 kg

117 9702/11/M/J/15/Q19

A conveyor belt is driven at velocity v by a motor. Sand drops vertically on to the belt at a rate of $m \text{ kg s}^{-1}$.

What is the additional power needed to keep the conveyor belt moving at a steady speed when the sand starts to fall on it?

- A** $\frac{1}{2} mv$ **B** mv **C** $\frac{1}{2} mv^2$ **D** mv^2

118 9702/12/M/J/15/Q17

A fisherman lifts a fish of mass 250 g from rest through a vertical height of 1.8 m. The fish gains a speed of 1.1 m s^{-1} .

What is the energy gained by the fish?

- A** 0.15 J **B** 4.3 J **C** 4.4 J **D** 4.6 J

119 9702/12/M/J/15/Q18

Water from a reservoir is fed to the turbine of a hydroelectric system at a rate of 500 kg s^{-1} . The reservoir is 300 m above the level of the turbine.

The electrical output from the generator driven by the turbine is 200 A at a potential difference of 6000 V.

What is the efficiency of the system?

- A** 8.0% **B** 8.2% **C** 80% **D** 82%

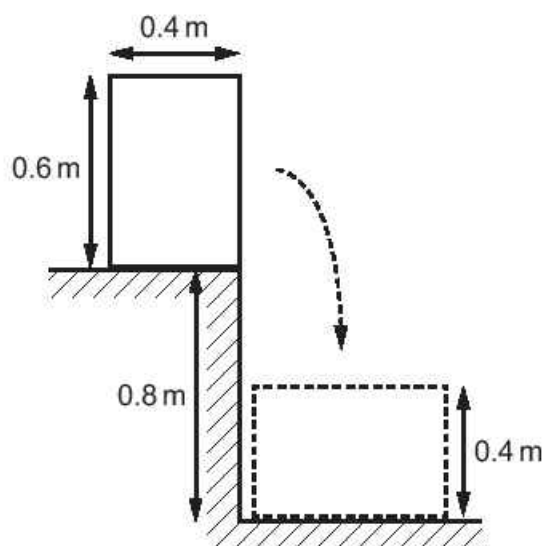
120 9702/12/M/J/15/Q15

A uniform solid block has weight 500 N, width 0.4 m and height 0.6 m. The block rests on the edge of a step of depth 0.8 m, as shown.

The block is knocked over the edge of the step and rotates through 90° before coming to rest with the 0.6 m edge horizontal.

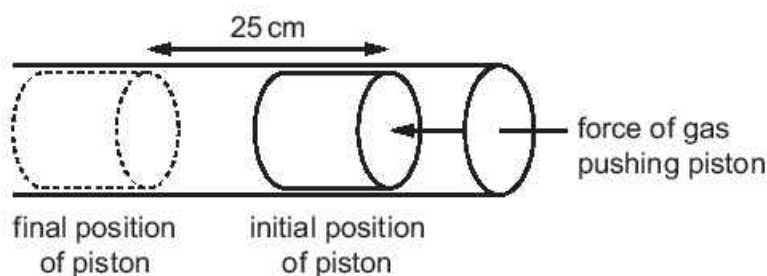
What is the change in gravitational potential energy of the block?

- A 300 J
- B 400 J
- C 450 J
- D 550 J



121 9702/13/M/J/15/Q17

The gas in an engine does work on a piston of cross-sectional area 80 cm^2 . The pressure on the piston has a constant value of $4.6 \times 10^5\text{ Pa}$.



How much work is done by the gas on the piston when it moves through a distance of 25 cm?

- A $9.2 \times 10^2\text{ J}$
- B $9.2 \times 10^4\text{ J}$
- C $9.2 \times 10^6\text{ J}$
- D $9.2 \times 10^8\text{ J}$

122 9702/13/M/J/15/Q18

A loaded aeroplane has a total mass of $1.2 \times 10^5\text{ kg}$ while climbing after take-off. It climbs at an angle of 23° to the horizontal with a speed of 50 m s^{-1} . What is the rate at which it is gaining potential energy at this time?

- A $2.3 \times 10^6\text{ J s}^{-1}$
- B $2.5 \times 10^6\text{ J s}^{-1}$
- C $2.3 \times 10^7\text{ J s}^{-1}$
- D $2.5 \times 10^7\text{ J s}^{-1}$

123 9702/12/O/N/11/Q15

When a horizontal force F is applied to a frictionless trolley over a distance s , the kinetic energy of the trolley changes from 4.0 J to 8.0 J.

If a force of $2F$ is applied to the trolley over a distance of $2s$, what will the original kinetic energy of 4.0 J become?

- A 16 J
- B 20 J
- C 32 J
- D 64 J

124 9702/11/M/J/17/Q15

A man climbs slowly at a steady speed to the top of a ladder.

What is the **main** energy transfer taking place for the man as he climbs?

- A** chemical potential to gravitational potential **C** kinetic to gravitational potential
B chemical potential to kinetic **D** thermal (heat) to kinetic

125 9702/11/M/J/17/Q16

During an interval of time, fuel supplies energy X to a car.

Some of this energy is converted into kinetic energy as the car accelerates.

The rest of the energy Y is lost as thermal energy.

What is the efficiency of the car?

- A** $\frac{X}{X-Y}$ **B** $\frac{Y}{X-Y}$ **C** $\frac{X-Y}{X}$ **D** $\frac{X-Y}{Y}$

126 9702/11/M/J/17/Q17

A railway engine accelerates a train of total mass 800 tonnes (1 tonne = 1000 kg) from rest to a speed of 50 m s^{-1} .

How much useful work must be done on the train to reach this speed?

- A** $1.0 \times 10^6 \text{ J}$ **B** $2.0 \times 10^6 \text{ J}$ **C** $1.0 \times 10^9 \text{ J}$ **D** $2.0 \times 10^9 \text{ J}$

127 9702/11/M/J/17/Q18

A mass is raised vertically. In time t , the increase in its gravitational potential energy is E_p and the increase in its kinetic energy is E_k .

What is the average power input to the mass?

- A** $(E_p - E_k)t$ **B** $(E_p + E_k)t$ **C** $\frac{E_p - E_k}{t}$ **D** $\frac{E_p + E_k}{t}$

128 9702/12/M/J/17/Q15

A cyclist is travelling at a constant speed up a hill. The frictional force resisting the cyclist's motion is 8.0 N.

The cyclist uses 450 J of energy to travel a distance of 20 m.

What is the increase in the gravitational potential energy of the cyclist?

- A** 160 J **B** 290 J **C** 440 J **D** 610 J

129 9702/12/M/J/17/Q17

A railway engine accelerates a train of total mass 1200 tonnes (1 tonne = 1000 kg) from rest to a speed of 75 m s^{-1} .

How much useful work must be done on the train to reach this speed?

- A** $3.4 \times 10^6 \text{ J}$ **B** $6.8 \times 10^6 \text{ J}$ **C** $3.4 \times 10^9 \text{ J}$ **D** $6.8 \times 10^9 \text{ J}$

130 9702/12/M/J/17/Q18

What is a correct derivation of the equation relating power, force and velocity?

A $\text{power} = \frac{\text{work done}}{\text{time taken}}$ and $\text{work done} = \text{force} \times \text{displacement}$

so $\text{power} = \frac{\text{force} \times \text{displacement}}{\text{time taken}}$

so $\text{power} = \text{force} \times \text{velocity}$

B $\text{power} = \frac{\text{work done}}{\text{time taken}}$ and $\text{work done} = \text{force} \times \text{distance}$

so $\text{power} = \frac{\text{force} \times \text{distance}}{\text{time taken}}$

so $\text{power} = \text{force} \times \text{velocity}$

C $\text{power} = \frac{\text{work done}}{\text{time taken}}$ and $\text{work done} = \frac{\text{force}}{\text{displacement}}$

so $\text{power} = \frac{\text{force}}{\text{displacement}} \times \text{time taken}$

so $\text{power} = \frac{\text{force}}{\text{velocity}}$

D $\text{power} = \frac{\text{work done}}{\text{time taken}}$ and $\text{work done} = \frac{\text{force}}{\text{distance}}$

so $\text{power} = \frac{\text{force}}{\text{distance}} \times \text{time taken}$

so $\text{power} = \frac{\text{force}}{\text{velocity}}$

131 9702/11/O/N/17/Q3

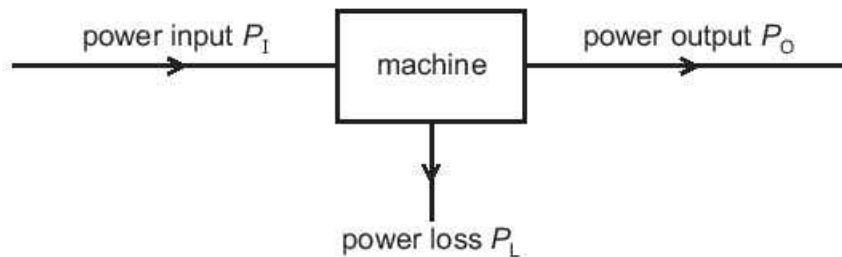
An ion is accelerated in a vacuum by a series of electrodes. A graph of the power supplied to the ion is plotted against time.

What is represented by the area under the graph between two times?

- A** the average force on the ion
- B** the change in kinetic energy of the ion
- C** the change in momentum of the ion
- D** the change in velocity of the ion

132 9702/11/O/N/17/Q16

Power is transferred through a machine as shown.



What is the efficiency of the machine?

- A $\frac{P_I}{P_O + P_L}$ B $\frac{P_L}{P_I}$ C $\frac{P_L}{P_O}$ D $\frac{P_O}{P_I}$

133 9702/11/O/N/17/Q17

A piston in a gas supply pump has an area of 400 cm^2 . The pump moves the gas against a fixed pressure of 3000 Pa .

During part of its stroke, the piston moves a distance of 25 cm in one direction. How much work is done by the piston during this movement?

- A 30 J B $3.0 \times 10^3 \text{ J}$ C $3.0 \times 10^5 \text{ J}$ D $3.0 \times 10^7 \text{ J}$

134 9702/11/O/N/17//Q18

A stone is projected vertically upwards from the ground at an initial speed of 15 m s^{-1} . Air resistance is negligible.

What is the maximum height reached by the stone?

- A 0.76 m B 11 m C 23 m D 110 m

135 9702/11/O/N/17/Q19

A turbine at a hydroelectric power station is situated 30 m below the level of the surface of a large lake. The water passes through the turbine at a rate of 340 m^3 per minute.

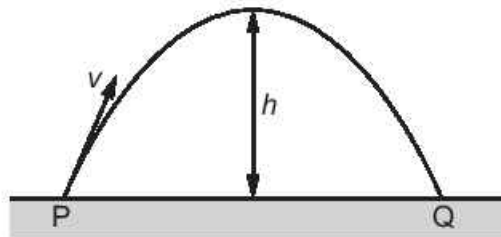
The overall efficiency of the turbine and generator system is 90% .

What is the output power of the power station? (The density of water is 1000 kg m^{-3} .)

- A 0.15 MW B 1.5 MW C 1.7 MW D 90 MW

136 9702/12/O/N/17/Q16

A ball of mass m is thrown up to height h in air with an initial velocity v , as shown.



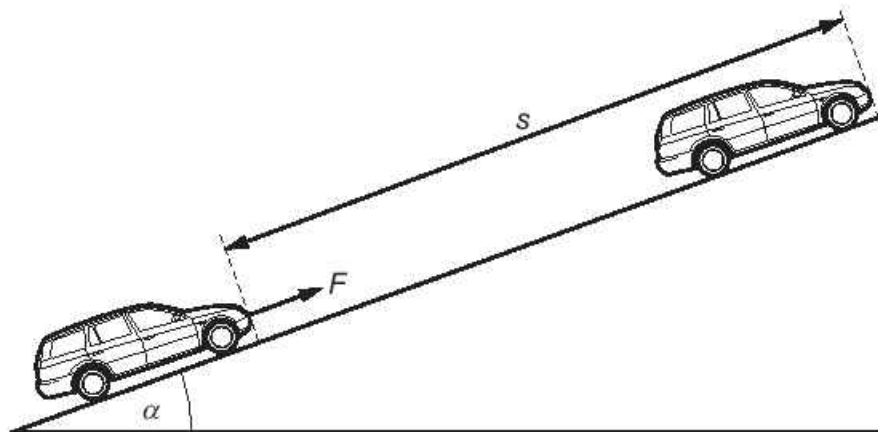
Air resistance is negligible. The acceleration of free fall is g .

What is the **total** work done by the gravitational force on the ball during its flight from P to Q?

- A zero B $\frac{1}{2}mv^2$ C mgh D $2mgh$

137 9702/12/O/N/17/Q17

A constant force F , acting on a car of mass m , moves the car up a slope through a distance s at constant velocity v . The angle of the slope to the horizontal is α .



What is the ratio $\frac{\text{gravitational potential energy gained by car}}{\text{work done by force } F}$?

- A $\frac{mgs \sin \alpha}{Fv}$ B $\frac{mv}{Fs}$ C $\frac{mv^2}{2Fs}$ D $\frac{mg \sin \alpha}{F}$

138 9702/12/O/N/17/Q18

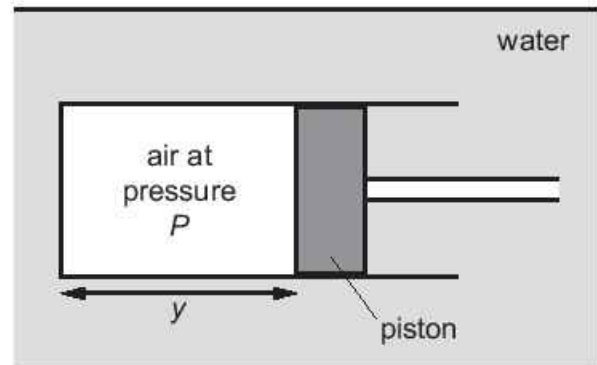
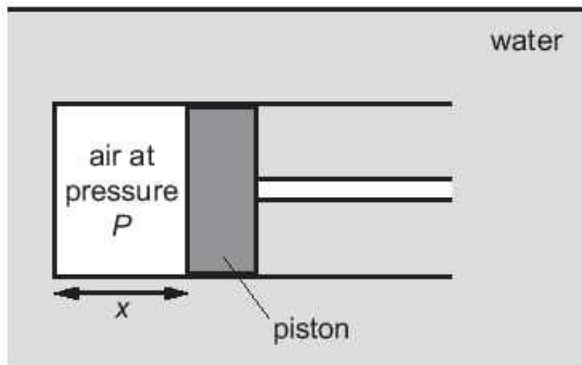
Car X is travelling at half the speed of car Y. Car X has twice the mass of car Y.

Which statement is correct?

- A Car X has half the kinetic energy of car Y.
 B Car X has one quarter of the kinetic energy of car Y.
 C Car X has twice the kinetic energy of car Y.
 D The two cars have the same kinetic energy.

139 9702/13/O/N/17/Q17

A horizontal cylinder of cross-sectional area A is fitted with a frictionless piston and contains air at pressure P . It is immersed in hot water and the length of the air column increases from x to y . The pressure P is constant.

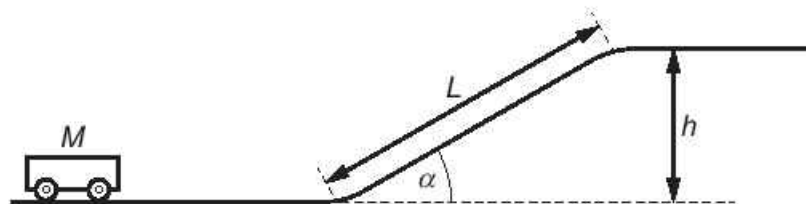


Which equation represents the work done by the trapped air during this process?

- A** PAy **B** $-PAy$ **C** $PA(y - x)$ **D** $-PA(y - x)$

140 9702/13/O/N/17/Q18

A trolley rolls along a horizontal surface and then travels up a slope before reaching a second horizontal surface. The slope is of length L . The trolley has mass M . The slope is at an angle α to the horizontal surface. The second horizontal surface is at height h above the first surface.



Assume negligible frictional forces. The acceleration of free fall is known.

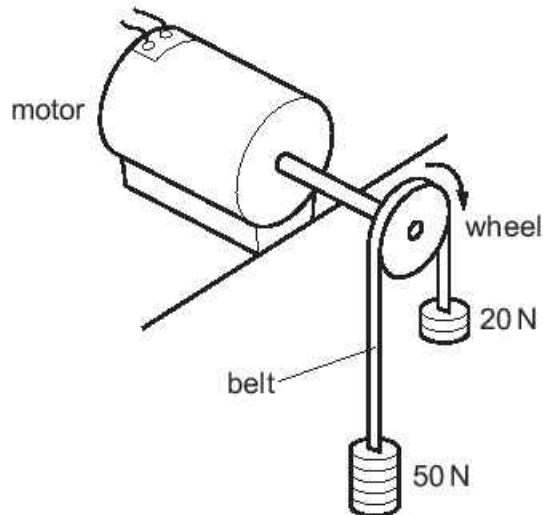
In order to determine the minimum initial velocity of the trolley for it to reach the top of the slope, which additional values are needed?

- A** h and M **B** M , L and h **C** α , L , M **D** h only

141 9702/13/O/N/17/Q19*

The diagram shows an arrangement used to find the output power of an electric motor.

The wheel attached to the motor's axle has a circumference of 0.5 m and the belt which passes over it is stationary when the weights have the values shown.



When the wheel is making 20 revolutions per second, what is the output power of the motor?

- A** 300 W **B** 500 W **C** 600 W **D** 700 W

142 9702/11/M/J/18/Q14

A train of mass 3.3×10^6 kg is moving at a constant speed up a slope inclined at an angle of 0.64° to the horizontal. The engine of the train is producing a useful output power of 14 MW.

Assume that there are no frictional forces opposing the motion of the train.

What is the speed of the train?

- A** 0.43 m s^{-1} **B** 4.2 m s^{-1} **C** 39 m s^{-1} **D** 380 m s^{-1}

143 9702/11/M/J/18/Q15

A cannon-ball of mass 3.50 kg is fired at a speed of 22.0 m s^{-1} from a gun on a ship at a height of 6.00 m above sea level.

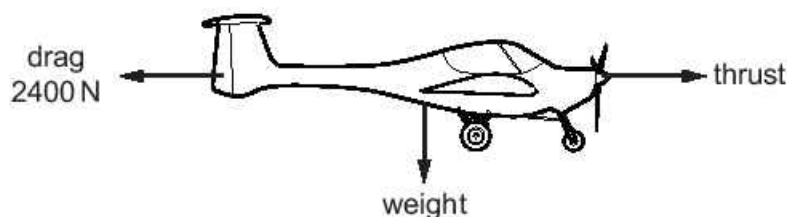
The total energy of the cannon-ball is the sum of the gravitational potential energy relative to the surface of the sea and the kinetic energy.

What is the total energy of the cannon-ball as it leaves the gun?

- A** 206 J **B** 641 J **C** 847 J **D** 1050 J

144 9702/11/M/J/18/Q16

An aircraft travels at a constant velocity of 90 m s^{-1} in horizontal flight. The diagram shows some of the forces acting on the aircraft.



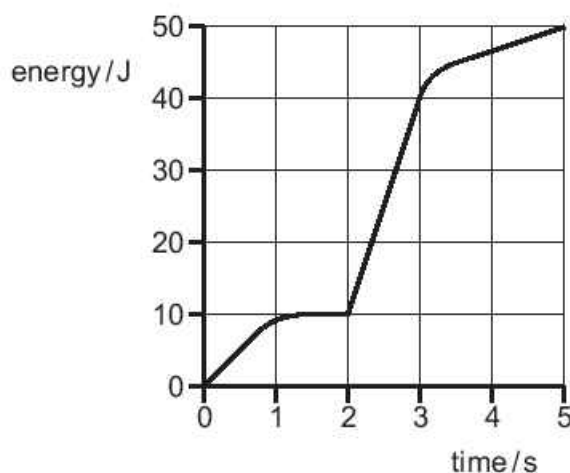
The mass of the aircraft is 2000 kg.

What is the power produced by the thrust force?

- A** $1.8 \times 10^5 \text{ W}$ **B** $2.2 \times 10^5 \text{ W}$ **C** $1.8 \times 10^6 \text{ W}$ **D** $2.0 \times 10^6 \text{ W}$

145 9702/11/M/J/18/Q18

An electrical generator is started at time zero. The total electrical energy generated during the first 5 seconds is shown in the graph.



What is the maximum electrical power generated at any instant during these first 5 seconds?

- A** 10 W **B** 13 W **C** 30 W **D** 50 W

146 9702/12/M/J/18/Q16

In 'normal driving conditions', an electric car has a range of 150 km. This uses all of the 200 MJ of energy stored in its batteries.

With the batteries initially fully charged, the car is driven 100 km in 'normal driving conditions'. The batteries are then recharged from a household electrical supply delivering a constant current of 13.0 A at a potential difference of 230 V.

What is the minimum time required to recharge the batteries?

- A** 0.95 hours **B** 12.4 hours **C** 18.6 hours **D** 27.9 hours

147 9702/12/M/J/18/Q18

A steel sphere is dropped vertically onto a horizontal metal plate. The sphere hits the plate with speed u , leaves it at speed v , and rebounds vertically to half of its original height. Ignore air resistance.

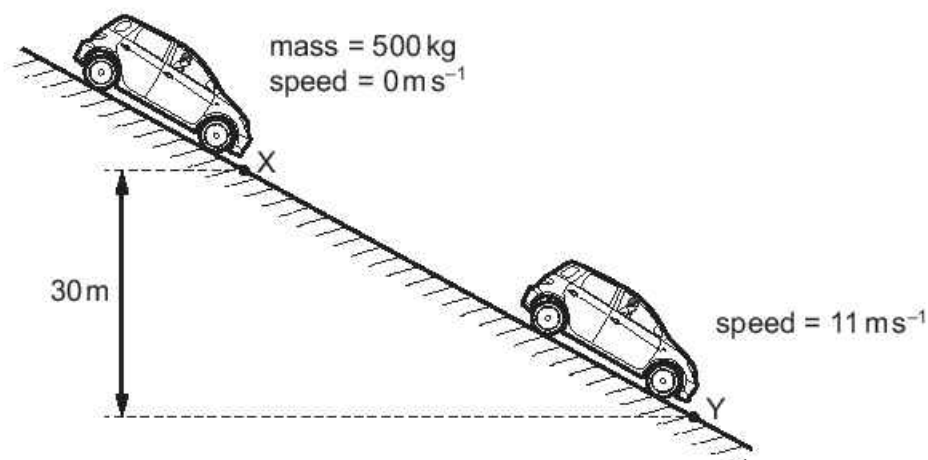
Which expression gives the value of $\frac{v}{u}$?

- A $\frac{1}{2^2}$ B $\frac{1}{2}$ C $\frac{1}{\sqrt{2}}$ D $1 - \frac{1}{\sqrt{2}}$

148 9702/12/M/J/18/Q19

A car of mass 500 kg is at rest at point X on a slope, as shown.

The car's brakes are released and the car rolls down the slope with its engine switched off. At point Y the car has moved through a vertical height of 30 m and has a speed of 11 m s^{-1} .



What is the energy dissipated by frictional forces when the car moves from X to Y?

- A $3.0 \times 10^4 \text{ J}$ B $1.2 \times 10^5 \text{ J}$ C $1.5 \times 10^5 \text{ J}$ D $1.8 \times 10^5 \text{ J}$

149 9702/13/M/J/18/Q15

Water from a reservoir is fed to the turbine of a hydroelectric system at a rate of 510 kg s^{-1} . The reservoir is 280 m above the level of the turbine.

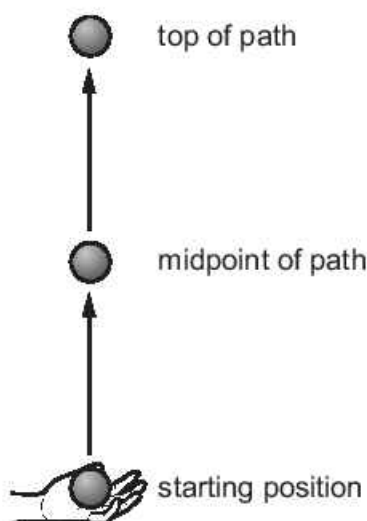
The electrical output from the generator driven by the turbine is a current of 205 A at a potential difference of 5800 V.

What is the efficiency of the system?

- A 8.3% B 12% C 83% D 85%

150 9702/13/M/J/18/Q16

A ball is thrown vertically up into the air. It rises to the top of its path before beginning to fall vertically downwards.



Assume that the gravitational potential energy of the ball is zero at its starting position.

Which statement about the ball is **not** correct?

- A As it rises, its kinetic energy is transferred to gravitational potential energy.
- B At the midpoint of its path, its gravitational potential energy is equal to its initial kinetic energy.
- C At the top of its path, its kinetic energy is zero.
- D At the top of its path, its total energy is less than its initial total energy.

151 9702/13/M/J/18/Q17

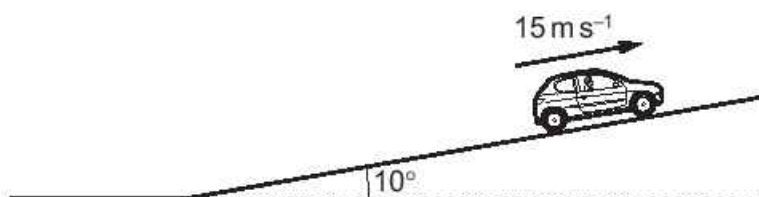
A force of 1000 N is needed to lift the hook of a crane at a constant velocity. The crane is then used to lift a load of mass 1000 kg at a constant velocity of 0.50 m s^{-1} .

What is the power needed to lift the hook and the load?

- A 4.9 kW
- B 5.4 kW
- C 20 kW
- D 22 kW

152 9702/12/F/M/18/Q14

A car of mass 1100 kg is travelling at a constant speed of 15 m s^{-1} up a slope inclined at 10° to the horizontal. The combined frictional forces acting on the car are directed down the slope and are equal to $\frac{W}{5}$, where W is the weight of the car.



What is the useful output power of the car's engine?

- A 28 kW
- B 32 kW
- C 60 kW
- D 190 kW

153 9702/12/F/M/18/Q15

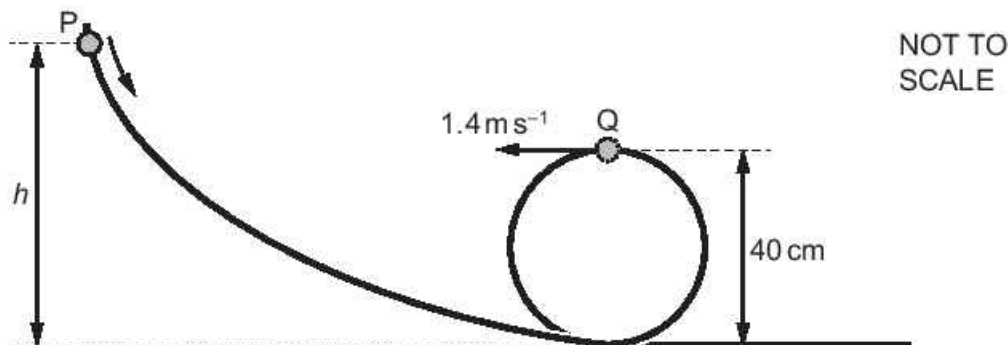
An old-fashioned 60 W lamp converts 95% of its energy supply into heat. A 4.0 W modern lamp has the same power output of light as the old-fashioned lamp.

What is the efficiency of the modern lamp?

- A** 5.0% **B** 6.7% **C** 75% **D** 95%

154 9702/12/F/M/18/Q16

A bead is released from rest at point P and slides along a wire, as shown.



The track loops around and forms a vertical circle of diameter 40 cm. At point Q, the bead has a speed of 1.4 m s^{-1} .

Air resistance and friction on the wire are negligible.

What is the height h from which the bead is released?

- A** 0.30 m **B** 0.40 m **C** 0.50 m **D** 0.60 m

155 9702/12/F/M/18/Q17

A small diesel engine uses a volume of $1.5 \times 10^4 \text{ cm}^3$ of fuel per hour to produce a useful power output of 40 kW. It may be assumed that 34 kJ of energy is transferred to the engine when it uses 1.0 cm^3 of fuel.

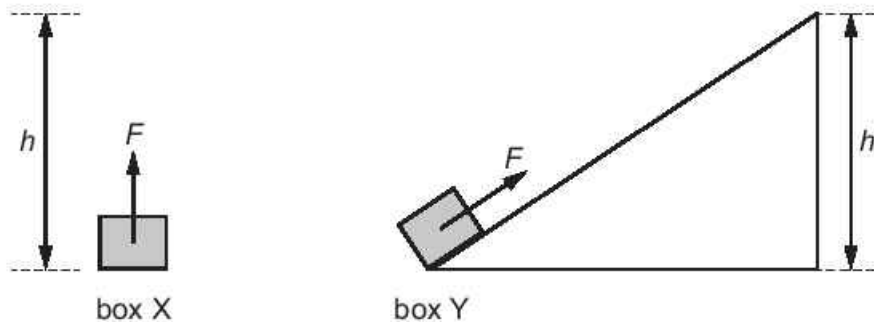
What is the rate of transfer from the engine of energy that is wasted?

- A** 102 kW **B** 142 kW **C** 182 kW **D** 470 kW

156 9702/12/F/M/18/Q18

Two boxes X and Y have the same mass. Box X is lifted vertically through a height h by a force of magnitude F .

Box Y is pulled along a slope by a force of the same magnitude to reach the same height, as shown.



Which statement is correct?

- A Both boxes gain the same amount of gravitational potential energy and the same amount of work is done by the two forces.
- B Both boxes gain the same amount of gravitational potential energy but more work is done by the force acting on box Y than by the force acting on box X.
- C Box Y gains less gravitational potential energy than box X because the weight of box Y is less than the weight of box X.
- D Box Y gains more gravitational potential energy than box X as more work is done by the force acting on box Y than by the force acting on box X.

157 9702/11/O/N/18/Q14

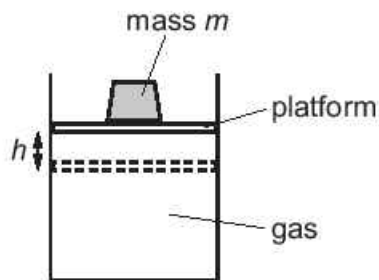
A rocket is fired upwards.

As it accelerates upwards after leaving the launch pad, which forms of energy are changing?

- A chemical energy, gravitational potential energy and kinetic energy
- B chemical energy and gravitational potential energy only
- C chemical energy and kinetic energy only
- D gravitational potential energy and kinetic energy only

158 9702/11/O/N/18/Q15

A mass m is on top of a platform that is supported by gas in a cylinder of cross-sectional area A , as shown.



The platform has negligible mass and can move freely up and down.

The gas is heated and expands so that the mass is raised through a height h . Atmospheric pressure is p .

What is the ratio $\frac{\text{gain in gravitational potential energy of the mass}}{\text{work done by the gas}}$?

- A $\frac{mg}{pA}$ B $\frac{mg}{mg + pA}$ C $\frac{pA}{mg}$ D $\frac{mg - pA}{mg}$

159 9702/11/O/N/18/Q17

The force resisting the motion of a car is proportional to the square of the car's speed. The magnitude of the force at a speed of 20.0 m s^{-1} is 800 N .

What useful output power is required from the car's engine to maintain a steady speed of 40.0 m s^{-1} ?

- A 32 kW B 64 kW C 128 kW D 512 kW

160 9702/12/O/N/18/Q15

A bungee jumper on a platform over a river is attached to an elastic rope that is 20 m long when unstretched. He falls towards the river and his lowest point is 30 m below the platform.

The initial gravitational potential energy of the jumper is transferred to other forms during the jump.

Which other forms of energy do the jumper and rope have when the jumper has fallen half-way and when he is at the lowest point of his jump?

	half-way	lowest point
A	kinetic energy and elastic potential energy	kinetic energy and elastic potential energy
B	kinetic energy and elastic potential energy	elastic potential energy only
C	kinetic energy only	kinetic energy and elastic potential energy
D	kinetic energy only	elastic potential energy only

161 9702/12/O/N/18/Q16

A cylinder contains a fixed mass of gas. The gas, at a constant pressure of $1.3 \times 10^5 \text{ Pa}$, expands from a volume of 900 cm^3 to a volume of 1100 cm^3 .

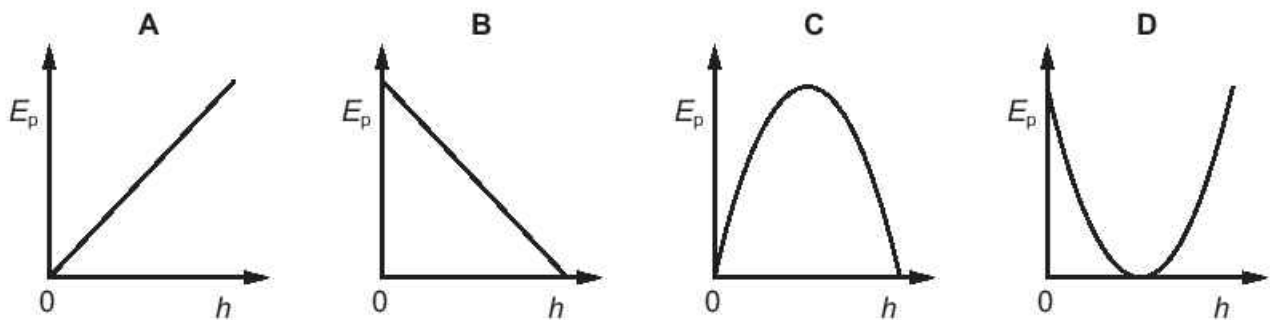
What is the work done by the gas during this expansion?

- A** 26 J **B** 130 J **C** 2600 J **D** 13 000 J

162 9702/12/O/N/18/Q17

An object is thrown into the air.

Which graph shows how the gravitational potential energy E_p of the object varies with height h above the ground?



163 9702/12/O/N/18/Q18

A car of mass 1800 kg accelerates along a horizontal road so that its speed increases from 20 m s^{-1} to 25 m s^{-1} in a time of 5.4 s .

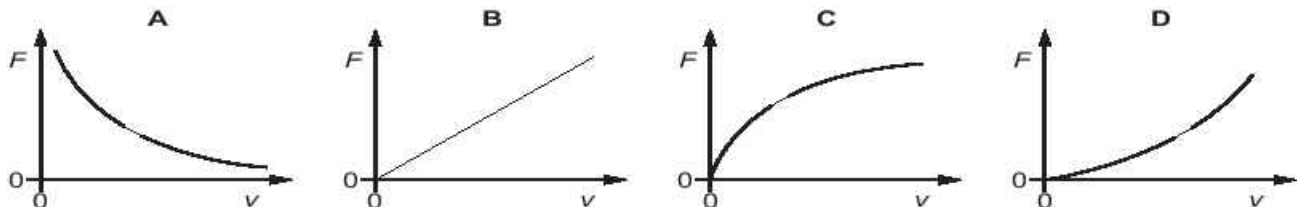
What is the average useful power output of the car's engine?

- A** 4.2 kW **B** 38 kW **C** 120 kW **D** 1100 kW

164 9702/12/O/N/18/Q19

A variable force is applied to ensure that a constant power is supplied to a train.

Which graph best shows the variation of the force F applied with the velocity v of the train?



165 9702/13/O/N/18/Q15

Which statement about energy is **not** correct?

- A Energy is never lost but it may be transferred between different forms.
- B In an inelastic collision, the total energy is constant.
- C The efficiency of a system is the ratio of the useful energy output to the total energy input.
- D When a machine does work, friction reduces the total energy.

166 9702/13/O/N/18/Q16

An electric kettle is rated as having an input power of 1.50 kW and an efficiency of 65.0%.
The kettle is switched on for 2.00 minutes.
How much energy is transferred to the water in the kettle?

- A 0.975 kJ B 117 kJ C 180 kJ D 277 kJ

167 9702/13/O/N/18/Q17

On a planet, a gravitational force F acts on a mass of 6.0 kg. The mass is moved by force F a distance of 30 m in the direction of the gravitational field. The work done by the field is 450 J.

What is the force F on the mass and what is the acceleration of free fall g on the planet?

	F/N	$g/\text{m s}^{-2}$
A	0.067	0.011
B	0.067	0.40
C	15	2.5
D	15	90

168 9702/13/O/N/18/Q18

A girl of mass 50 kg runs up a flight of 20 steps in 7.0 seconds. Each step is 25 cm high.

What is the useful average output power provided by the girl to climb the flight of steps?

- A 18 W B 36 W C 350 W D 2500 W

169 9702/11/M/J/19/Q8

A positive charge of $2.6 \times 10^{-8} \text{ C}$ is in a uniform electric field of field strength $300\,000 \text{ V m}^{-1}$.

How much work must be done on the charge in order to move it a distance of 4.0 mm in the opposite direction to the direction of the field?

- A $3.1 \times 10^{-5} \text{ J}$ C $3.1 \times 10^{-2} \text{ J}$
B $2.0 \times 10^{-3} \text{ J}$ D 2.0 J

169 9702/11/M/J/19/Q8

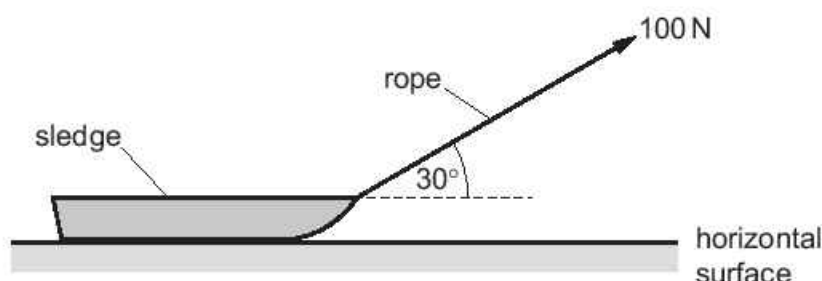
A positive charge of $2.6 \times 10^{-8} \text{ C}$ is in a uniform electric field of field strength $300\,000 \text{ V m}^{-1}$.

How much work must be done on the charge in order to move it a distance of 4.0 mm in the opposite direction to the direction of the field?

- A $3.1 \times 10^{-5} \text{ J}$ C $3.1 \times 10^{-2} \text{ J}$
 B $2.0 \times 10^{-3} \text{ J}$ D 2.0 J

170 9702/11/M/J/19/Q17

A rope is attached to a sledge and a boy uses the rope to pull the sledge along a horizontal surface with a constant velocity. The tension in the rope is 100 N and the rope is held at 30° to the horizontal.



How much work does the boy do on the sledge when he pulls it a distance of 5.0 m along the surface?

- A 250 J B 290 J C 430 J D 500 J

171 9702/11/M/J/19/Q18

The kinetic energy E_k of an object of mass m moving at speed v is given by the equation shown.

$$E_k = \frac{1}{2} mv^2$$

Which equation is **not** used in the derivation of this equation?

- A $F = ma$ B $s = vt$ C $v^2 = u^2 + 2as$ D $W = Fs$

172 9702/11/M/J/19/Q19

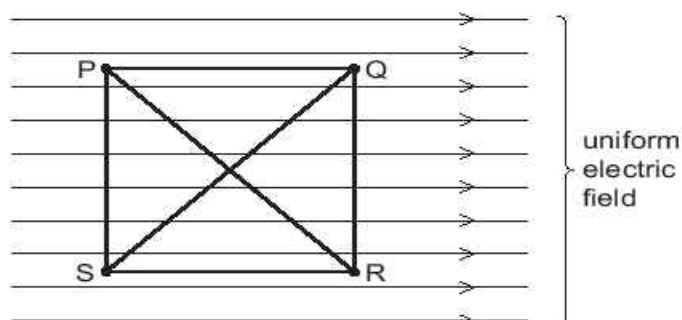
A grasshopper of mass 0.12 g jumps vertically. It uses its back legs over a time of 0.020 s to jump, leaving the ground with a velocity of 3.0 m s^{-1} .

What is the average power developed by the legs of the grasshopper?

- A $9.0 \times 10^{-3} \text{ W}$ B $1.8 \times 10^{-2} \text{ W}$ C $2.7 \times 10^{-2} \text{ W}$ D 37 W

173 9702/12/M/J/19/Q13

A small positive charge can move inside a uniform electric field.



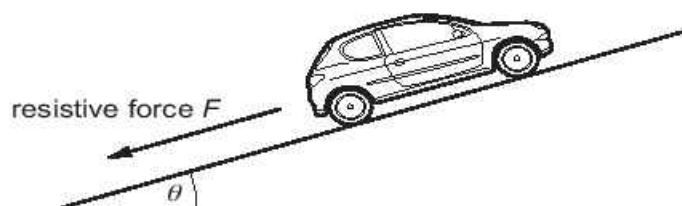
The charge moves along different straight paths between points P, Q, R and S.

Which row gives two paths that result in the same total work done on the charge?

	path 1	path 2
A	P to R	Q to S
B	P to R	P to S
C	S to Q	S to R
D	S to Q	R to P

174 9702/12/M/J/19/Q14

A car of mass m travels at constant speed up a slope at an angle θ to the horizontal, as shown in the diagram. Air resistance and friction provide a resistive force F . The acceleration of free fall is g .



What is the force needed to propel the car at this constant speed?

- | | |
|---------------------------|-------------------------------|
| A $mg \cos \theta$ | C $mg \cos \theta + F$ |
| B $mg \sin \theta$ | D $mg \sin \theta + F$ |

175 9702/12/M/J/19/Q16

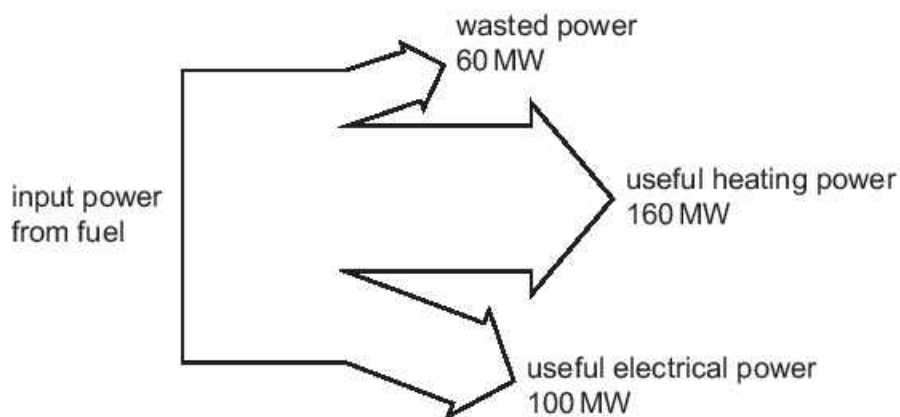
A parachutist is falling at constant (terminal) velocity.

Which statement is **not** correct?

- A** Gravitational potential energy is converted into kinetic energy of the air.
- B** Gravitational potential energy is converted into kinetic energy of the parachutist.
- C** Gravitational potential energy is converted into thermal energy of the air.
- D** Gravitational potential energy is converted into thermal energy of the parachutist.

176 9702/12/M/J/19/Q17

A combined heat and power (CHP) station generates electrical power and useful heat. The diagram shows the input and output powers for a CHP station.



What is the efficiency of the CHP station for producing useful power?

- A** 31% **B** 38% **C** 50% **D** 81%

177 9702/12/M/J/19/Q19

A train on a mountain railway is carrying 200 people of average mass 70 kg up a slope at an angle of 30° to the horizontal and at a speed of 6.0 m s^{-1} . The train itself has a mass of 80 000 kg. The percentage of the power from the engine which is used to raise the passengers and the train is 40%.

What is the power of the engine?

- A** 1.1 MW **B** 2.8 MW **C** 6.9 MW **D** 14 MW

178 9702/13/M/J/19/Q15

Which statement about the principle of conservation of energy is correct?

- A** Energy conversion helps to conserve energy sources.
B Energy is conserved only in systems with an efficiency of 100%.
C The supply of energy is limited so energy should be conserved.
D The total amount of energy in a closed system is constant.

179 9702/13/M/J/19/Q16

A student can run or walk up the stairs to her classroom.

Which statement describes the power required and the gravitational potential energy gained while running up the stairs compared to walking up them?

- A** Running provides more gravitational potential energy and uses more power.
B Running provides more gravitational potential energy and uses the same power.
C Running provides the same gravitational potential energy and uses more power.
D Running provides the same gravitational potential energy and uses the same power.

180 9702/13/M/J/19/Q18

On the surface of a planet, 30 J of work is done against gravity to raise a mass of 1.0 kg through a height of 10 m.

How much work must be done to raise a mass of 4.0 kg through a height of 5.0 m on this planet?

- A 15 J B 60 J C 120 J D 200 J

181 9702/11/O/N/19/Q15

2

An electric motor produces 120 W of useful mechanical output power. The efficiency of the motor is 60%.

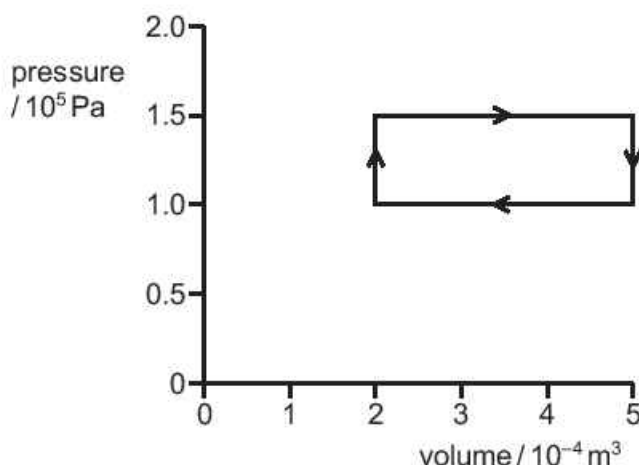
Which row is correct?

	electrical power input / W	waste heat power output / W
A	72	48
B	192	72
C	200	72
D	200	80

182 9702/11/O/N/19/Q16

A fixed amount of a gas undergoes a series of changes to its pressure and volume.

In two of the changes, no work is done by or on the gas. In one change work is done by the gas on its surroundings. In another change work is done on the gas by its surroundings.



During the change when work is done on the gas by its surroundings, how much work is done on the gas?

- A 15 J B 25 J C 30 J D 45 J

183 9702/11/O/N/19/Q17

An object travelling with a speed of 10 m s^{-1} has kinetic energy 1500 J.

The speed of the object is increased to 40 m s^{-1} .

What is the new kinetic energy of the object?

- A** 4500 J **B** 6000 J **C** 24 000 J **D** 1 350 000 J

184 9702/11/O/N/19/Q18

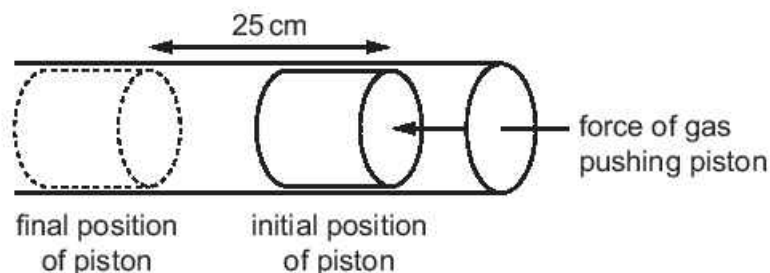
The engine of a car exerts a force of 600 N in moving the car 1.0 km in 150 seconds.

What is the average useful output power of the engine?

- A** 4.0 W **B** 4.0 kW **C** 90 kW **D** 90 MW

185 9702/12/O/N/19/Q14

The gas in an engine does work on a piston of cross-sectional area 80 cm^2 . The pressure on the piston has a constant value of $4.6 \times 10^5 \text{ Pa}$.



How much work is done by the gas on the piston when it moves through a distance of 25 cm?

- A** $9.2 \times 10^2 \text{ J}$ **B** $9.2 \times 10^4 \text{ J}$ **C** $9.2 \times 10^6 \text{ J}$ **D** $9.2 \times 10^8 \text{ J}$

186 9702/12/O/N/19/Q15

A power station using coal as fuel has an average power output of 3000 MW. Coal is supplied by 20 trains each day. The efficiency of the station in converting the thermal energy released from the coal to electrical energy is 26%.

A mass of 1.0 kg of coal will release 33 MJ of thermal energy when burnt.

Which mass of coal does each train bring?

- A** $2.5 \times 10^4 \text{ kg}$ **B** $6.3 \times 10^4 \text{ kg}$ **C** $1.5 \times 10^6 \text{ kg}$ **D** $3.0 \times 10^7 \text{ kg}$

187 9702/12/O/N/19/Q16

A wooden cylinder floats partially submerged in a bath of water. A force F is applied to the cylinder until it is just fully submerged.



Which statement is **not** correct?

- A Some of the water gains gravitational potential energy.
- B The cylinder loses gravitational potential energy.
- C Work is done by force F on the cylinder.
- D Work is done by the upthrust on the cylinder.

188 9702/12/O/N/19/Q17

The motor of a crane lifts a load of mass 600 kg. The load rises vertically at a constant speed of 12m per minute.

What is the useful power output of the motor?

- A 0.12 kW B 1.2 kW C 7.2 kW D 71 kW

189 9702/13/O/N/19/Q14

Trains supply coal to a power station. The table shows quantities describing the operation of the power station.

	symbol	unit
power station average output	P	W
number of trains per day	N	
mass of coal on a train	M	kg
energy from 1 kg of coal	E	J
number of seconds in one day	S	

Which expression gives the efficiency of the power station?

- A $\frac{PS}{NME}$ B $\frac{PSN}{ME}$ C $\frac{NME}{PS}$ D $\frac{NM}{PSE}$

190 9702/13/O/N/19/Q15

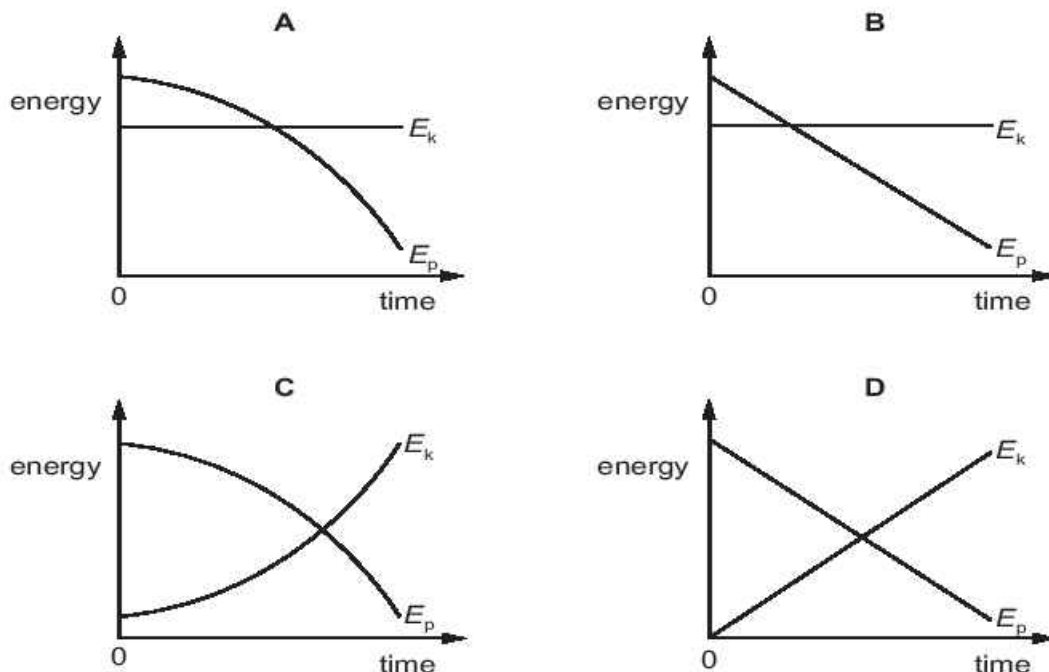
A piston in a gas supply pump has an area of 500 cm^2 and it moves a distance of 30 cm. The pump moves the gas against a fixed pressure of 4000 Pa. How much work is done by the piston?

- A 60 J B $6.0 \times 10^3 \text{ J}$ C $6.0 \times 10^5 \text{ J}$ D $6.0 \times 10^7 \text{ J}$

191 9702/13/O/N/19/Q16

A steel ball is falling at constant speed in oil.

Which graph shows the variation with time of the gravitational potential energy E_p and the kinetic energy E_k of the ball?



192 9702/13/O/N/19/Q17

The maximum useful output power of a car travelling on a horizontal road is P . The total resistive force acting on the car is kv^2 , where v is the speed of the car and k is a constant.

Which equation is correct when the car is travelling at maximum speed?

A $v^3 = \frac{P}{k}$
 B $v^2 = \frac{P}{k}$
 C $v = \left(\frac{P}{k}\right)^2$
 D $v = \left(\frac{P}{k}\right)^3$

193 9702/13/O/N/19/Q18

Initially, four identical uniform blocks, each of mass m and thickness h , are spread on a table.



The acceleration of free fall is g .

How much work is done on the blocks in stacking them on top of one another?

A $3 mgh$
 B $6 mgh$
 C $8 mgh$
 D $10 mgh$

194. 9702/12/F/M/20/Q15

A ball is thrown vertically upwards from the surface of the Earth.

Which statement describes the energy of the ball as it rises through the air?

- A** The kinetic energy of the ball decreases as the gravitational potential energy decreases.
- B** The kinetic energy of the ball decreases as the gravitational potential energy increases.
- C** The kinetic energy of the ball increases as the gravitational potential energy decreases.
- D** The total energy of the ball increases.

195. 9702/12/F/M/20/Q16

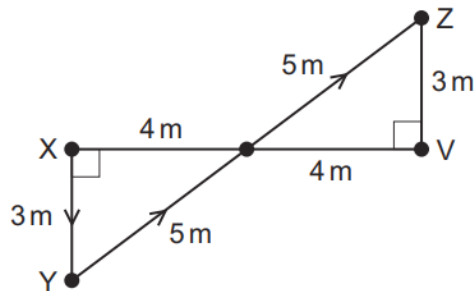
A sledge of mass 50 kg sits on a snowy surface. It is pulled horizontally for 10 m against a frictional force of 200 N, then it is pulled horizontally across ice for 10 m. There is no friction between the ice and the sledge. It is lifted up vertically by 1 m and finally carried back at a constant speed to where it started.

During which stage of its journey is most work done on the sledge?

- A** being carried back 20 m at constant speed
- B** being lifted up 1 m
- C** being pulled 10 m across ice
- D** being pulled 10 m across snow

196. 9702/12/F/M/20/Q17

An object is moved in a vertical plane from X to Y, and then from Y to Z, as shown in the diagram.



The distances between various points are indicated on the diagram.

Lines XY and VZ are vertical.

The object weighs 20 N.

How much gravitational potential energy does the object gain by moving from X to Z?

- A** 60 J
- B** 120 J
- C** 140 J
- D** 260 J

197. 9702/12/F/M/20/Q18

A car travels at a constant speed of 25 m s^{-1} up a slope. The wheels driven by the engine exert a forward force of 3000 N. The total force due to air resistance and friction is 2100 N. The weight of the car has a component down the slope of 900 N.

What is the rate at which thermal energy is dissipated?

- A** zero **B** $2.3 \times 10^4 \text{ W}$ **C** $5.3 \times 10^4 \text{ W}$ **D** $7.5 \times 10^4 \text{ W}$

198. 9702/11/M/J/20/Q16

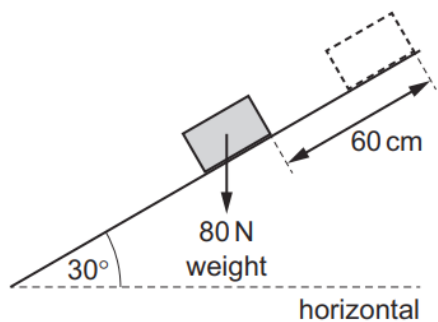
A mass attached to the lower end of a spring bounces up and down.

At which points in the path of the mass do the gravitational potential energy of the mass (GPE), the elastic potential energy in the spring (EPE) and the kinetic energy of the mass (KE) have their highest values?

	GPE	EPE	KE
A	bottom	middle	top
B	bottom	top	middle
C	top	bottom	middle
D	top	bottom	top

199. 9702/11/M/J/20/Q17

A block of weight 80 N is pushed a distance of 60 cm up a slope inclined at 30° to the horizontal. There is a frictional force of 25 N between the block and the surface of the slope.



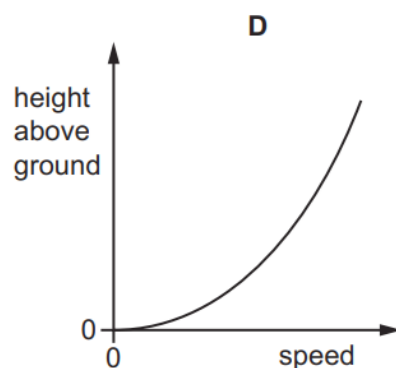
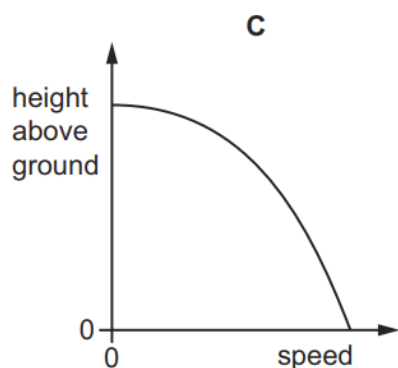
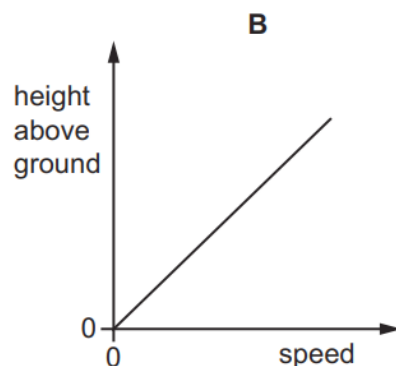
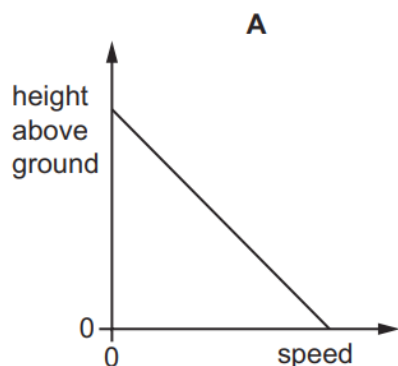
What is the work W_g done against the gravitational force and the work W_f done against the frictional force?

	W_g / J	W_f / J
A	24	7.5
B	24	15
C	48	7.5
D	48	15

200. 9702/11/M/J/20/Q18

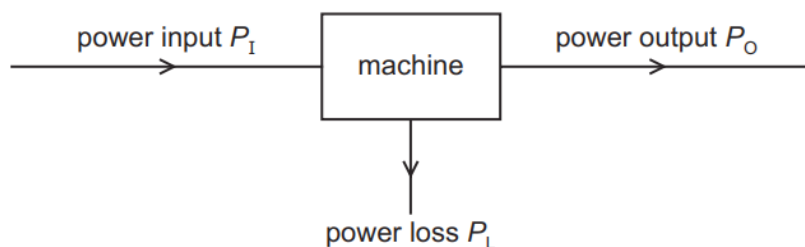
A ball is dropped from rest and falls towards the ground. Air resistance is negligible.

Which graph shows the variation with speed of the height of the ball above the ground?



201. 9702/11/M/J/20/Q19

Power is transferred through a machine as shown.



What is the efficiency of the machine?

A $\frac{P_I}{P_O + P_L}$

B $\frac{P_L}{P_I}$

C $\frac{P_L}{P_O}$

D $\frac{P_O}{P_I}$

202. 9702/12/M/J/20/Q14

During an interval of time, fuel supplies energy X to a car.

Some of this energy is converted into kinetic energy as the car accelerates.

The rest of the energy Y is lost as thermal energy.

What is the efficiency of the car?

- A** $\frac{X}{X-Y}$ **B** $\frac{Y}{X-Y}$ **C** $\frac{X-Y}{X}$ **D** $\frac{X-Y}{Y}$

203. 9702/12/M/J/20/Q15

During an interval of time, fuel supplies energy X to a car.

Some of this energy is converted into kinetic energy as the car accelerates.

The rest of the energy Y is lost as thermal energy.

What is the efficiency of the car?

- A** $\frac{X}{X-Y}$ **B** $\frac{Y}{X-Y}$ **C** $\frac{X-Y}{X}$ **D** $\frac{X-Y}{Y}$

In which situation is work done on an object?

- A** The object slides with a constant velocity along a horizontal frictionless surface in a vacuum.
B A person holds the object at arm's length and at a fixed height above the ground.
C A person pushes the object up a frictionless ramp.
D The stationary object floats partially submerged in water.

204. 9702/12/M/J/20/Q17

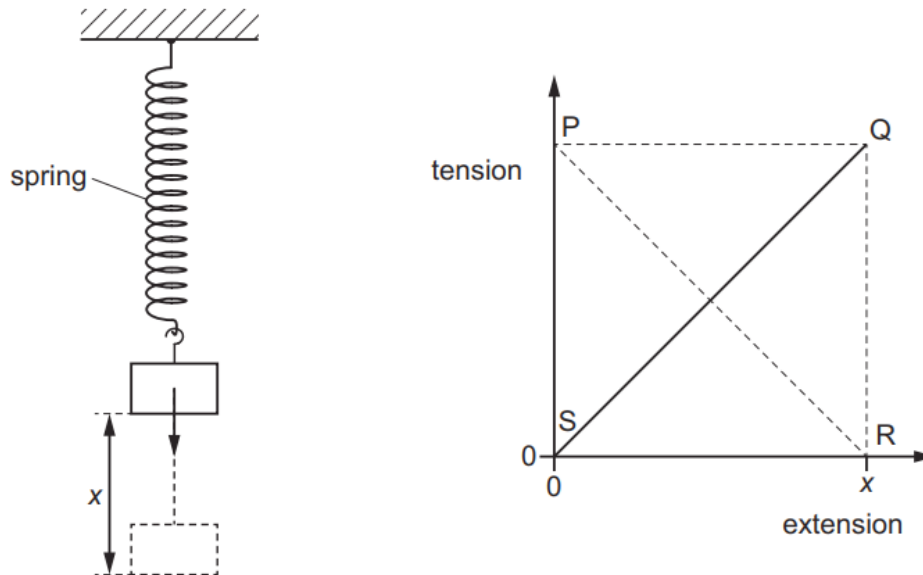
An escalator in an underground station has 25 people standing on it and is moving with a speed of 4.3 ms^{-1} . The average mass of a person is 78 kg and the angle of the escalator to the horizontal is 40° .

What is the minimum power required to lift these people?

- A** 5.4 kW **B** 6.4 kW **C** 53 kW **D** 63 kW

205. 9702/12/M/J/20/Q16

A spring is attached at one end to a fixed point. A mass is then hung from the other end of the spring. The spring has extension x when the system is in equilibrium.



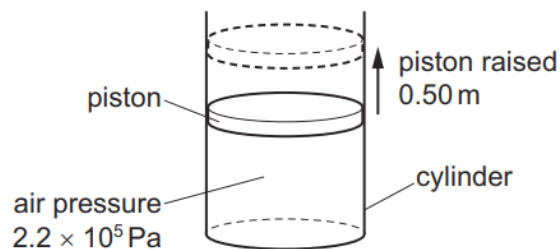
The variation of the tension in the spring with its extension is shown on the graph.

Which statement is correct?

- A** Area SPR represents the energy stored in the spring which cannot be recovered.
- B** Area SPQR represents the energy stored in the spring which can be recovered.
- C** Area SPQ represents the loss of gravitational potential energy of the mass due to the extension of the spring.
- D** Area SQR represents the elastic potential energy stored in the spring.

206. 9702/13/M/J/20/Q17

A cylinder is heated, causing the air inside to expand at a constant pressure of $2.2 \times 10^5 \text{ Pa}$.



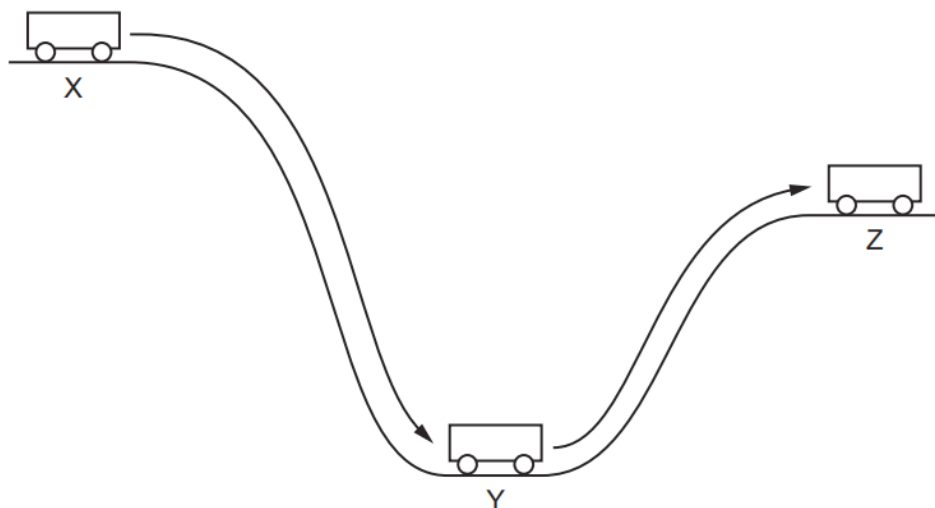
The expansion of the air causes the piston to rise through a vertical distance of 0.50 m, doing 11 kJ of work. Frictional forces are negligible.

What is the cross-sectional area of the piston?

- A** $1.0 \times 10^{-4} \text{ m}^2$
- B** $2.5 \times 10^{-2} \text{ m}^2$
- C** $5.0 \times 10^{-2} \text{ m}^2$
- D** $1.0 \times 10^{-1} \text{ m}^2$

207. 9702/13/M/J/20/Q16

A trolley starts from rest at X. It rolls down to Y and eventually comes to rest at Z.



Which row is a possible summary of the energy changes during this process?

	X to Y	Y to Z
A	PE $\rightarrow$ KE	KE $\rightarrow$ PE
B	PE $\rightarrow$ KE	KE $\rightarrow$ PE + heat
C	PE $\rightarrow$ KE + heat	KE $\rightarrow$ PE
D	PE $\rightarrow$ KE + heat	KE $\rightarrow$ PE + heat

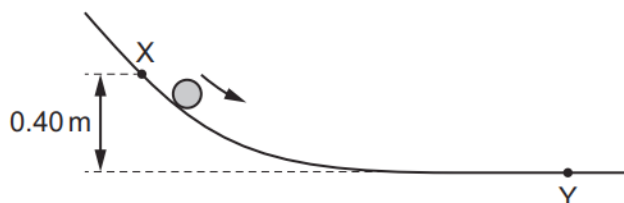
key

PE = potential energy

KE = kinetic energy

208. 9702/13/M/J/20/Q18

A ball slides down a curved track, as shown.



Point X is at a height of 0.40 m above point Y. The speed of the ball at point X is 2.5 m s^{-1} .

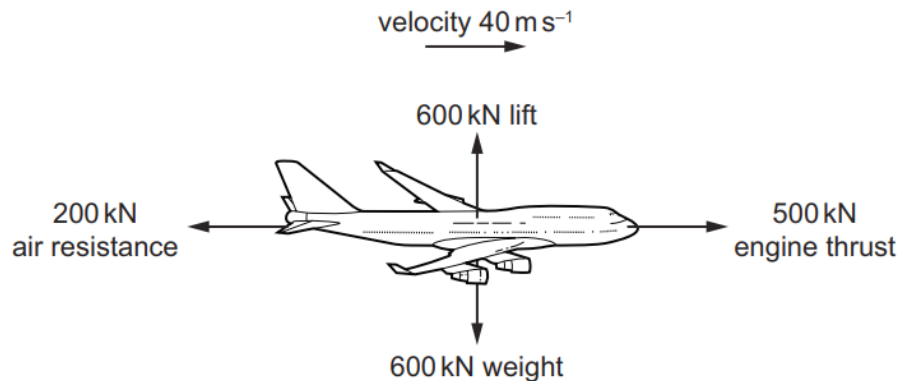
Frictional forces are negligible.

What is the speed of the ball at point Y?

- A** 2.8 m s^{-1} **B** 3.2 m s^{-1} **C** 3.8 m s^{-1} **D** 14 m s^{-1}

209. 9702/13/M/J/20/Q19

The force diagram shows an aircraft accelerating. At the instant shown, the velocity of the aircraft is 40 m s^{-1} .

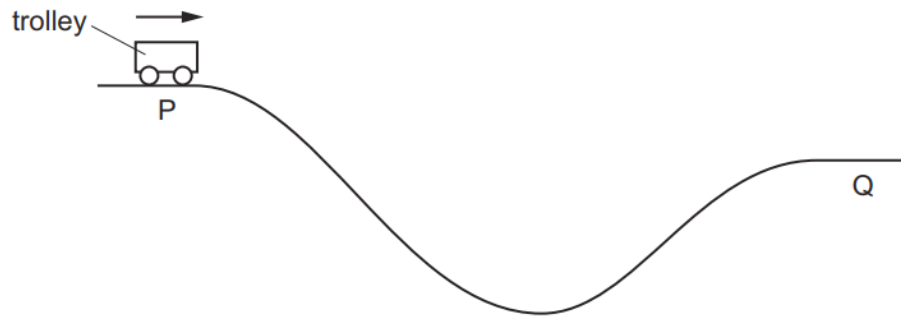


At which rate is its kinetic energy increasing?

- A** 2.4 MW **B** 8.0 MW **C** 12 MW **D** 20 MW

210. 9702/11/O/N/20/Q15

A trolley runs from P to Q along a track. At Q its potential energy is 50 kJ less than at P.



At P, the kinetic energy of the trolley is 5 kJ. Between P and Q, the trolley does 10 kJ of work against friction.

What is the kinetic energy of the trolley at Q?

- A** 35 kJ **B** 45 kJ **C** 55 kJ **D** 65 kJ

211. 9702/11/O/N/20/Q16

A hydroelectric power station uses the gravitational potential energy of water to generate electrical energy.

In one particular power station, the mass of water flowing per unit time is $1.5 \times 10^5 \text{ kg s}^{-1}$. The water falls through a vertical height of 120 m.

The electrical power generated is 100 MW.

What is the efficiency of the power station?

- A** 5.6% **B** 43% **C** 57% **D** 77%

212. 9702/11/O/N/20/Q17

Which amount of energy is **not** 2400 J?

- A the decrease in gravitational potential energy of a mass of 60 kg when it moves vertically downwards through 40 m near the Earth's surface
- B the energy transferred in 15 s by a machine of power 160 W
- C the kinetic energy of a mass of 12 kg moving at a speed of 20 m s^{-1}
- D the work done by a gas expanding against a constant external pressure of 120 kPa when its volume increases by 0.020 m^3

213. 9702/11/O/N/20/Q18

A train of mass 300 000 kg is accelerating at 0.80 m s^{-2} . At one instant, the speed of the train is 5.0 m s^{-1} and the resistive force to its motion is 15 kN.

At this instant, what is the rate of increase of kinetic energy of the train?

- A 0.075 MW B 1.2 MW C 1.3 MW D 3.8 MW

214. 9702/12/O/N/20/Q16

A ball is thrown vertically upwards. Air resistance is negligible.

Which statement is correct?

- A By the principle of conservation of energy, the total energy of the ball is constant throughout its motion.
- B By the principle of conservation of momentum, the momentum of the ball is constant throughout its motion.
- C The kinetic energy of the ball is greatest at the greatest height attained.
- D The potential energy of the ball increases at a constant rate during its ascent.

215. 9702/12/O/N/20/Q17

A hammer with 10 J of kinetic energy hits a nail and pushes it 5.0 mm into a plank.

Both the hammer and nail come to rest after the collision.

What is the approximate average force that acts on the nail while it moves through 5.0 mm?

- A 0.050 N B 2.0 N C 50 N D 2000 N

216. 9702/12/O/N/20/Q18

The change in gravitational potential energy ΔE of an object of mass m when moving through height Δh near the surface of the Earth is given by the equation shown.

$$\Delta E = mg\Delta h$$

Which equation is needed as part of the derivation of this expression?

- A kinetic energy = $\frac{1}{2} \times \text{mass} \times (\text{speed})^2$
- B moment = force $\times$ distance
- C weight = mass $\times$ acceleration of free fall
- D work done = power $\times$ time

217. 9702/12/O/N/20/Q19

A racing car has an output power of 300 kW when travelling at a constant speed of 60 m s^{-1} .

What is the total resistive force acting on the car?

- A** 5 kN **B** 10 kN **C** 50 kN **D** 100 kN

218. 9702/13/O/N/20/Q16

A ball drops onto a horizontal surface and bounces elastically.

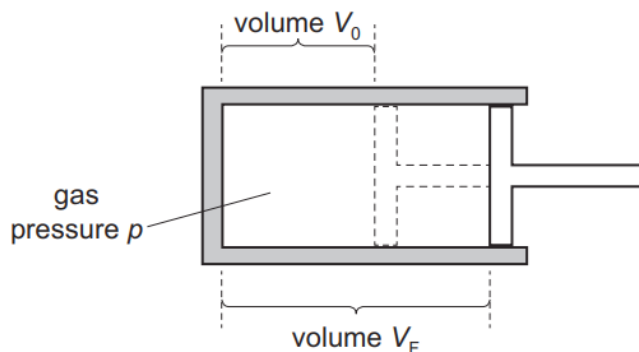
What happens to the kinetic energy of the ball during the very short time that it is in contact with the surface?

- A** Most of the kinetic energy is lost as heat and sound.
B The kinetic energy decreases to zero and then returns to its original value.
C The kinetic energy remains constant because it is an elastic collision.
D The kinetic energy remains constant in magnitude but changes direction.

219. 9702/13/O/N/20/Q17

Some gas in a cylinder is supplied with thermal energy q .

The gas does useful work in expanding at constant pressure p from volume V_0 to volume V_F , as shown.



Which expression gives the efficiency of this process?

- A** $\frac{pV_0}{q}$ **B** $\frac{V_F}{V_0 q}$ **C** $\frac{p(V_F - V_0)}{q}$ **D** $\frac{(V_F - V_0)}{V_0 q}$

220. 9702/13/O/N/20/Q18

An object of mass 0.30 kg is thrown vertically upwards from the ground with an initial velocity of 8.0 m s^{-1} . The object reaches a maximum height of 1.9 m.

How much work is done against air resistance as the object rises to its maximum height?

- A** 4.0 J **B** 5.6 J **C** 9.6 J **D** 15 J

221. 9702/13/O/N/20/Q19

A water pump raises a mass of $27 \times 10^3 \text{ kg}$ of water through a vertical distance of 80 m in a time of 1.0 hour.

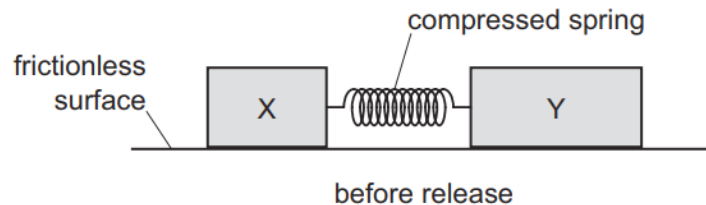
What is the average useful output power of the pump?

- A** 0.60 kW **B** 5.9 kW **C** 36 kW **D** 350 kW

222. 9702/12/F/M/21/Q15

Two blocks, X and Y, are on a horizontal frictionless surface. The mass of block Y is greater than that of block X. Block Y has a spring attached to its end.

The blocks are pushed together so that the spring is compressed between them and the blocks are held stationary as shown.



When released, the blocks move in opposite directions.

Which statement is correct?

- A** After release, the kinetic energy of block X must equal the kinetic energy of block Y.
B After release, the sum of the kinetic energies of the blocks is equal to zero.
C The total energy of the spring and blocks immediately before release is zero.
D The total energy of the spring and blocks is equal to the energy needed to bring the blocks together.

223. 9702/12/F/M/21/Q17

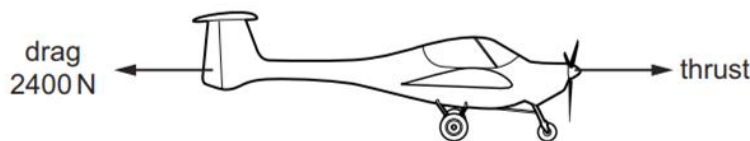
An egg of mass 25 g falls vertically downwards from the surface of a table which is 900 mm above the ground. Air resistance is negligible.

What is the kinetic energy of the egg when it hits the ground?

- A** 0.023 J **B** 0.22 J **C** 23 J **D** 220 J

224. 9702/12/F/M/21/Q18

An aircraft travels at a constant velocity of 90 ms^{-1} in horizontal flight. The diagram shows the horizontal forces acting on the aircraft.



The mass of the aircraft is 2000 kg.

What is the power produced by the thrust force?

- A** $1.8 \times 10^5 \text{ W}$ **B** $2.2 \times 10^5 \text{ W}$ **C** $1.8 \times 10^6 \text{ W}$ **D** $2.0 \times 10^6 \text{ W}$

225. 9702/11/M/J/21/Q15

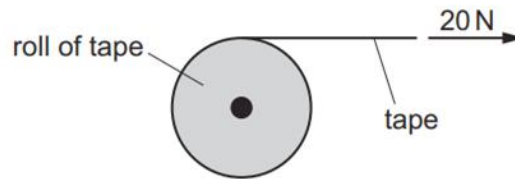
A rocket is fired upwards.

As it accelerates upwards after leaving the launch pad, which forms of energy are changing?

- A** chemical energy, gravitational potential energy and kinetic energy
- B** chemical energy and gravitational potential energy only
- C** chemical energy and kinetic energy only
- D** gravitational potential energy and kinetic energy only

226. 9702/11/M/J/21/Q16

A roll of tape of length 50 m requires a constant force of 20 N to unwrap it.

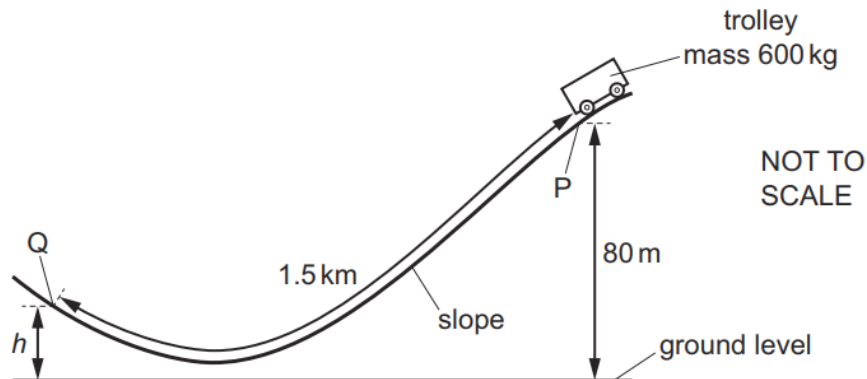


What is the work done in unwrapping the whole roll?

- A** 0.4 J **B** 2.5 J **C** 500 J **D** 1000 J

227. 9702/11/M/J/21/Q17

A trolley of mass 600 kg is initially at point P on a slope, at a height of 80 m above ground level, as shown. The trolley is released from rest and moves along the slope, first coming to rest at point Q, at height h above ground level.



The total distance PQ moved by the trolley along the slope is 1.5 km. A constant resistive force of 300 N opposes the motion of the trolley on the slope.

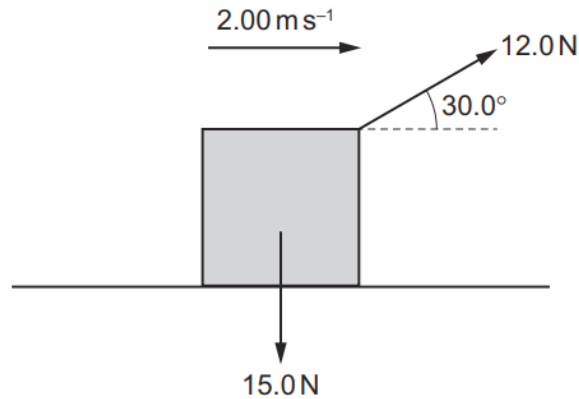
What is h ?

- A** 3.5 m **B** 76 m **C** 79 m **D** 80 m

228. 9702/11/M/J/21/Q18

An object of weight 15.0 N is pulled along a horizontal surface at a constant velocity of 2.00 m s^{-1} .

The force pulling the object is 12.0 N at 30.0° to the horizontal, as shown.



What is the power used to move the object?

- A** 12.0 W **B** 20.8 W **C** 24.0 W **D** 30.0 W

229. 9702/12/M/J/21/Q15

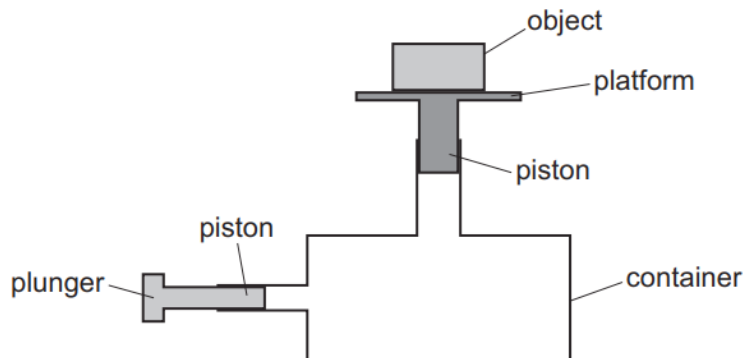
A stone is falling vertically through the air at a constant (terminal) velocity.

Which energy change is occurring?

- A** gravitational potential energy to thermal energy
B gravitational potential energy to kinetic energy of the stone
C kinetic energy to gravitational potential energy of the stone
D kinetic energy of the stone to thermal energy

230. 9702/12/M/J/21/Q16

An object of weight 12 N rests on a platform on top of a container with two pistons, as shown. The container contains a fixed mass of gas, and the pistons are free to move.



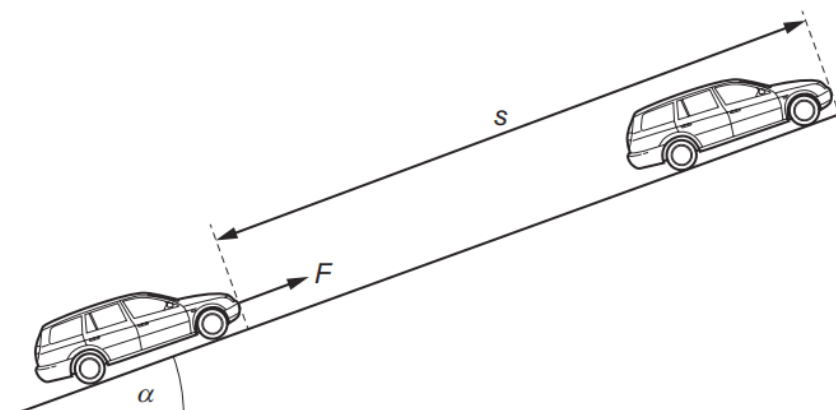
The plunger is slowly pushed 4.0 cm to the right. As a result, the object slowly moves upwards a distance 0.50 cm .

How much work is done on the object?

- A** 0.060 J **B** 0.48 J **C** 6.0 J **D** 48 J

231. 9702/12/M/J/21/Q17

A constant force F , acting on a car of mass m , moves the car up a slope through a distance s at constant velocity v . The angle of the slope to the horizontal is α .



The acceleration of free fall is g .

What is the ratio $\frac{\text{gravitational potential energy gained by car}}{\text{work done by force } F}$?

- A** $\frac{mgs \sin \alpha}{Fv}$ **B** $\frac{mv}{Fs}$ **C** $\frac{mv^2}{2Fs}$ **D** $\frac{mg \sin \alpha}{F}$

232. 9702/12/M/J/21/Q18

What is the definition of power?

- A** Power is the product of force and velocity.
B Power is the product of force and work done per unit time.
C Power is the product of force per unit time and velocity.
D Power is the rate at which work is done.

233. 9702/13/M/J/21/Q16

A spring is initially neither compressed nor extended.

A force can be applied to this spring so that it is either compressed to a shorter length or extended to a longer length.

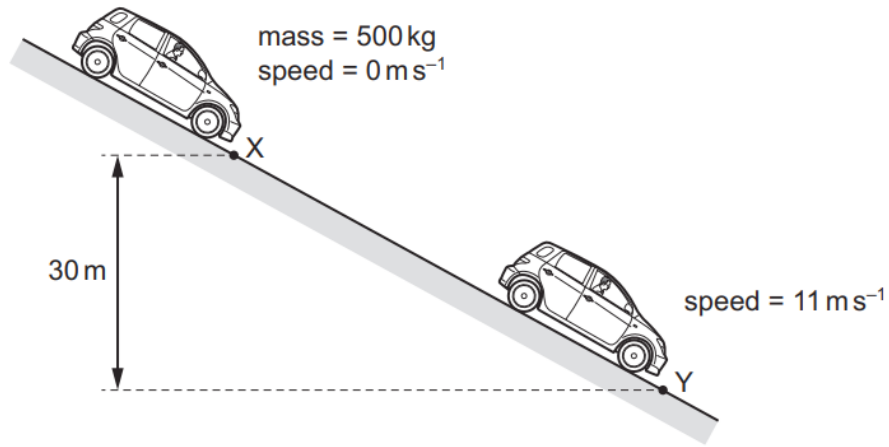
What is the change in the elastic potential energy in the spring when it is extended and when it is compressed?

	change in the elastic potential energy	
	spring is extended	spring is compressed
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

234. 9702/13/M/J/21/Q18

A car of mass 500 kg is at rest at point X on a slope, as shown.

The car's brakes are released and the car rolls down the slope with its engine switched off. At point Y the car has moved through a vertical height of 30 m and has a speed of 11 m s^{-1} .



What is the energy dissipated by frictional forces when the car moves from X to Y?

- A** $3.0 \times 10^4 \text{ J}$ **B** $1.2 \times 10^5 \text{ J}$ **C** $1.5 \times 10^5 \text{ J}$ **D** $1.8 \times 10^5 \text{ J}$

235. 9702/13/M/J/21/Q19

Which expression **cannot** be used to calculate power?

- A** $\frac{(\text{force} \times \text{displacement})}{\text{time}}$
- B** $\text{force} \times \text{velocity}$
- C** $\frac{\text{work done}}{\text{time}}$
- D** $\text{work done} \times \text{velocity}$

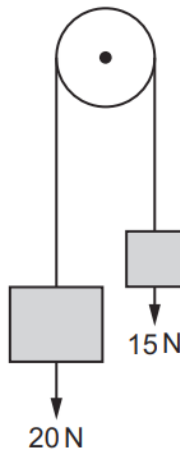
236. 9702/11/O/N/21/Q15

Which statement about energy is **not** correct?

- A** Energy is never lost but it may be transferred between different forms.
- B** In an inelastic collision, the total energy is constant.
- C** The efficiency of a system is the ratio of the useful energy output to the total energy input.
- D** When a machine does work, friction reduces the total energy.

237. 9702/11/O/N/21/Q16

A pulley of radius 0.40 m supports weights of 20 N and 15 N by means of a thin string, as shown.



The weights are moved by slowly rotating the pulley clockwise through an angle of 60° .

What is the increase in the total gravitational potential energy of the weights?

- A** 0.33 J **B** 2.0 J **C** 2.1 J **D** 15 J

238. 9702/11/O/N/21/Q17

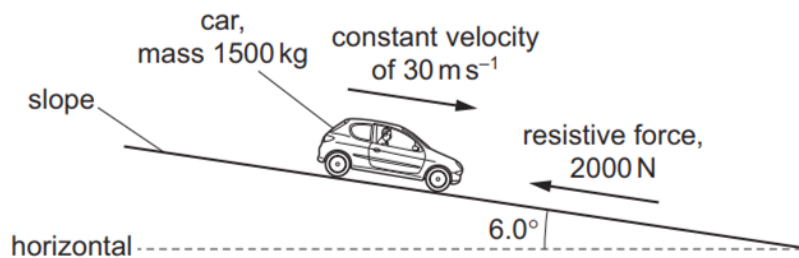
A car of mass 1500 kg accelerates from an initial speed of 15 m s^{-1} . This acceleration causes the car to gain $3.0 \times 10^5 \text{ J}$ of kinetic energy.

What is the change in the speed of the car?

- A** 5.4 m s^{-1} **B** 10 m s^{-1} **C** 20 m s^{-1} **D** 25 m s^{-1}

239. 9702/11/O/N/21/Q18

A car of mass 1500 kg travels at a constant velocity of 30 m s^{-1} down a slope. The slope is at an angle of 6.0° to the horizontal, as shown.



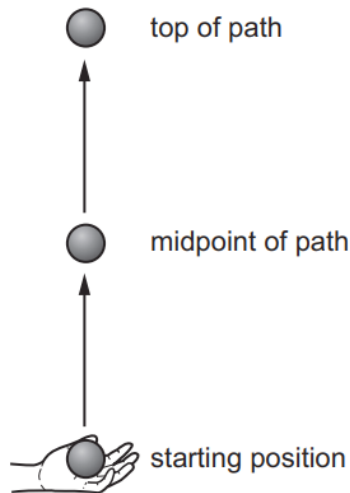
The magnitude of the total resistive force acting on the car is 2000 N.

What is the power output of the car's engine?

- A** 14 kW **B** 60 kW **C** 110 kW **D** 380 kW

240. 9702/12/O/N/21/Q16

A ball is thrown vertically upwards into the air. It rises to the top of its path before beginning to fall vertically downwards.



Assume that the gravitational potential energy of the ball is zero at its starting position.

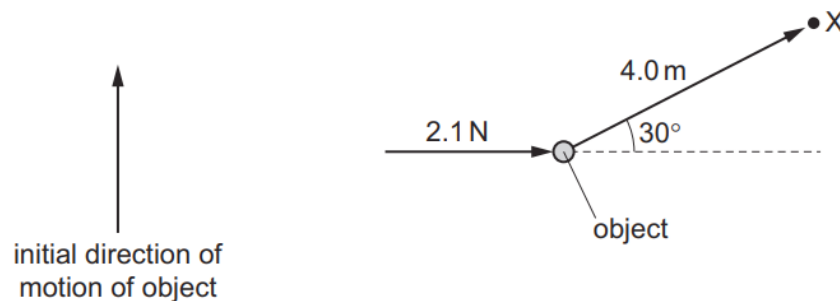
Which statement about the ball is **not** correct?

- A** As it rises, its kinetic energy is transferred to gravitational potential energy.
- B** At the midpoint of its path, its gravitational potential energy is equal to its initial kinetic energy.
- C** At the top of its path, its kinetic energy is zero.
- D** At the top of its path, its total energy is less than its initial total energy.

241. 9702/12/O/N/21/Q17

An object slides with constant velocity across a horizontal sheet of ice. Friction is negligible.

A constant horizontal force of 2.1 N is then applied to the object as shown.



A short time after applying the force, the object reaches point X at a displacement of 4.0 m from its position when the force was first applied.

What is the work done by the force on the object as it travels to point X?

- A** 4.2 J
- B** 4.8 J
- C** 7.3 J
- D** 8.4 J

242. 9702/12/O/N/21/Q18

The energy conversions inside a power station burning fossil fuel can be simplified as shown.

chemical energy $W \rightarrow$ thermal energy $X \rightarrow$ electrical energy Y

Which expression gives the efficiency of the power station?

- A $\frac{Y}{W}$ B $\frac{Y}{(W + X)}$ C $\frac{Y}{X}$ D $\frac{Y}{(W + X + Y)}$

243. 9702/12/O/N/21/Q19

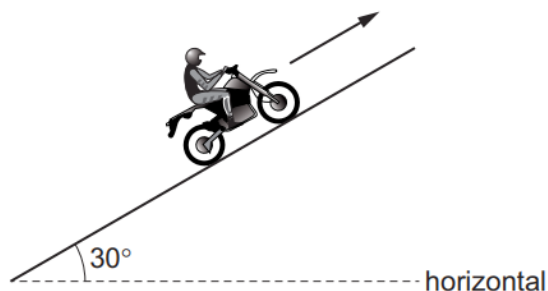
Car X is travelling at half the speed of car Y. Car X has twice the mass of car Y.

Which statement is correct?

- A Car X has half the kinetic energy of car Y.
B Car X has one quarter of the kinetic energy of car Y.
C Car X has twice the kinetic energy of car Y.
D The two cars have the same kinetic energy.

244. 9702/12/O/N/21/Q20

The total weight of a motorbike and rider is 1800 N. The motorbike travels in a straight line at constant speed up a hill at an angle of 30° to the horizontal.



The useful output power of the motorbike is 36 000 W. The total resistive force due to air resistance and friction on the motorbike and rider is 2400 N.

What is the speed of the motorbike?

- A 8.6 ms^{-1} B 11 ms^{-1} C 15 ms^{-1} D 24 ms^{-1}

245. 9702/13/O/N/21/Q17

A mechanical device does useful work at rate X and wastes energy at rate Y .

Which expression gives the efficiency of this device?

- A $\frac{X}{Y}$ B $\frac{(X - Y)}{Y}$ C $\frac{X}{(X + Y)}$ D $\frac{(X - Y)}{(X + Y)}$

246. 9702/13/O/N/21/Q18

Car P has kinetic energy 240 kJ.

Car Q has half the mass and twice the speed of car P.

What is the kinetic energy of car Q?

- A** 120 kJ **B** 240 kJ **C** 480 kJ **D** 960 kJ

247. 9702/13/O/N/21/Q19

A water pump is driven by an engine. The pump raises a volume of 0.50 m^3 of water in 1.0 minute from a depth of 30 m. The pump has an efficiency of 70%.

The density of water is 1000 kg m^{-3} .

What is the useful output power from the engine?

- A** 2.5 kW **B** 3.5 kW **C** 150 kW **D** 210 kW

248. 9702/12/F/M/22/Q15

The total energy input E_{in} in a process is partly transferred to useful energy output U and partly transferred to energy that is wasted W .

What is the efficiency of the process?

- A** $\frac{U}{E_{\text{in}}} \times 100\%$
- B** $\frac{W}{E_{\text{in}}} \times 100\%$
- C** $\frac{U}{W} \times 100\%$
- D** $\frac{U+W}{E_{\text{in}}} \times 100\%$

249. 9702/12/F/M/22/Q17

A rock of mass 40 kg is released from rest from a height of 20 m above the surface of a planet.

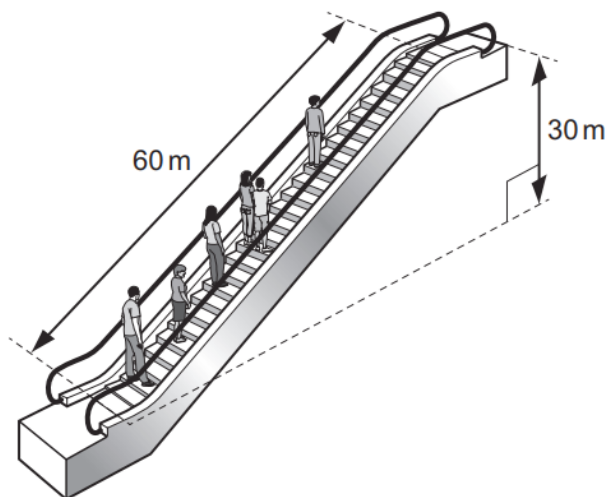
The rock has a kinetic energy of 32 kJ when it hits the surface of the planet. The planet does not have an atmosphere.

What is the weight of the rock on the surface of the planet?

- A** 1.6 N **B** 390 N **C** 1.6 kN **D** 64 kN

250. 9702/12/F/M/22/Q16

An escalator is 60 m long and lifts passengers through a vertical height of 30 m, as shown.



To drive the escalator against the forces of friction when there are no passengers requires a power of 2.0 kW.

The escalator is used by passengers of average mass 60 kg and the power to overcome friction remains constant.

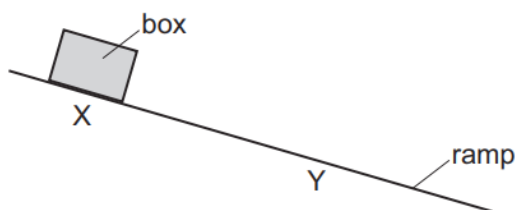
How much power is required to drive the escalator when it is carrying 20 passengers and is travelling at 0.75 m s^{-1} ?

- A** 4.4 kW **B** 6.4 kW **C** 8.8 kW **D** 10.8 kW

251. 9702/11/M/J/22/Q16

A box slides down a rough ramp.

The change in the gravitational potential energy of the box is 16 J as it moves between positions X and Y. The box has 24 J of kinetic energy at X and 35 J of kinetic energy at Y.



How much work is done against the frictional force?

- A** 5 J **B** 19 J **C** 27 J **D** 43 J

252. 9702/11/M/J/22/Q17

The total energy supplied to an electric motor is E . Energy Q is wasted and the remaining energy does useful work.

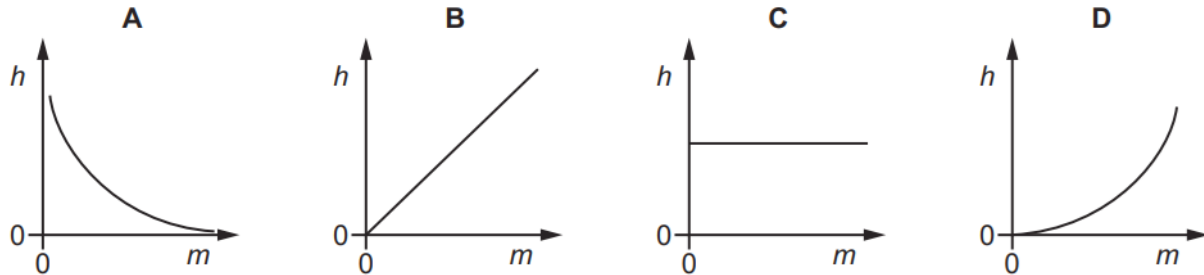
What is the efficiency of the motor?

- A** $\frac{Q}{E}$ **B** $\left(\frac{Q}{E}\right) - 1$ **C** $1 - \left(\frac{Q}{E}\right)$ **D** $\frac{(1-Q)}{E}$

253. 9702/11/M/J/22/Q18

- Objects with different masses are placed on the horizontal surface of a table. The objects are then raised to different heights above the table. The gain in gravitational potential energy of each object is the same.

Which graph best shows the variation of the height h of the objects above the table with their mass m ?



254. 9702/12/M/J/22/Q16

Which product of two quantities is equal to power?

- A** force $\times$ distance
- B** force $\times$ velocity
- C** work done $\times$ time
- D** work done $\times$ velocity

255. 9702/12/M/J/22/Q17

Researchers have developed a new type of filament lamp with an efficiency of 40%. Old-type filament lamps have an efficiency of 5.0%. The two types of lamp produce the same useful output power.

What is the ratio $\frac{\text{input power to new type of lamp}}{\text{input power to old type of lamp}}$?

- A** 0.13 **B** 0.63 **C** 1.6 **D** 8.0

256. 9702/12/M/J/22/Q18

A student attempts to derive the formula for kinetic energy E_K . She begins by considering an object of mass m which is initially at rest. A constant force F applied to the object causes it to accelerate to final velocity v in displacement s . The kinetic energy gained by the object is equal to the work done on the object by the force F .

Which equation would the student **not** need in order to derive the formula for E_K ?

- A** $F = ma$ **B** $W = Fs$ **C** $E = \frac{1}{2}Fs$ **D** $v^2 = u^2 + 2as$

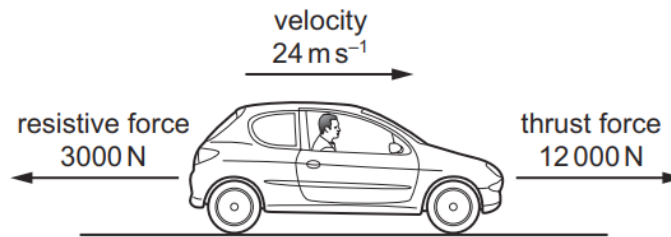
257. 9702/13/M/J/22/Q17

What is meant by the efficiency of a system?

- A** the difference between the useful energy output from the system and the total energy input
- B** the difference between the useful energy output from the system and the wasted energy output
- C** the ratio of the useful energy output from the system to the total energy input
- D** the ratio of the useful energy output from the system to the wasted energy output

258. 9702/13/M/J/22/Q18

A car of weight 15 000 N is travelling along a horizontal road.



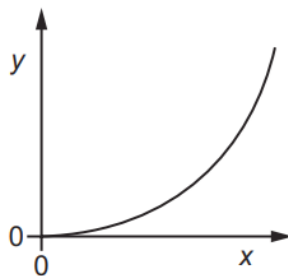
At one instant, the thrust force acting on the car from the engine is 12 000 N and the resistive force acting on the car is 3000 N. The velocity of the car at this instant is 24 m s⁻¹.

What is the power output from the engine?

- A** 72 kW
- B** 220 kW
- C** 290 kW
- D** 360 kW

259. 9702/13/M/J/22/Q19

The diagram shows the variation of a quantity y with a quantity x for objects in a uniform gravitational field.

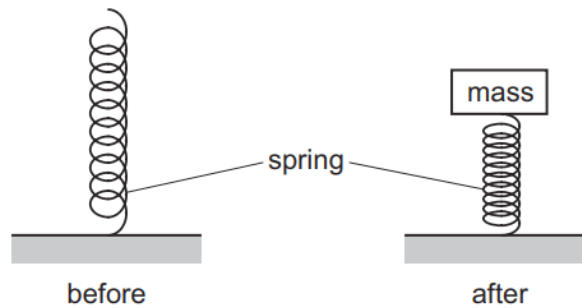


What could x and y represent?

	x	y
A	mass for different objects moving at the same speed	kinetic energy
B	speed for an object of constant mass	kinetic energy
C	vertical distance fallen for an object of constant mass	change of gravitational potential energy
D	mass for different objects falling the same vertical distance	change of gravitational potential energy

260. 9702/11//O/N/22/Q16

A spring is compressed by a mass, as shown.



Which statement describes the changes to the energy of the spring when it is compressed by the mass?

- A** The spring gains both gravitational potential energy and elastic potential energy.
- B** The spring gains gravitational potential energy and loses elastic potential energy.
- C** The spring loses both gravitational potential energy and elastic potential energy.
- D** The spring loses gravitational potential energy and gains elastic potential energy.

261. 9702/11//O/N/22/Q17

A man of mass 75 kg runs up a staircase consisting of 30 steps. Each step is 20 cm high.

The man takes a time of 7.0 s to run from the bottom of the staircase to the top.

What is the average rate of increase of gravitational potential energy of the man?

- A** 64 W **B** 450 W **C** 630 W **D** 4400 W

262. 9702/11//O/N/22/Q18

An alpha-particle has 2.2×10^{-13} J of kinetic energy.

What is the speed of the alpha-particle?

- A** $4.1 \times 10^6 \text{ m s}^{-1}$
- B** $5.8 \times 10^6 \text{ m s}^{-1}$
- C** $8.1 \times 10^6 \text{ m s}^{-1}$
- D** $1.2 \times 10^7 \text{ m s}^{-1}$

263. 9702/12//O/N/22/Q18

A lamp is switched on for 2.0 hours. The power input to the lamp is 1.0 W. The energy given out by the lamp as light is 7.0×10^3 J.

How much energy is converted to other forms by the lamp?

- A** 120 J **B** 200 J **C** 3400 J **D** 7200 J

264. 9702/12/O/N/22/Q17

A motor is used to lift a load vertically upwards.

The load has weight W .

The motor produces useful power output P .

The load is lifted at constant velocity v .

Which expression gives the time taken for the motor to lift the load vertically upwards through a distance d ?

A $\frac{P}{Wd}$

B $\frac{Wv}{P}$

C $\frac{Wd}{P}$

D $\frac{Pv}{W}$

265. 9702/12/O/N/22/Q19

An object of mass m is dropped onto the surface of two planets, X and Y, which have no atmosphere.

The height from which the object is dropped and the change in gravitational potential energy of the object, for each planet, are given in the table.

	height/m	change in gravitational potential energy
planet X	3	ΔE
planet Y	4	$4\Delta E$

The acceleration of free fall near the surface of planet X is g_x .

What is the acceleration of free fall near the surface of planet Y?

A $\frac{3}{4}g_x$

B $\frac{4}{3}g_x$

C $3g_x$

D $4g_x$

266. 9702/13/O/N/22/Q16

Which expression could be used to calculate power?

A $\frac{(\text{current})^2}{\text{resistance}}$

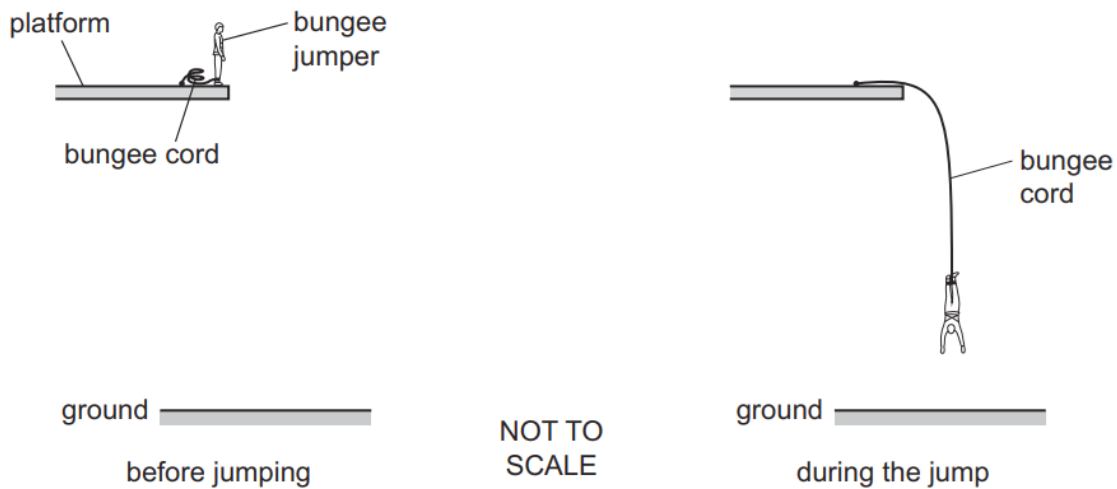
B $\frac{\text{force} \times \text{displacement}}{\text{time}}$

C $\text{current} \times (\text{resistance})^2$

D $\frac{\text{weight}}{\text{time}}$

267. 9702/13/O/N/22/Q17

A bungee jumper jumps from a platform and is decelerated by an elastic bungee cord, as shown.



When the jumper makes the jump, his initial gravitational potential energy is converted into his kinetic energy and into elastic potential energy in the cord.

At which part of the jump are all three types of energy non-zero?

- A on the platform before the jump
- B on the way down before the cord has started to extend
- C on the way down as he decelerates
- D at the bottom of the jump when he is stationary

268. 9702/13/O/N/22/Q18

An object is displaced horizontally to the right in a uniform vertical gravitational field.

Which statement describes the change in the gravitational potential energy of the object?

- A It decreases in direct proportion to the displacement.
- B It does not change with the displacement.
- C It increases in direct proportion to the displacement.
- D It increases in direct proportion to the square of the displacement.

269. 9702/12/F/M/23/Q16

A system with an efficiency of 74% wastes 230 W of power.

What is the useful output power of the system?

- A 170 W
- B 310 W
- C 650 W
- D 880 W

270. 9702/12/F/M/23/Q17

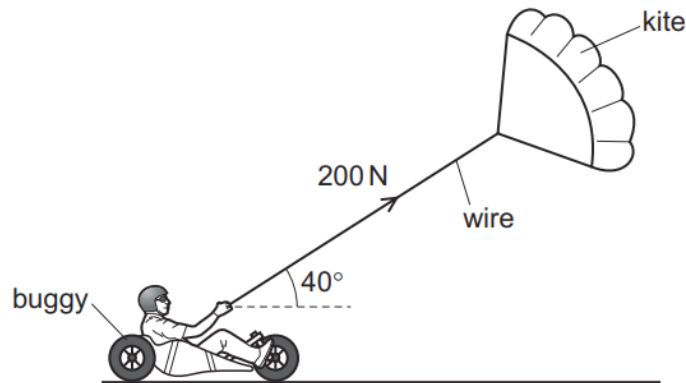
A projectile of mass 0.25 kg is at a height of 30 m above horizontal ground and travelling at a speed of 15 ms^{-1} . A short time later, it is at a height of 35 m above the horizontal ground and travelling at a speed of 5.0 ms^{-1} .

How much work is done against air resistance during this time?

- A** 0 J **B** 13 J **C** 25 J **D** 37 J

271. 9702/11/M/J/23/Q16

A man sits on a buggy that is pulled along by a wire attached to a kite. The wire is at an angle of 40° to the horizontal and has a constant tension of 200 N . The man and buggy travel a distance of 20 m along a straight horizontal path. The wire and the path of the buggy are in the same vertical plane.



What is the work done by the tension force on the man and buggy?

- A** 2.6 kJ **B** 3.1 kJ **C** 3.4 kJ **D** 4.0 kJ

272. 9702/11/M/J/23/Q17

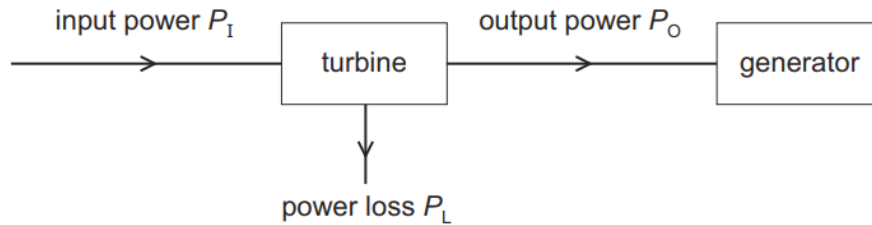
A ball is thrown vertically upwards from the surface of the Earth.

Which statement describes the energy of the ball as it rises through the air?

- A** The kinetic energy of the ball decreases as the gravitational potential energy decreases.
- B** The kinetic energy of the ball decreases as the gravitational potential energy increases.
- C** The kinetic energy of the ball increases as the gravitational potential energy decreases.
- D** The total energy of the ball increases.

273. 9702/12/M/J/23/Q15

A steam turbine is used to drive a generator. The input power to the turbine is P_I and the output power is P_O . The power loss in the turbine is P_L , as shown.



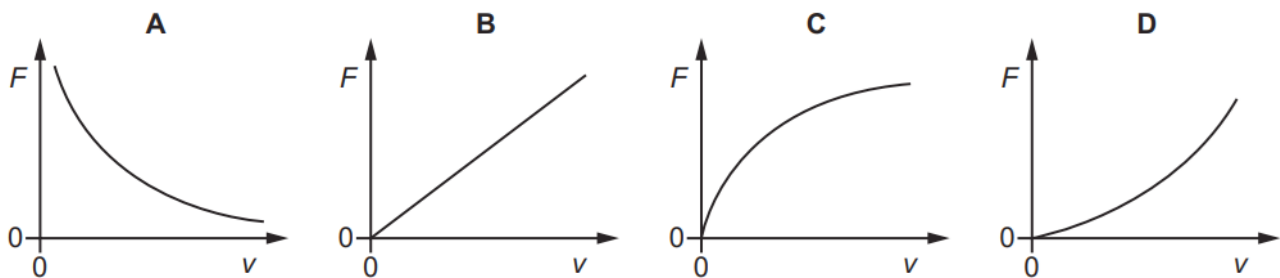
What is the efficiency of the turbine?

- A** $\frac{P_L}{P_O}$
 B $\frac{P_I}{P_O}$
 C $\frac{P_L}{P_I}$
 D $\frac{P_O}{P_I}$

274. 9702/12/M/J/23/Q16

A variable force is applied to ensure that a constant power is supplied to a train.

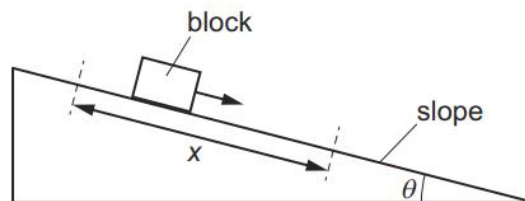
Which graph best shows the variation of the force F applied with the velocity v of the train?



275. 9702/12/M/J/23/Q17

A block is released from rest and slides a distance x down a straight slope.

The slope is at an angle θ to the horizontal.



The slope is frictionless and air resistance is negligible.

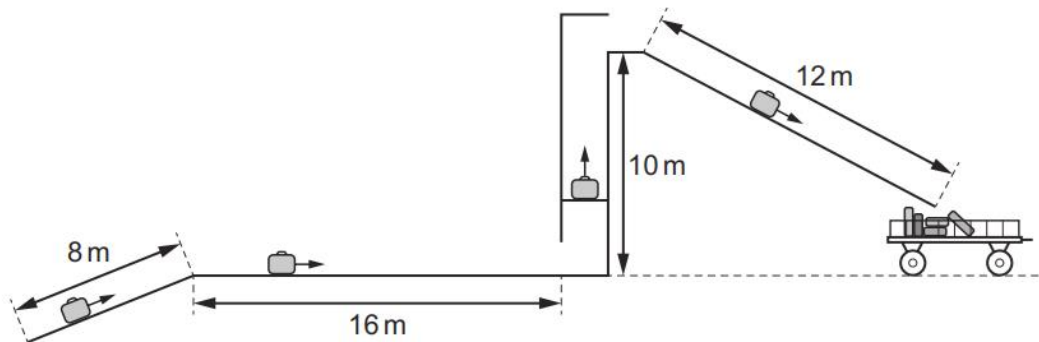
The acceleration of free fall is g .

Which expression gives the final speed of the block?

- A** $\sqrt{\frac{gx \sin \theta}{2}}$
 B $\sqrt{2gx \sin \theta}$
 C $\sqrt{2gx}$
 D $2gx \sin \theta$

276. 9702/13/M/J/23/Q15

An airport has a mechanical system for moving luggage. The system uses a horizontal conveyor belt, a sloping conveyor belt, a lift and a frictionless slide. A suitcase is moved around the airport using this system.



Resistive forces opposing the motion of the suitcase are negligible.

For which movement of the suitcase is the net work done on the suitcase greatest?

- A** moving the suitcase a distance of 8 m at a constant speed up the sloping conveyor belt
- B** moving the suitcase a distance of 16 m at a constant speed along the horizontal conveyor belt
- C** moving the suitcase a distance of 10 m at a constant speed vertically upwards on the lift
- D** moving the suitcase a distance of 12 m at increasing speed downwards on the slide

277. 9702/13/M/J/23/Q16

A car moves along a horizontal road with a constant velocity v against a resistive force F .

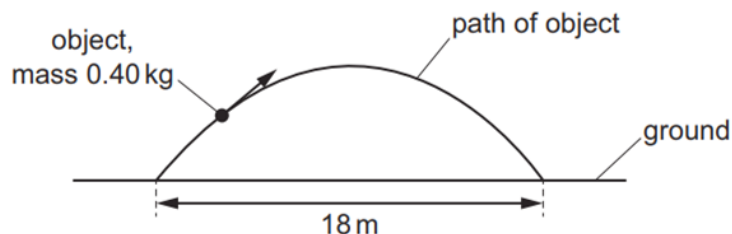
The engine of the car has an efficiency of 25%.

What is the input power to the engine?

- A** $\frac{Fv}{4.0}$ **B** $\frac{4.0}{Fv}$ **C** $4.0Fv$ **D** $\frac{4.0F}{v}$

278. 9702/13/M/J/23/Q17

An object of mass 0.40 kg is projected into the air and follows a curved path above horizontal ground.



The object takes a time of 1.5 s to move along its path. The object lands a horizontal distance of 18 m from its initial position. Air resistance is negligible.

What is the kinetic energy of the object at its maximum height?

- A** 0 J **B** 2.4 J **C** 11 J **D** 29 J

279. 9702/11/O/N/23/Q15

A crate of mass 50 kg is pushed a distance of 6.0 m along a horizontal surface against a constant resistive force of 70 N. The crate moves at a constant speed. It is then lifted, at a constant speed, through a vertical distance of 1.2 m onto the back of a lorry.

What is the total work done in this process?

- A** 420 J **B** 480 J **C** 590 J **D** 1000 J

280. 9702/11/O/N/23/Q16

The input power to a television is P_{in} . The useful sound and light power emitted by the television is P_{out} .

What is the efficiency of the television?

- A** $\frac{P_{\text{out}}}{P_{\text{in}}}$ **B** $\frac{P_{\text{in}} - P_{\text{out}}}{P_{\text{in}}}$ **C** $\frac{P_{\text{in}}}{P_{\text{out}}}$ **D** $\frac{P_{\text{out}}}{P_{\text{in}} - P_{\text{out}}}$

281. 9702/11/O/N/23/Q17

A builder holding a brick of mass 3000 g drops the brick on his foot.

What is a reasonable estimate of the change in gravitational potential energy of the brick?

- A** 30 J **B** 300 J **C** 3000 J **D** 30 000 J

282. 9702/12/O/N/23/Q15

An electric car travels at a constant speed of 70 km h^{-1} for 80 km on a straight horizontal road and uses energy E from its battery.

The total resistive force acting on the car is proportional to $(\text{speed})^2$. Assume that the electric motor is 100% efficient.

How much energy is used from the battery when the car travels at a constant speed of 60 km h^{-1} for 80 km on the straight horizontal road?

- A** $0.73E$ **B** $0.86E$ **C** $1.2E$ **D** $1.4E$

283. 9702/12/O/N/23/Q16

What is meant by the efficiency of a system?

- A** the total energy input to the system divided by the useful energy output by the system
B the useful energy output from the system divided by the energy wasted by the system
C the useful energy output from the system divided by the total energy input to the system
D the energy wasted by the system divided by the total energy input to the system

284. 9702/12/O/N/23/Q17

When an object of mass m is raised through a vertical height Δh , the gain of its gravitational potential energy is ΔE_p .

ΔE_p and Δh are related by the equation

$$\Delta E_p = mg\Delta h,$$

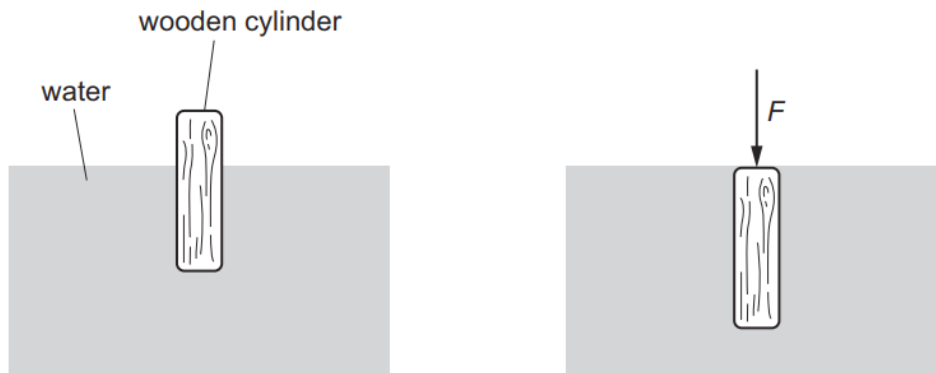
where g is the acceleration of free fall.

The definition of which physical quantity is needed to derive this equation?

- A** acceleration
- B** momentum
- C** power
- D** work done

285. 9702/13/O/N/23/Q15

A wooden cylinder floats partially submerged in a bath of water. A force F is applied to the cylinder until it is just fully submerged.



Which statement is **not** correct?

- A** Some of the water gains gravitational potential energy.
- B** The cylinder loses gravitational potential energy.
- C** Work is done by force F on the cylinder.
- D** Work is done by the upthrust on the cylinder.

286. 9702/13/O/N/23/Q16

A system has a useful power output of 4.0 W and a wasted power of 16 W.

What is the efficiency of the system?

- A** 5.0%
- B** 20%
- C** 25%
- D** 80%

287. 9702/13/O/N/23/Q17

A parachutist is falling towards the ground at a constant speed v . The rate at which she is losing gravitational potential energy is R .

The acceleration of free fall is g .

What is the mass of the parachutist?

A $\frac{gv}{R}$

B $\frac{R}{gv}$

C $\frac{2R}{v^2}$

D $\frac{v^2}{2R}$

ANSWERS

1. C	49. B	97. A	145. C	192. A
2. B	50. D	98. B	146. B	193. B
3. B	51. D	99. B	147. C	194. B
4. D	52. D	100. D	148. B	195. D
5. C	53. D	101. D	149. D	196. A
6. D	54. B	102. B	150. B	197. C
7. C	55. C	103. C	151. B	198. C
8. C	56. C	104. B	152. C	199. B
9. C	57. B	105. B	153. C	200. C
10. C	58. A	106. C	154. C	201. D
11. B	59. C	107. C	155. A	202. C
12. B	60. C	108. A	156. B	203. C
13. B	61. D	109. D	157. A	204. C
14. A	62. D	110. C	158. B	205. D
15. D	63. A	111. A	159. C	206. D
16. B	64. C	112. A	160. D	207. D
17. A	65. C	113. D	161. A	208. C
18. C	66. A	114. C	162. A	209. C
19. A	67. C	115. A	163. B	210. B
20. A	68. A	116. B	164. A	211. C
21. A	69. B	117. D	165. D	212. A
22. B	70. B	118. D	166. B	213. B
23. C	71. A	119. D	167. C	214. A
24. B	72. D	120. C	168. C	215. D
25. C	73. B	121. A	169. A	216. C
26. B	74. B	122. C	170. C	217. A
27. D	75. D	123. B	171. B	218. B
28. D	76. C	124. A	172. C	219. C
29. D	77. B	125. C	173. C	220. A
30. A	78. D	126. C	174. D	221. B
31. B	79. B	127. D	175. B	222. D
32. B	80. A	128. B	176. D	223. B
33. C	81. D	129. C	177. C	224. B
34. B	82. D	130. A	178. D	225. A
35. C	83. A	131. B	179. C	226. D
36. C	84. B	132. D	180. B	227. A
37. B	85. D	133. A	181. D	228. B
38. C	86. B	134. B	182. C	229. A
39. C	87. B	135. B	183. C	230. A
40. A	88. B	136. A	184. B	231. D
41. A	89. B	137. D	185. A	232. D
42. C	90. B	138. A	186. C	233. D
43. D	91. B	139. C	187. D	234. B
44. B	92. D	140. D	189. A	235. D
45. A	93. C	141. A	190. A	236. D
46. D	94. A	142. C	191. B	237. C
47. B	95. A	143. D		
48. D	96. A	144. B		

238. B	280. A
239. A	281. A
240. B	282. A
241. C	283. C
242. A	284. D
243. A	285. D
244. B	286. B
245. C	287. B
246. C	
247. B	
248. A	
249. C	
250. B	
251. A	
252. C	
253. A	
254. B	
255. A	
256. C	
257. C	
258. C	
259. B	
260. D	
261. C	
262. C	
263. B	
264. C	
265. C	
266. B	
267. C	
268. B	
269. C	
270. B	
271. B	
272. B	
273. D	
274. A	
275. B	
276. C	
277. C	
278. D	
279. D	

- 1 (a) The variation with extension x of the tension F in a spring is shown in Fig. 3.1.

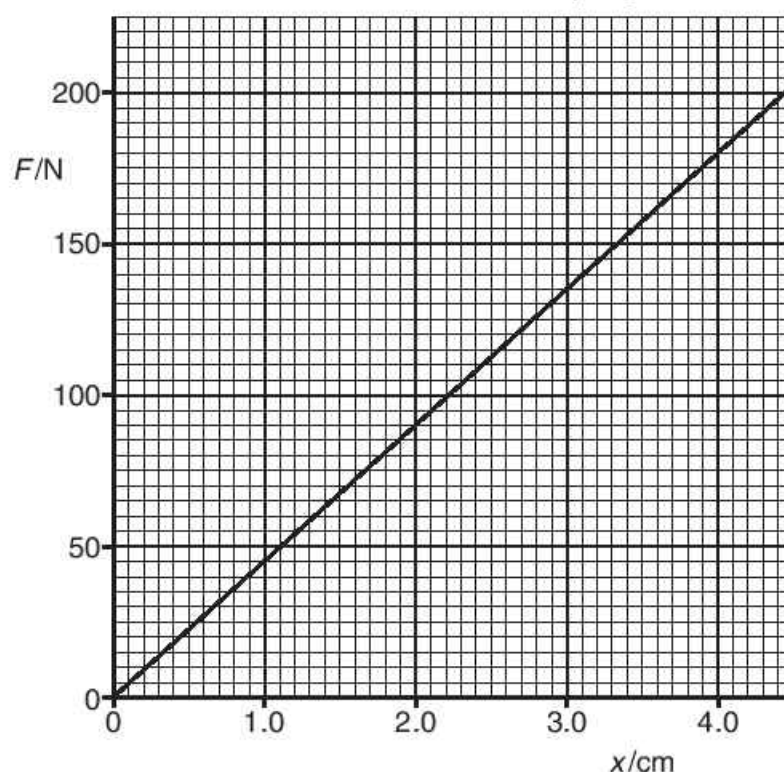


Fig. 3.1

Use Fig. 3.1 to calculate the energy stored in the spring for an extension of 4.0 cm.
Explain your working.

energy = J [3]

- (b) The spring in (a) is used to join together two frictionless trolleys A and B of mass M_1 and M_2 respectively, as shown in Fig. 3.2.

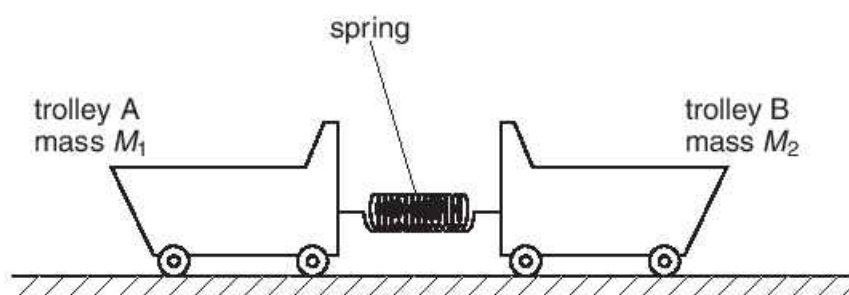


Fig. 3.2

The trolleys rest on a horizontal surface and are held apart so that the spring is extended.

The trolleys are then released.

- (i) Explain why, as the extension of the spring is reduced, the momentum of trolley A is equal in magnitude but opposite in direction to the momentum of trolley B.

.....
.....
.....
..... [2]

- (ii) At the instant when the extension of the spring is zero, trolley A has speed V_1 and trolley B has speed V_2 .
Write down

1. an equation, based on momentum, to relate V_1 and V_2 ,

.....
..... [1]

2. an equation to relate the initial energy E stored in the spring to the final energies of the trolleys.

.....
..... [1]

- (iii) 1. Show that the kinetic energy E_K of an object of mass m is related to its momentum p by the expression

$$E_K = \frac{p^2}{2m}.$$

[1]

2. Trolley A has a larger mass than trolley B.
Use your answer in **(ii) part 1** to deduce which trolley, A or B, has the larger kinetic energy at the instant when the extension of the spring is zero.

.....
..... [1]

- 2 (a) Explain what is meant by *work done*.

9702/21/M/J/11/Q2

.....
..... [1]

- (b) A car is travelling along a road that has a uniform downhill gradient, as shown in Fig. 2.1.

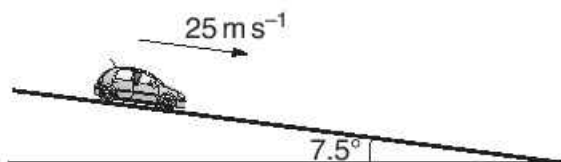


Fig. 2.1

The car has a total mass of 850 kg. The angle of the road to the horizontal is 7.5° . Calculate the component of the weight of the car down the slope.

component of weight = N [2]

- (c) The car in (b) is travelling at a constant speed of 25 m s^{-1} . The driver then applies the brakes to stop the car. The constant force resisting the motion of the car is 4600 N.

- (i) Show that the deceleration of the car with the brakes applied is 4.1 m s^{-2} .

[2]

- (ii) Calculate the distance the car travels from when the brakes are applied until the car comes to rest.

distance = m [2]

- (iii) Calculate

1. the loss of kinetic energy of the car,

loss of kinetic energy = J [2]

2. the work done by the resisting force of 4600 N.

work done = J [1]

- (iv) The quantities in (iii) part 1 and in (iii) part 2 are not equal. Explain why these two quantities are not equal.

.....
..... [1]

3 (a) (i) Explain what is meant by *work done*.

9702/22/M/J/11/Q3

.....
..... [1]

(ii) Define *power*.

.....
..... [1]

(b) Fig. 3.1 shows part of a fairground ride with a carriage on rails.

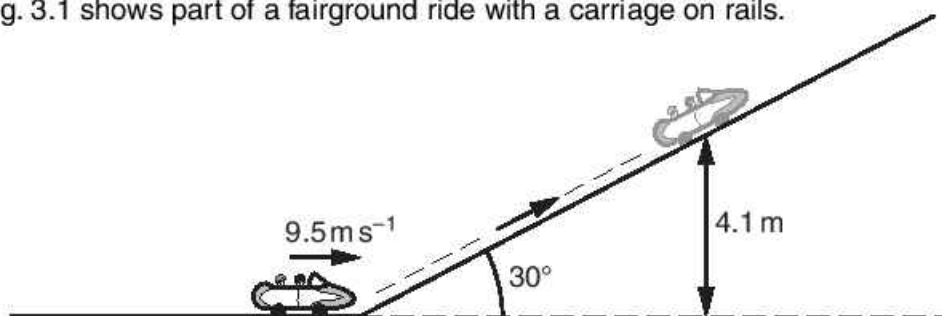


Fig. 3.1

The carriage and passengers have a total mass of 600 kg. The carriage is travelling at a speed of 9.5 m s^{-1} towards a slope inclined at 30° to the horizontal. The carriage comes to rest after travelling up the slope to a vertical height of 4.1 m.

(i) Calculate the kinetic energy, in kJ, of the carriage and passengers as they travel towards the slope.

kinetic energy = kJ [3]

(ii) Show that the gain in potential energy of the carriage and passengers is 24 kJ.

[2]

(iii) Calculate the work done against the resistive force as the carriage moves up the slope.

work done = kJ [1]

(iv) Use your answer in (iii) to calculate the resistive force acting against the carriage as it moves up the slope.

resistive force = N [2]

- 4 (a) Distinguish between *gravitational potential energy* and *electric potential energy*.

.....

 [2]

- (b) A body of mass m moves vertically through a distance h near the Earth's surface. Use the defining equation for work done to derive an expression for the gravitational potential energy change of the body.

[2]

- (c) Water flows down a stream from a reservoir and then causes a water wheel to rotate, as shown in Fig. 4.1.

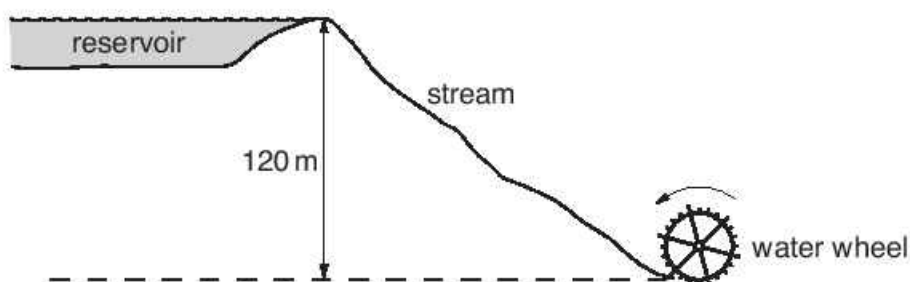


Fig. 4.1

As the water falls through a vertical height of 120 m, gravitational potential energy is converted to different forms of energy, including kinetic energy of the water. At the water wheel, the kinetic energy of the water is only 10% of its gravitational potential energy at the reservoir.

- (i) Show that the speed of the water as it reaches the wheel is 15 ms^{-1} .

[2]

- (ii) The rotating water wheel is used to produce 110 kW of electrical power. Calculate the mass of water flowing per second through the wheel, assuming that the production of electric energy from the kinetic energy of the water is 25% efficient.

mass of water per second = kg s^{-1} [3]

5 (a) Define

9702/22/O/N/11/Q2

(i) *force*,

.....
..... [1]

(ii) *work done*.

.....
..... [1]

(b) A force F acts on a mass m along a straight line for a distance s . The acceleration of the mass is a and the speed changes from an initial speed u to a final speed v .

(i) State the work W done by F .

[1]

(ii) Use your answer in (i) and an equation of motion to show that kinetic energy of a mass can be given by the expression

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times (\text{speed})^2.$$

[3]

(c) A resultant force of 3800 N causes a car of mass of 1500 kg to accelerate from an initial speed of 15 m s^{-1} to a final speed of 30 m s^{-1} .

(i) Calculate the distance moved by the car during this acceleration.

distance = m [2]

(ii) The same force is used to change the speed of the car from 30 m s^{-1} to 45 m s^{-1} . Explain why the distance moved is not the same as that calculated in (i).

.....
.....
..... [1]

- 6 (a) Show that the pressure P due to a liquid of density ρ is proportional to the depth h below the surface of the liquid.

[4]

- (b) The pressure of the air at the top of a mountain is less than that at the foot of the mountain. Explain why the difference in air pressure is not proportional to the difference in height as suggested by the relationship in (a).

.....

.....

.....

..... [2]

7 (a) Define *power*.

9702/23/O/N/12/Q3

.....
..... [1]

(b) A cyclist travels along a horizontal road. The variation with time t of speed v is shown in Fig. 3.1.

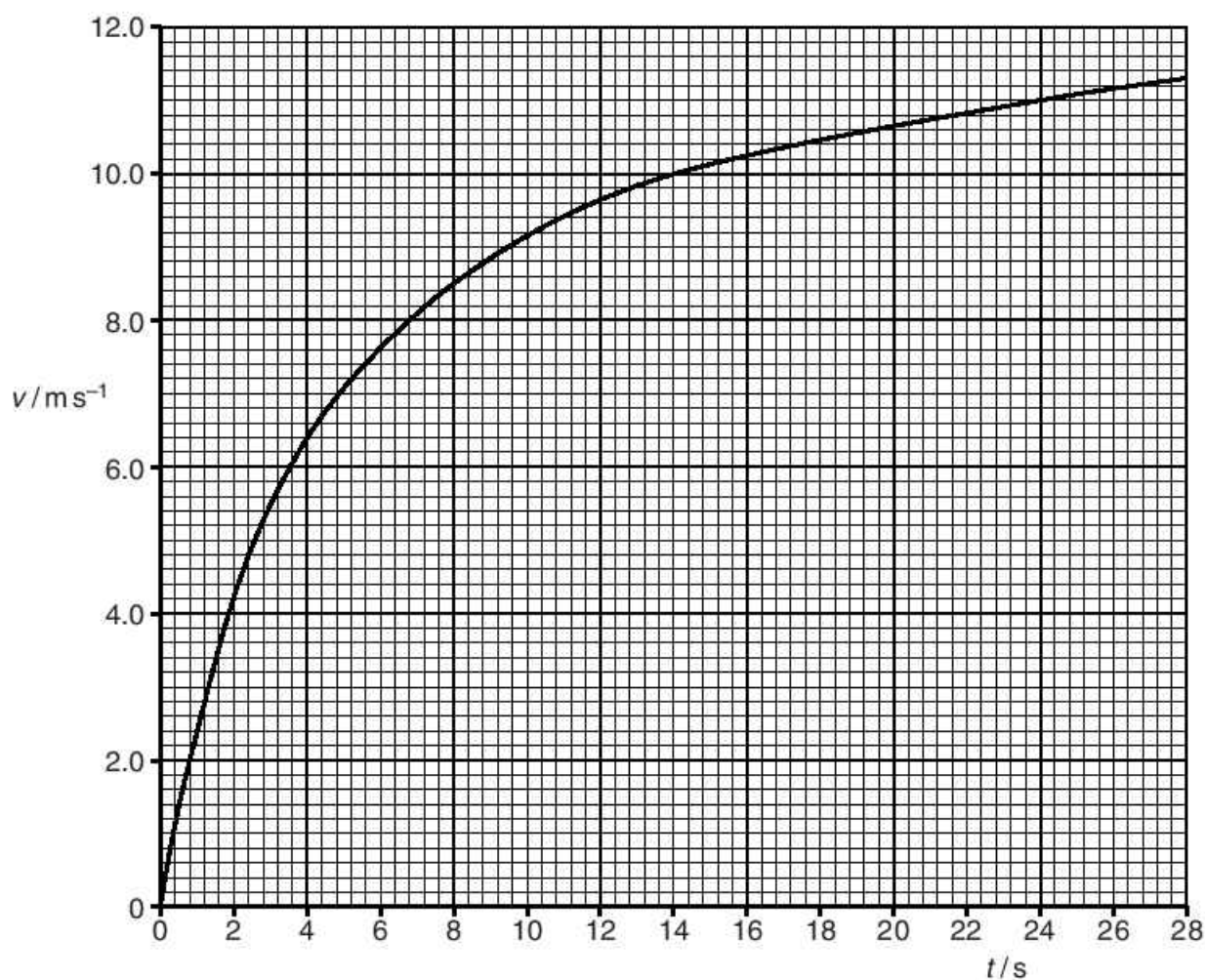


Fig. 3.1

The cyclist maintains a constant power and after some time reaches a constant speed of 12 ms^{-1} .

(i) Describe and explain the motion of the cyclist.

.....
.....
.....
..... [3]

- (ii) When the cyclist is moving at a constant speed of 12 ms^{-1} the resistive force is 48 N . Show that the power of the cyclist is about 600 W . Explain your working.

[2]

- (iii) Use Fig. 3.1 to show that the acceleration of the cyclist when his speed is 8.0 ms^{-1} is about 0.5 ms^{-2} .

[2]

- (iv) The total mass of the cyclist and bicycle is 80 kg . Calculate the resistive force R acting on the cyclist when his speed is 8.0 ms^{-1} . Use the value for the acceleration given in (iii).

$R = \dots\dots\dots\text{ N}$ [3]

- (v) Use the information given in (ii) and your answer to (iv) to show that, in this situation, the resistive force R is proportional to the speed v of the cyclist.

[1]

- 8 (a)** An object falls vertically from rest through air. State and explain the energy conversions that occur as the object falls.

.....

.....

.....

..... [3]

- (b)** A ball of mass 150 g is thrown vertically upwards with an initial speed of 25 ms^{-1} .

- (i)** Calculate the initial kinetic energy of the ball.

kinetic energy = J [3]

- (ii)** The ball reaches a height of 21 m above the point of release.

For the ball rising to this height, calculate

- 1.** the loss of energy of the ball to air resistance,

energy loss = J [3]

- 2.** the average force due to the air resistance.

force = N [2]

- 9 (a) Determine the SI base units of power.

9702/22/M/J/13/Q1

SI base units of power [3]

- (b) Fig. 1.1 shows a turbine that is used to generate electrical power from the wind.

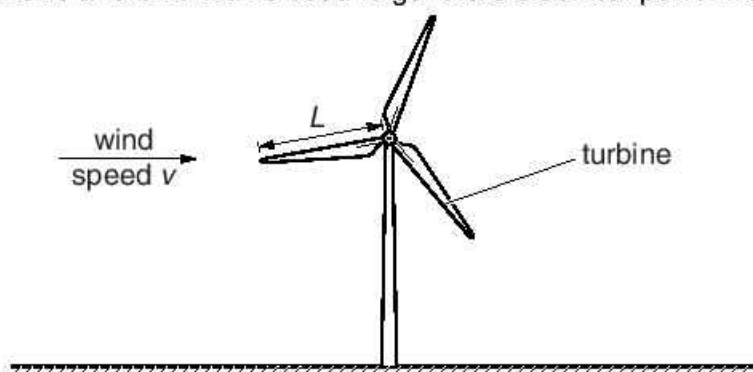


Fig. 1.1

The power P available from the wind is given by

$$P = CL^2\rho v^3$$

where L is the length of each blade of the turbine,
 ρ is the density of air,
 v is the wind speed,
 C is a constant.

- (i) Show that C has no units.

[3]

- (ii) The length L of each blade of the turbine is 25.0 m and the density ρ of air is 1.30 in SI units. The constant C is 0.931.
 The efficiency of the turbine is 55% and the electric power output P is 3.50×10^5 W.
 Calculate the wind speed.

wind speed = ms^{-1} [3]

- (iii) Suggest two reasons why the electrical power output of the turbine is less than the power available from the wind.

1.

2.

.....

[2]

10 (a) State what is meant by *work done*.

9702/22/O/N/13/Q3

.....
..... [1]

- (b) A trolley of mass 400 g is moving at a constant velocity of 2.5 m s^{-1} to the right as shown in Fig. 3.1.

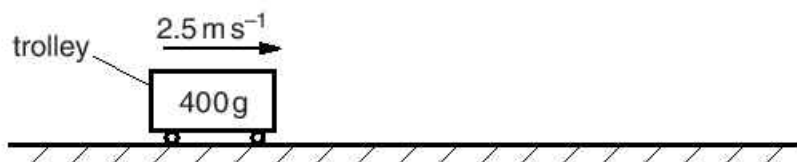


Fig. 3.1

Show that the kinetic energy of the trolley is 1.3 J.

[2]

- (c) The trolley in (b) moves to point P as shown in Fig. 3.2.

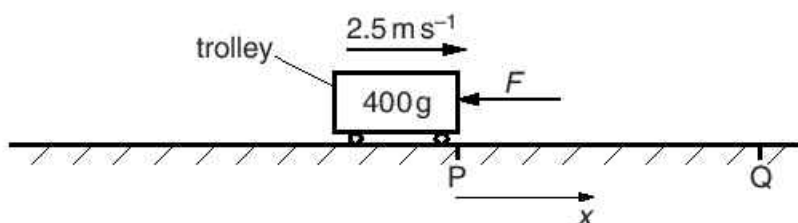


Fig. 3.2

At point P the speed of the trolley is 2.5 m s^{-1} .

A variable force F acts to the left on the trolley as it moves between points P and Q.

The variation of F with displacement x from P is shown in Fig. 3.3.

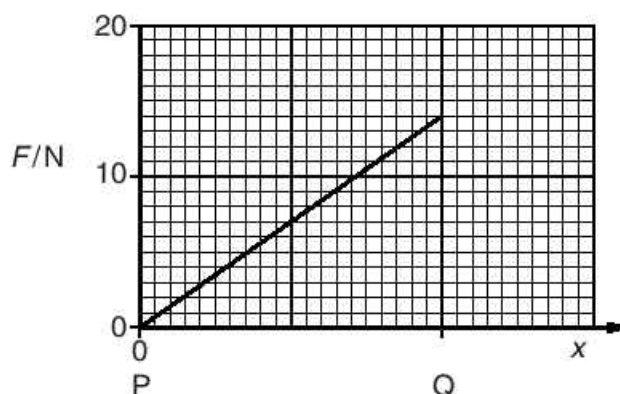


Fig. 3.3

The trolley comes to rest at point Q.

- (i) Calculate the distance PQ.

distance PQ = m [3]

- (ii) On Fig. 3.4, sketch the variation with x of velocity v for the trolley moving between P and Q.

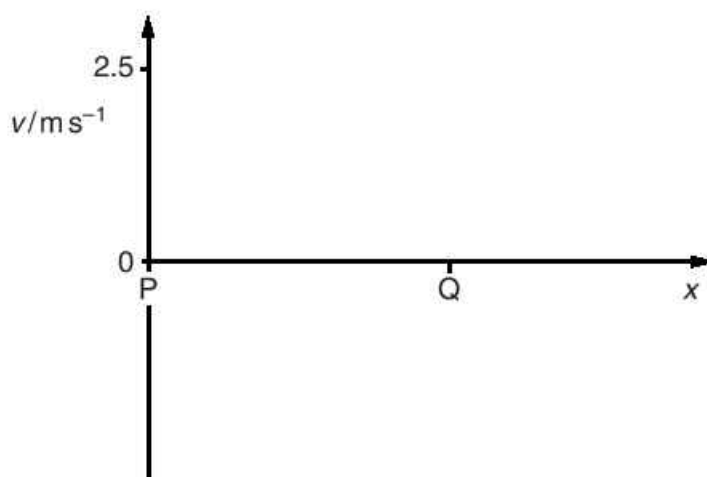


Fig. 3.4

[2]

- 11 (a) Distinguish between gravitational potential energy and elastic potential energy.

.....

[2]

- (b) A ball of mass 65 g is thrown vertically upwards from ground level with a speed of 16 m s^{-1} . Air resistance is negligible.

- (i) Calculate, for the ball,

1. the initial kinetic energy,

kinetic energy = J [2]

2. the maximum height reached.

maximum height = m [2]

- (ii) The ball takes time t to reach maximum height. For time $\frac{t}{2}$ after the ball has been thrown, calculate the ratio

$$\frac{\text{potential energy of ball}}{\text{kinetic energy of ball}}.$$

ratio = [3]

- (iii) State and explain the effect of air resistance on the time taken for the ball to reach maximum height.

.....
[1]

12 (a) (i) Define *power*.

..... [1]

(ii) Use your definition in (i) to show that power may also be expressed as the product of force and velocity.

[2]

(b) A lorry moves up a road that is inclined at 9.0° to the horizontal, as shown in Fig. 2.1.

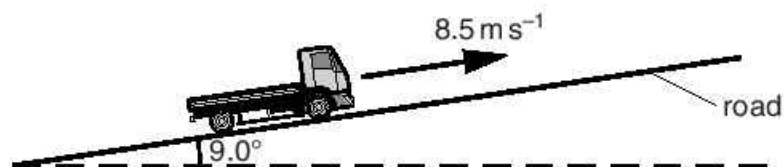


Fig. 2.1

The lorry has mass 2500 kg and is travelling at a constant speed of 8.5 m s^{-1} . The force due to air resistance is negligible.

(i) Calculate the useful power from the engine to move the lorry up the road.

power = kW [3]

(ii) State two reasons why the rate of change of potential energy of the lorry is equal to the power calculated in (i).

1.

.....

2.

.....

[2]

- 13 A metal ball of mass 40 g falls vertically onto a spring, as shown in Fig. 4.1.

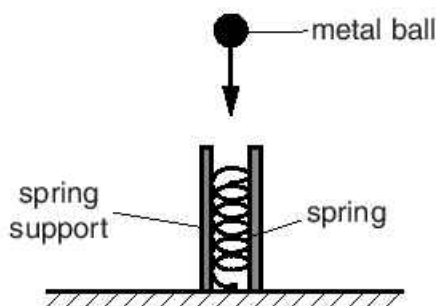


Fig. 4.1 (not to scale)

The spring is supported and stands vertically. The ball has a speed of 2.8 ms^{-1} as it makes contact with the spring. The ball is brought to rest as the spring is compressed.

- (a) Show that the kinetic energy of the ball as it makes contact with the spring is 0.16 J.

[2]

- (b) The variation of the force F acting on the spring with the compression x of the spring is shown in Fig. 4.2.

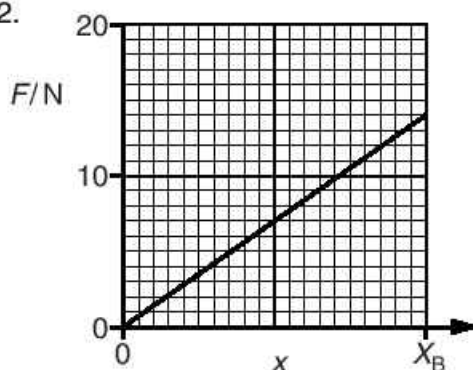


Fig. 4.2

The ball produces a maximum compression X_B when it comes to rest. The spring has a spring constant of 800 N m^{-1} .

Use Fig. 4.2 to

- (i) calculate the compression X_B .

$$X_B = \dots\dots\dots \text{ m [2]}$$

- (ii) show that not all the kinetic energy in (a) is converted into elastic potential energy in the spring.

[2]

- 14 (a) Explain what is meant by *gravitational potential* energy and *kinetic* energy.

gravitational potential energy:

.....

kinetic energy:

.....

[2]

- (b) A ball of mass 400 g is thrown with an initial velocity of 30.0 ms^{-1} at an angle of 45.0° to the horizontal, as shown in Fig. 4.1.

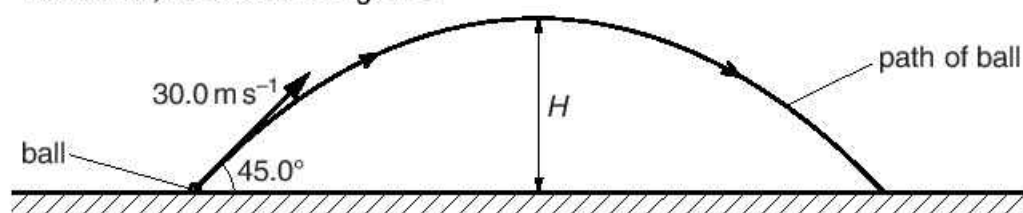


Fig. 4.1

Air resistance is negligible. The ball reaches a maximum height H after a time of 2.16 s.

- (i) Calculate

1. the initial kinetic energy of the ball,

kinetic energy = J [3]

2. the maximum height H of the ball,

$H =$ m [2]

3. the gravitational potential energy of the ball at height H .

potential energy = J [2]

- (ii) 1. Determine the kinetic energy of the ball at its maximum height.

kinetic energy = J [1]

2. Explain why the kinetic energy of the ball at maximum height is not zero.

.....

..... [1]

15 (a) Explain what is meant by *work done*.

9702/23/M/J/14/Q3

.....
..... [1]

(b) A boy on a board B slides down a slope, as shown in Fig. 3.1.

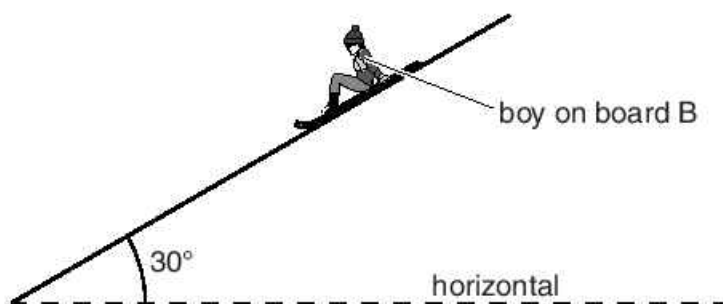


Fig. 3.1

The angle of the slope to the horizontal is 30° . The total resistive force F acting on B is constant.

(i) State a word equation that links the work done by the force F on B to the changes in potential and kinetic energy.

.....
..... [1]

(ii) The boy on the board B moves with velocity v down the slope. The variation with time t of v is shown in Fig. 3.2.

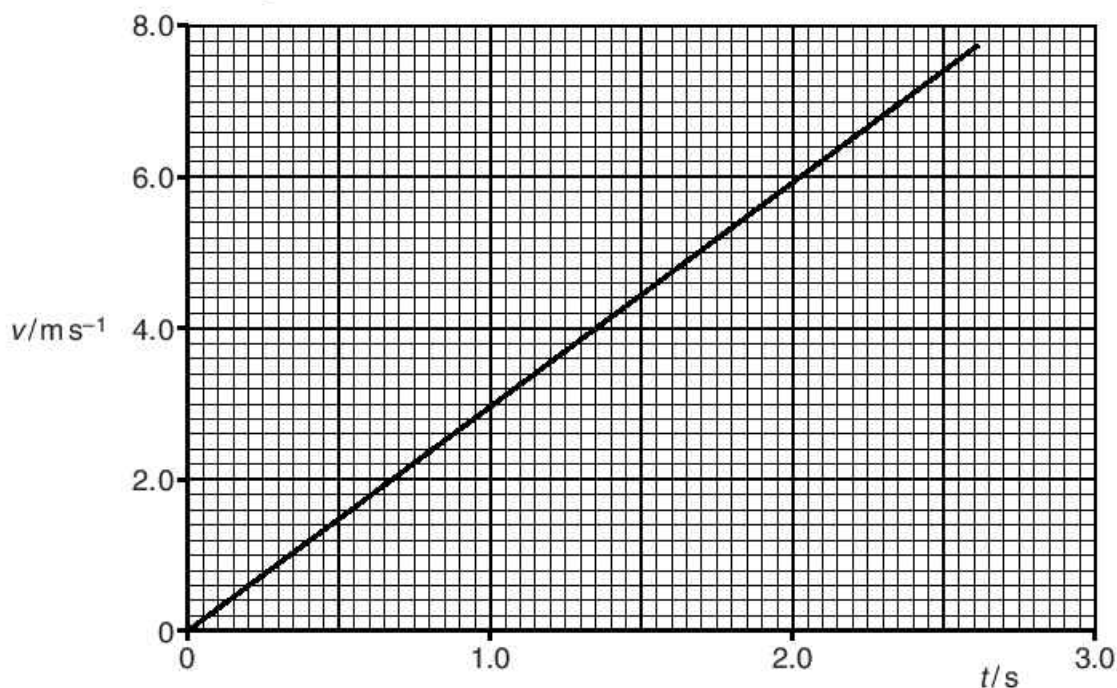


Fig. 3.2

The total mass of B is 75 kg.

For B, from $t = 0$ to $t = 2.5$ s,

1. show that the distance moved down the slope is 9.3 m,

[2]

2. calculate the gain in kinetic energy,

gain in kinetic energy = J [3]

3. calculate the loss in potential energy,

loss in potential energy = J [3]

4. calculate the resistive force F .

$F =$ N [3]

16 A motor is used to move bricks vertically upwards, as shown in Fig. 5.1.

9702/21/O/N/14/Q5

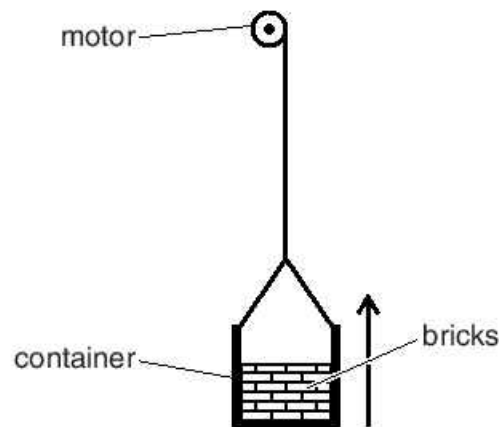


Fig. 5.1

The bricks start from rest and accelerate for 2.0 s. The bricks then travel at a constant speed of 0.64 m s^{-1} for 25 s. Finally the bricks are brought to rest in a further 3.0 s.

The total mass of the bricks is 25 kg.

(a) Determine the change in kinetic energy of the bricks

(i) in the first 2.0 s,

change in kinetic energy = J [2]

(ii) in the next 25 s,

change in kinetic energy = J [1]

(iii) in the final 3.0 s.

change in kinetic energy = J [1]

(b) The bricks are in a container. The weight of the container and bricks is 350 N.
Calculate, for the lifting of the bricks and container when travelling at constant speed,

(i) the gain in potential energy,

energy gain = J [3]

(ii) the power required.

power = W [2]

17 (a) Define *power*.

9702/23/M/J/15/Q3

.....
.....[1]

(b) Fig. 3.1 shows a car travelling at a speed of 22 m s^{-1} on a horizontal road.

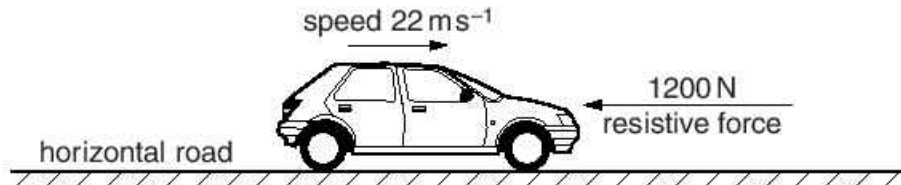


Fig. 3.1

The car has a mass of 1500 kg . A resistive force of 1200 N acts on the car.
Calculate

(i) the force F required from the car to produce an acceleration of 0.82 m s^{-2} ,

$F =$ N [3]

(ii) the power required to produce this acceleration.

power = W [2]

(c) The resistive force on the car is proportional to v^2 , where v is the speed of the car.
Suggest why the car has a maximum speed.

.....
.....[1]

18 (a) State what is meant by

9702/22/F/M/16/Q3

(i) *work done*,

.....
..... [1]

(ii) *elastic potential energy*.

.....
..... [1]

(b) A block of mass 0.40 kg slides in a straight line with a constant speed of 0.30 m s^{-1} along a horizontal surface, as shown in Fig. 3.1.



Fig. 3.1

The block hits a spring and decelerates. The speed of the block becomes zero when the spring is compressed by 8.0 cm .

(i) Calculate the initial kinetic energy of the block.

kinetic energy = J [2]

- (ii) The variation of the compression x of the spring with the force F applied to the spring is shown in Fig. 3.2.

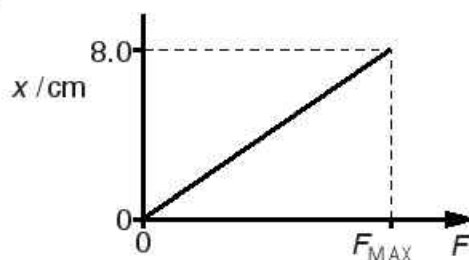


Fig. 3.2

Use your answer in (b)(i) to determine the maximum force F_{MAX} exerted on the spring by the block.
Explain your working.

$$F_{\text{MAX}} = \dots\dots\dots \text{ N [3]}$$

- (iii) Calculate the maximum deceleration of the block.

$$\text{deceleration} = \dots\dots\dots \text{ ms}^{-2} [1]$$

- (iv) State and explain whether the block is in equilibrium

1. before it hits the spring,

.....

2. when its speed becomes zero.

.....

[2]

(c) The energy E stored in a spring is given by

$$E = \frac{1}{2}kx^2$$

where k is the spring constant of the spring and x is its compression.

The mass m of the block in **(b)** is now varied. The initial speed of the block remains constant and the spring continues to obey Hooke's law.

On Fig. 3.3, sketch the variation of the maximum compression x_0 of the spring with mass m .

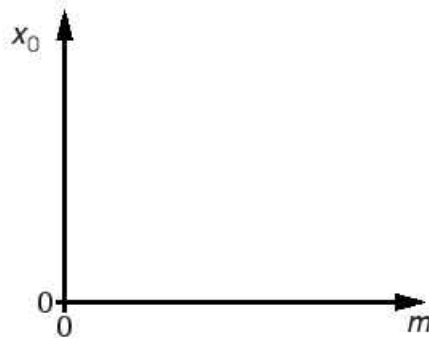


Fig. 3.3

[2]

19 (a) State what is meant by *elastic potential energy*.

9702/21/M/J/16/Q4

.....
.....[1]

(b) A spring is extended by applying a force. The variation with extension x of the force F is shown in Fig. 4.1 for the range of values of x from 20 cm to 40 cm.

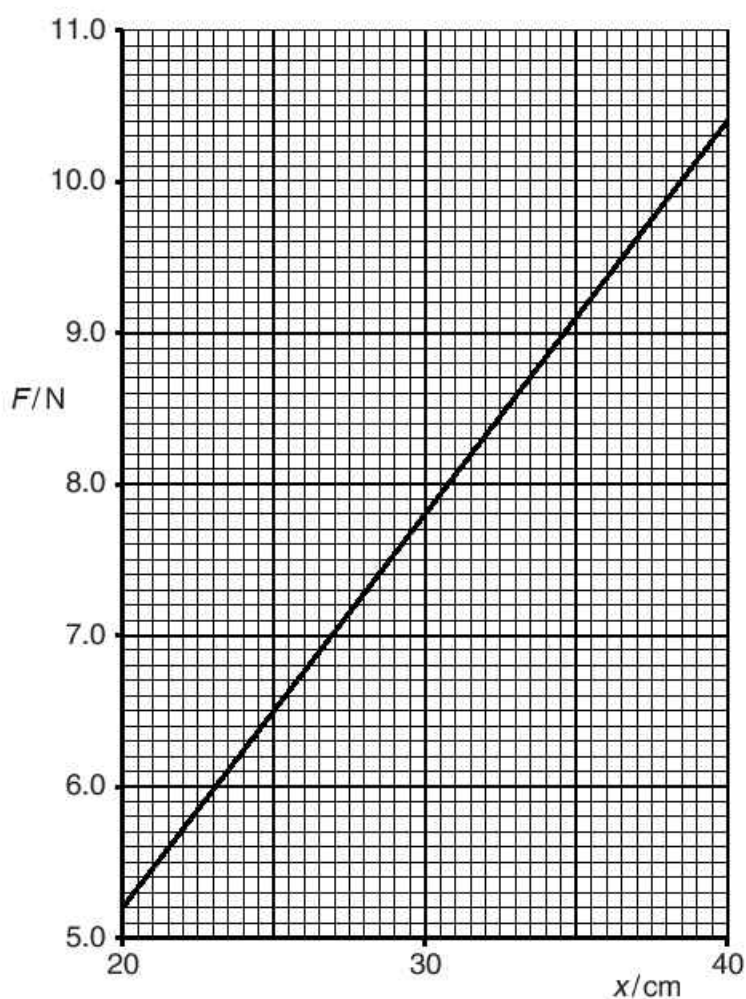


Fig. 4.1

(i) Use data from Fig. 4.1 to show that the spring obeys Hooke's law for this range of extensions.

.....
.....
.....[2]

(ii) Use Fig. 4.1 to calculate

1. the spring constant,

spring constant = Nm^{-1} [2]

2. the work done extending the spring from $x = 20\text{ cm}$ to $x = 40\text{ cm}$.

work done = J [3]

(c) A force is applied to the spring in **(b)** to give an extension of 50 cm.

State how you would check that the spring has not exceeded its elastic limit.

.....
.....[1]

20 (a) Define *acceleration*.

.....
[1]

(b) A man travels on a toboggan down a slope covered with snow from point A to point B and then to point C. The path is illustrated in Fig. 1.1.

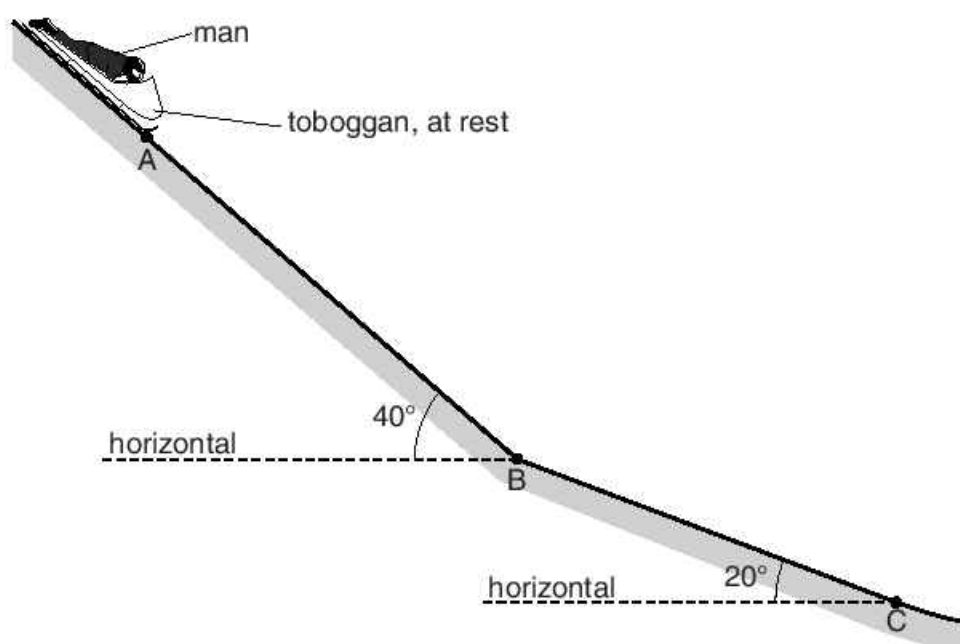


Fig. 1.1 (not to scale)

The slope AB makes an angle of 40° with the horizontal and the slope BC makes an angle of 20° with the horizontal. Friction is not negligible.

The man and toboggan have a combined mass of 95 kg.

The man starts from rest at A and has constant acceleration between A and B. The man takes 19 s to reach B. His speed is 36 ms^{-1} at B.

(i) Calculate the acceleration from A to B.

acceleration = ms^{-2} [2]

(ii) Show that the distance moved from A to B is 340 m.

[1]

(iii) For the man and toboggan moving from A to B, calculate

1. the change in kinetic energy,

change in kinetic energy = J [2]

2. the change in potential energy.

change in potential energy = J [2]

(iv) Use your answers in **(iii)** to determine the average frictional force that acts on the toboggan between A and B.

frictional force = N [2]

(v) A parachute opens on the toboggan as it passes point B. There is a constant deceleration of 3.0 ms^{-2} from B to C.

Calculate the frictional force that produces this deceleration between B and C.

frictional force = N [2]

- 21 (a) Explain what is meant by *gravitational potential energy* and by *kinetic energy*.

gravitational potential energy:

.....

kinetic energy:

.....

[2]

- (b) A motion sensor is used to measure the velocity of a ball falling vertically towards the ground, as illustrated in Fig. 3.1.

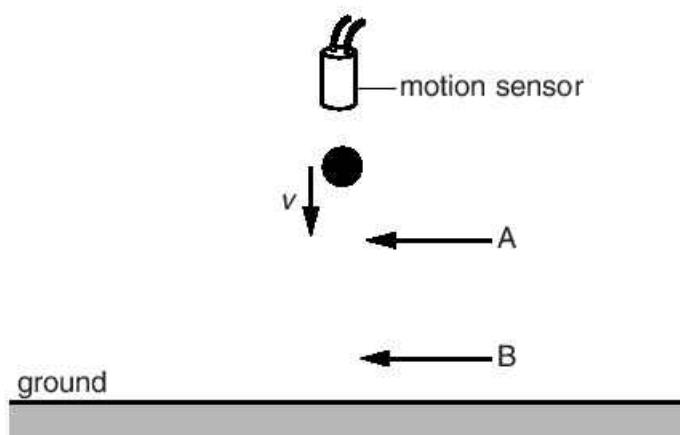


Fig. 3.1

The ball passes through points A and B as it falls. The ball has a mass of 1.5 kg.

The variation with time t of the velocity v of the ball as it falls from A to B is shown in Fig. 3.2.

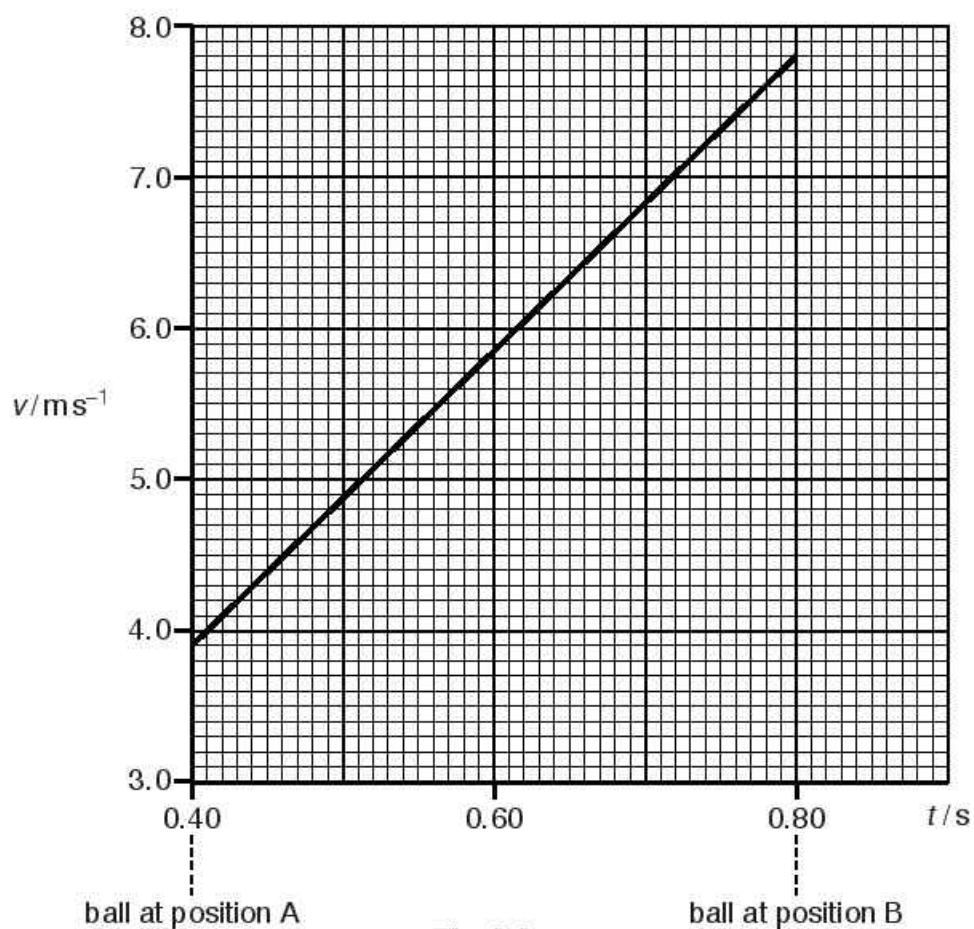


Fig. 3.2

Use Fig. 3.2 to calculate, for the ball falling from A to B,

(i) the displacement,

displacement = m [3]

(ii) the acceleration,

acceleration = ms^{-2} [2]

(iii) the change in kinetic energy.

change in kinetic energy =J [3]

(c) Show that the work done by the gravitational field on the ball in **(b)** as it moves from A to B is equal to the change in kinetic energy.

[2]

22 A ball of mass 0.030 kg moves along a curved track, as shown in Fig. 2.1.

9702/22/O/N/16/Q2

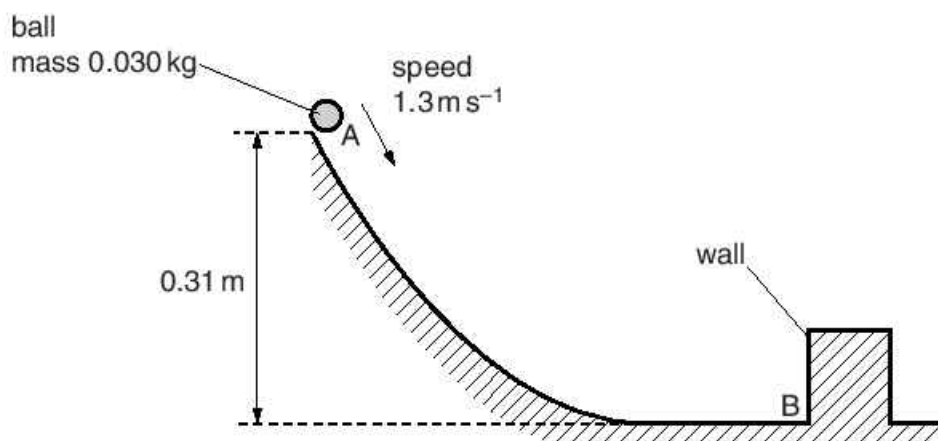


Fig. 2.1

The speed of the ball is 1.3 m s^{-1} when it is at point A at a height of 0.31 m. The ball moves down the track and collides with a vertical wall at point B. The ball then rebounds back up the track. It may be assumed that frictional forces are negligible.

- (a)** Calculate the change in gravitational potential energy of the ball in moving from point A to point B.

change in gravitational potential energy = J [2]

- (b)** Show that the ball hits the wall at B with a speed of 2.8 m s^{-1} .

- (c)** The change in momentum of the ball due to the collision with the wall is $0.096 \text{ kg m s}^{-1}$. The ball is in contact with the wall for a time of 20 ms.

Determine, for the ball colliding with the wall,

- (i)** the speed immediately after the collision,

speed = ms^{-1} [2]

- (ii)** the magnitude of the average force on the ball.

force = N [2]

- (d)** State and explain whether the collision is elastic or inelastic.

.....
.....[1]

- (e)** In practice, frictional effects are significant so that the actual increase in kinetic energy of the ball in moving from A to B is 76 mJ. The length of the track between A and B is 0.60 m.

Use your answer in **(a)** to determine the average frictional force acting on the ball as it moves from A to B.

frictional force = N [2]

- 23 A ball is thrown vertically upwards towards a ceiling and then rebounds, as illustrated in Fig. 3.1.

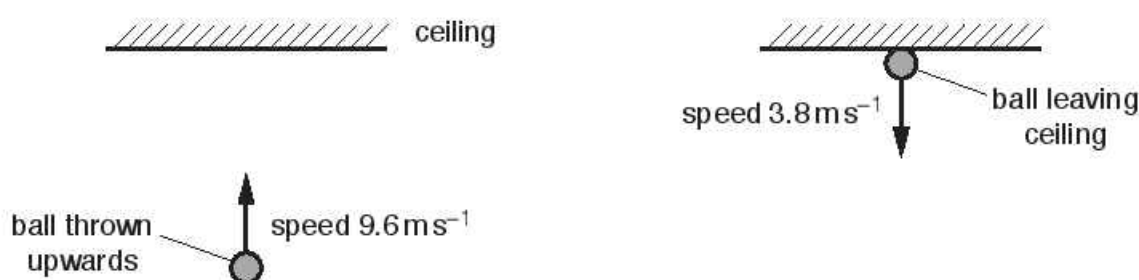


Fig. 3.1

The ball is thrown with speed 9.6 ms^{-1} and takes a time of 0.37 s to reach the ceiling. The ball is then in contact with the ceiling for a further time of 0.085 s until leaving it with a speed of 3.8 ms^{-1} . The mass of the ball is 0.056 kg . Assume that air resistance is negligible.

- (a) Show that the ball reaches the ceiling with a speed of 6.0 ms^{-1} .

[1]

- (b) Calculate the height of the ceiling above the point from which the ball was thrown.

height = m [2]

- (c) Calculate

- (i) the increase in gravitational potential energy of the ball for its movement from its initial position to the ceiling,

increase in gravitational potential energy = J [2]

- (ii) the decrease in kinetic energy of the ball while it is in contact with the ceiling.

decrease in kinetic energy = J [2]

- (d) State how Newton's third law applies to the collision between the ball and the ceiling.

.....
.....
.....
..... [2]

- (e) Calculate the change in momentum of the ball during the collision.

change in momentum = N s [2]

- (f) Determine the magnitude of the average force exerted by the ceiling on the ball during the collision.

average force = N [2]

24 (a) Define

(i) *displacement*,

.....
 [1]

(ii) *acceleration*.

.....
 [1]

(b) A remote-controlled toy car moves up a ramp and travels across a gap to land on another ramp, as illustrated in Fig. 1.1.

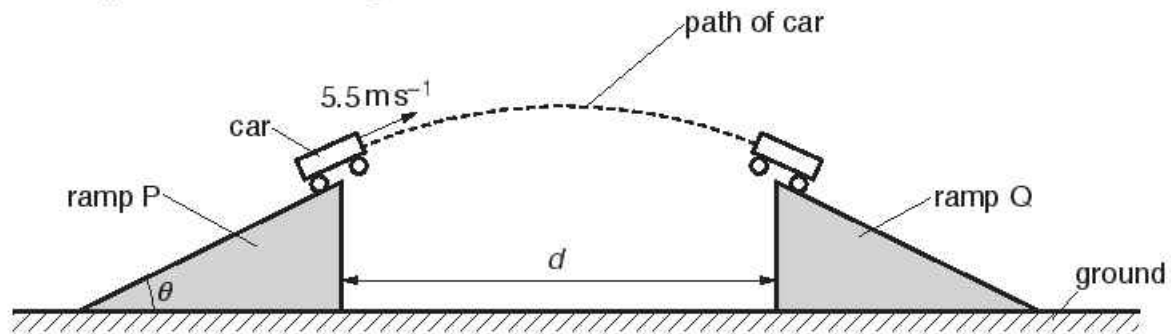


Fig. 1.1

The car leaves ramp P with a velocity of 5.5 ms^{-1} at an angle θ to the horizontal. The horizontal component of the car's velocity as it leaves the ramp is 4.6 ms^{-1} . The car lands at the top of ramp Q. The tops of both ramps are at the same height and are distance d apart. Air resistance is negligible.

(i) Show that the car leaves ramp P with a vertical component of velocity of 3.0 ms^{-1} .

[1]

(ii) Determine the time taken for the car to travel between the ramps.

time taken = s [2]

- (iii) Calculate the horizontal distance d between the tops of the ramps.

$$d = \dots\dots\dots \text{ m [1]}$$

- (iv) Calculate the ratio

$$\frac{\text{kinetic energy of the car at its maximum height}}{\text{kinetic energy of the car as it leaves ramp P}}.$$

$$\text{ratio} = \dots\dots\dots [3]$$

- (c) Ramp Q is removed. The car again leaves ramp P as in (b) and now lands directly on the ground. The car leaves ramp P at time $t = 0$ and lands on the ground at time $t = T$.

On Fig. 1.2, sketch the variation with time t of the vertical component v_y of the car's velocity from $t = 0$ to $t = T$. Numerical values of v_y and t are not required.

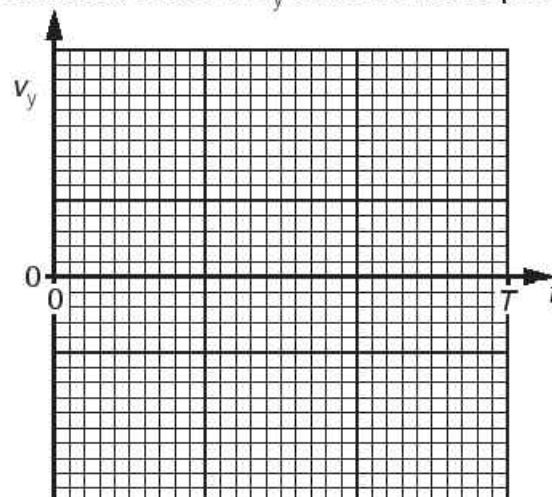


Fig. 1.2

25 (a) (i) Define *power*.

.....
 [1]

(ii) State what is meant by *gravitational potential energy*.

.....
 [1]

(b) An aircraft of mass 1200 kg climbs upwards with a constant velocity of 45 ms^{-1} , as shown in Fig. 3.1.

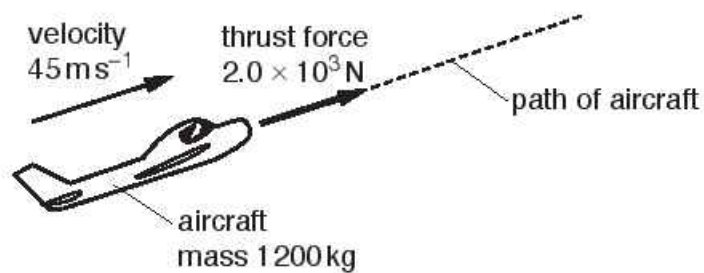


Fig. 3.1 (not to scale)

The aircraft's engine produces a thrust force of $2.0 \times 10^3\text{ N}$ to move the aircraft through the air. The rate of increase in height of the aircraft is 3.3 ms^{-1} .

(i) Calculate the power produced by the thrust force.

power = W [2]

- (ii) Determine, for a time interval of 3.0 minutes,
1. the work done by the thrust force to move the aircraft,

work done = J [2]

2. the increase in gravitational potential energy of the aircraft,

increase in gravitational potential energy = J [2]

3. the work done against air resistance.

work done = J [1]

- (iii) Use your answer in (b)(ii) part 3 to calculate the force due to air resistance acting on the aircraft.

force = N [1]

- (iv) With reference to the motion of the aircraft, state and explain whether the aircraft is in equilibrium.

.....

.....

..... [2]

- 26 A golfer strikes a ball so that it leaves horizontal ground with a velocity of 6.0 m s^{-1} at an angle θ to the horizontal, as illustrated in Fig. 1.1.

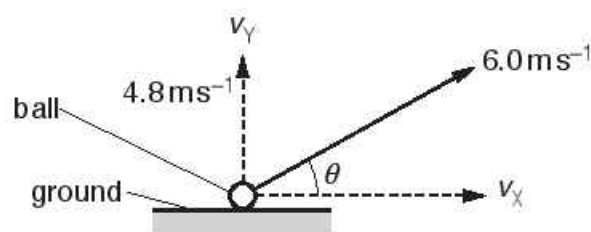


Fig. 1.1 (not to scale)

The magnitude of the initial vertical component v_y of the velocity is 4.8 m s^{-1} . Assume that air resistance is negligible.

- (a) Show that the magnitude of the initial horizontal component v_x of the velocity is 3.6 m s^{-1} .

- (b) The ball leaves the ground at time $t = 0$ and reaches its maximum height at $t = 0.49\text{ s}$.

On Fig. 1.2, sketch separate lines to show the variation with time t , until the ball returns to the ground, of

(i) the vertical component v_y of the velocity (label this line Y), [2]

(ii) the horizontal component v_x of the velocity (label this line X). [2]

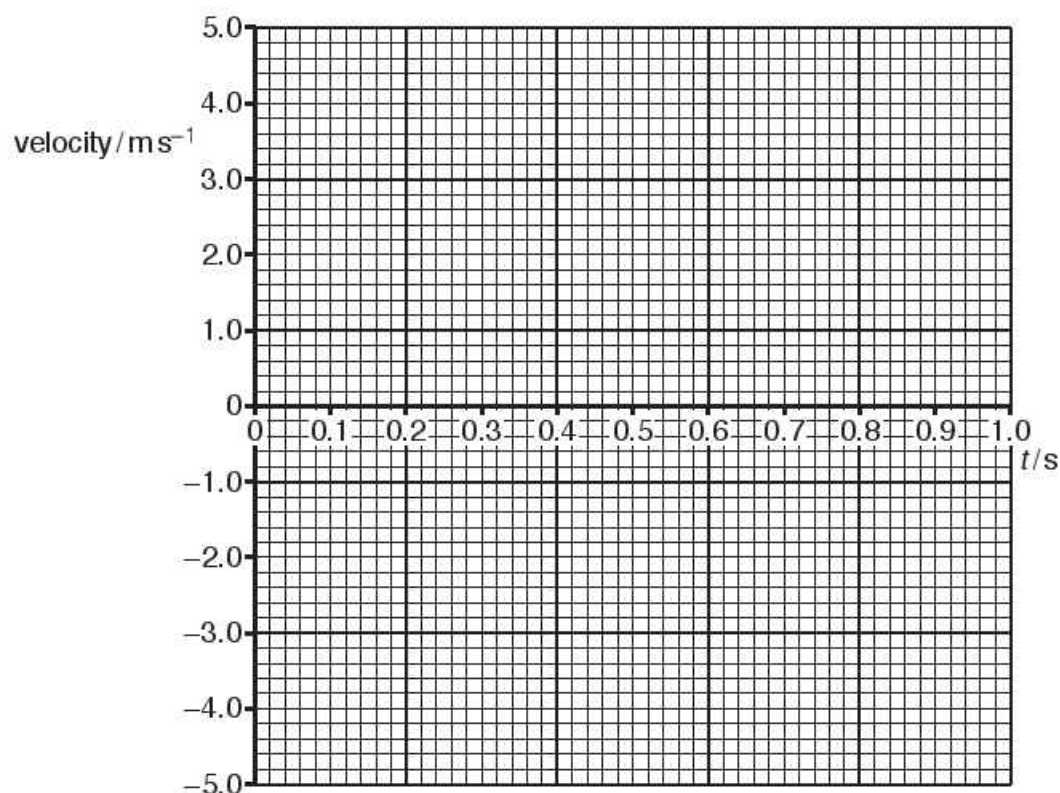


Fig. 1.2

- (c) Calculate the maximum height reached by the ball.

maximum height = m [2]

- (d) For the movement of the ball from the ground to its maximum height, determine the ratio
- $$\frac{\text{kinetic energy at maximum height}}{\text{change in gravitational potential energy}}.$$

ratio = [4]

- (e) In practice, significant air resistance acts on the ball. Explain why the actual time taken for the ball to reach maximum height is less than the time calculated when air resistance is assumed to be negligible.

.....
.....
..... [1]

- 27 (a) State what is meant by *kinetic energy*.

.....
 [1]

- (b) A cannon fires a shell vertically upwards. The shell leaves the cannon with a speed of 80 m s^{-1} and a kinetic energy of 480 J . The shell then rises to a maximum height of 210 m . The effect of air resistance is significant.

- (i) Show that the mass of the shell is 0.15 kg .

[2]

- (ii) For the movement of the shell from the cannon to its maximum height, calculate

1. the gain in gravitational potential energy,

gain in gravitational potential energy = J [2]

2. the work done against air resistance.

work done = J [1]

- (iii) Determine the average force due to the air resistance acting on the shell as it moves from the cannon to its maximum height.

force = N [2]

- (iv) The shell leaves the cannon at time $t = 0$ and reaches maximum height at time $t = T$.

On Fig. 2.1, sketch the variation with time t of the velocity v of the shell from time $t = 0$ to time $t = T$. Numerical values of v and t are not required.

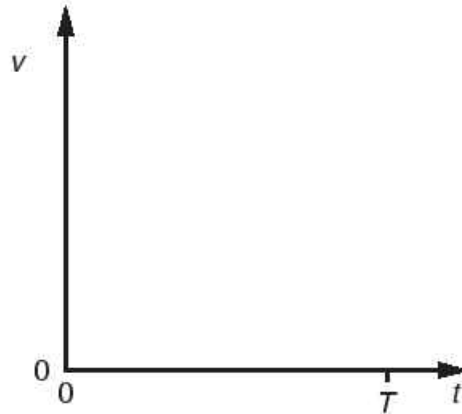


Fig. 2.1

[2]

- (v) The force due to the air resistance is a vector quantity.

Compare the force due to the air resistance acting on the shell as it rises with the force due to the air resistance as it falls.

.....

.....

.....

.....

..... [2]

- 28 The variation with extension x of the force F acting on a spring is shown in Fig. 3.1.

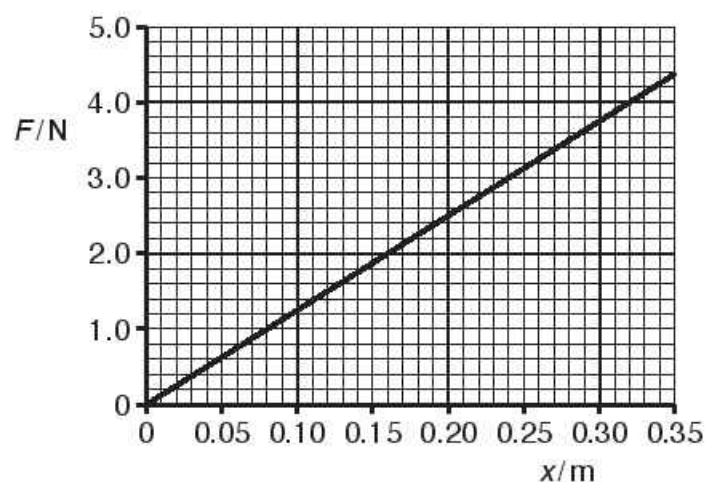


Fig. 3.1

The spring of unstretched length 0.40 m has one end attached to a fixed point, as shown in Fig. 3.2.

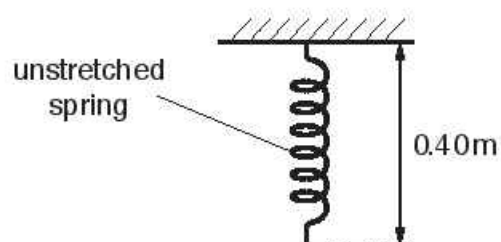


Fig. 3.2

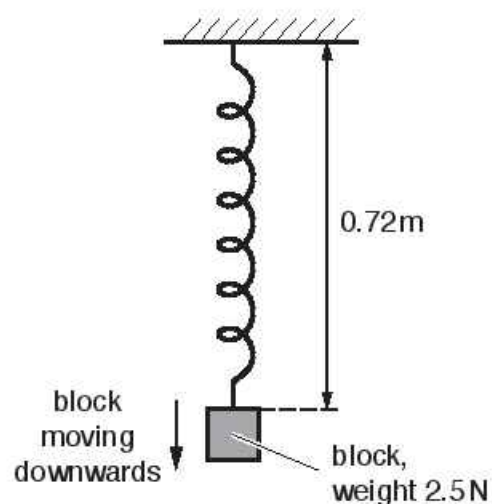


Fig. 3.3

A block of weight 2.5 N is then attached to the spring. The block is then released and begins to move downwards. At one instant, as the block is continuing to move downwards, the spring has a length of 0.72 m, as shown in Fig. 3.3.

Assume that the air resistance and the mass of the spring are both negligible.

(a) For the change in length of the spring from 0.40 m to 0.72 m:

(i) use Fig. 3.1 to show that the increase in elastic potential energy of the spring is 0.64 J

[2]

(ii) calculate the decrease in gravitational potential energy of the block of weight 2.5 N.

decrease in potential energy = J [2]

(b) Use the information in **(a)(i)** and your answer in **(a)(ii)** to determine, for the instant when the length of the spring is 0.72 m:

(i) the kinetic energy of the block

kinetic energy = J [1]

(ii) the speed of the block.

speed = ms^{-1} [2]

- 29 (a) State what is meant by the *centre of gravity* of a body.

.....
 [1]

- (b) A uniform square sign with sides of length 0.68 m is fixed at its corner points A and B to a wall. The sign is also supported by a wire CD, as shown in Fig. 3.1.

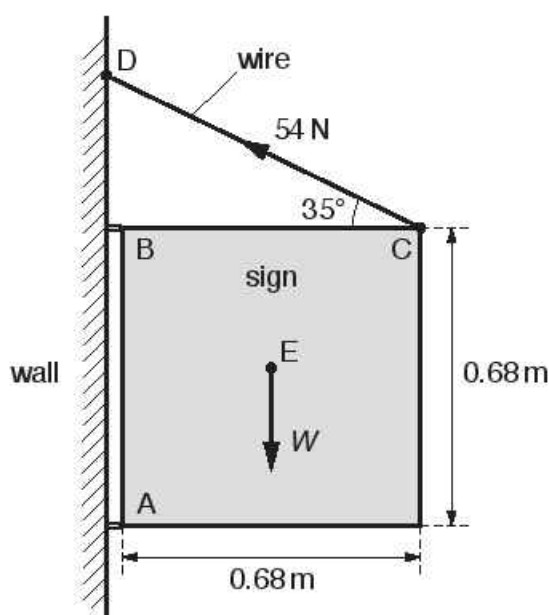


Fig. 3.1 (not to scale)

The sign has weight W and centre of gravity at point E. The sign is held in a vertical plane with side BC horizontal. The wire is at an angle of 35° to side BC. The tension in the wire is 54 N .

The force exerted on the sign at B is only in the vertical direction.

- (i) Calculate the vertical component of the tension in the wire.

vertical component of tension = N [1]

- (ii) Explain why the force on the sign at B does not have a moment about point A.

.....
 [1]

(iii) By taking moments about point A, show that the weight W of the sign is 150 N.

[2]

(iv) Calculate the total vertical force exerted by the wall on the sign at points A and B.

total vertical force = N [1]

(c) The sign in (b) is held together by nuts and bolts. One of the nuts falls vertically from rest through a distance of 4.8 m to the pavement below. The nut lands on the pavement with a speed of 9.2 m s^{-1} .

Determine, for the nut falling from the sign to the pavement, the ratio

$$\frac{\text{change in gravitational potential energy}}{\text{final kinetic energy}}.$$

ratio = [4]

- 30 (a) A resultant force F moves an object of mass m through distance s in a straight line. The force gives the object an acceleration a so that its speed changes from initial speed u to final speed v .

(i) State an expression for:

1. the work W done by the force, in terms of a , m and s

$$W = \dots\dots\dots [1]$$

2. the distance s , in terms of a , u and v .

$$s = \dots\dots\dots [1]$$

(ii) Use your answers in (i) to show that the kinetic energy of the object is given by

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times (\text{speed})^2.$$

Explain your working.

[2]

- (b) A ball of mass 0.040 kg is projected into the air from horizontal ground, as illustrated in Fig. 2.1.

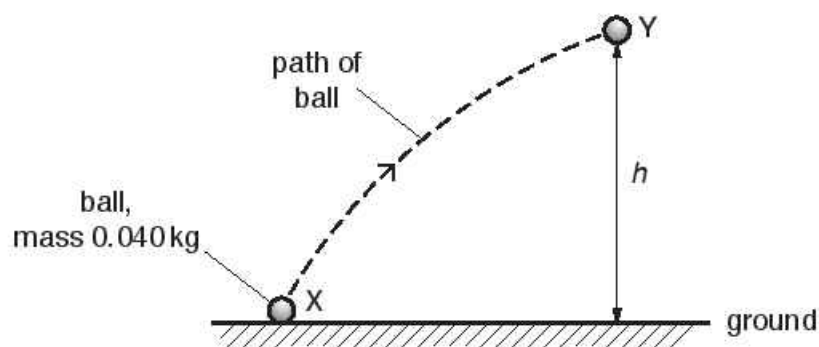


Fig. 2.1

The ball is launched from a point X with a kinetic energy of 4.5 J . At point Y, the ball has a speed of 9.5 m s^{-1} . Air resistance is negligible.

(i) For the movement of the ball from X to Y, draw a solid line on Fig. 2.1 to show:

1. the distance moved (label this line D)
2. the displacement (label this line S).

[2]

(ii) By consideration of energy transfer, determine the height h of point Y above the ground.

$h = \dots\dots\dots$ m [3]

(iii) On Fig. 2.2, sketch the variation of the kinetic energy of the ball with its vertical height above the ground for the movement of the ball from X to Y. Numerical values are not required.

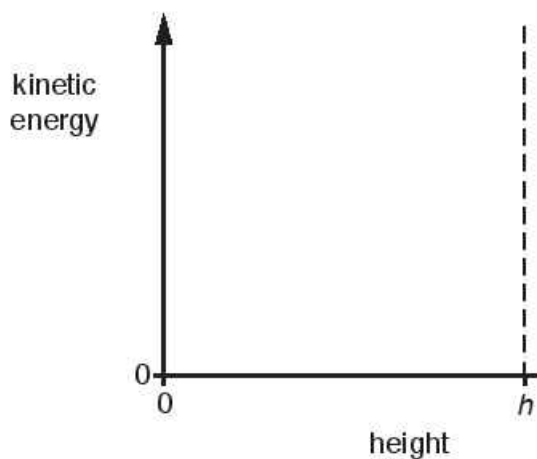


Fig. 2.2

[2]

- 31 The variation with extension x of the force F applied to a spring is shown in Fig. 4.1.

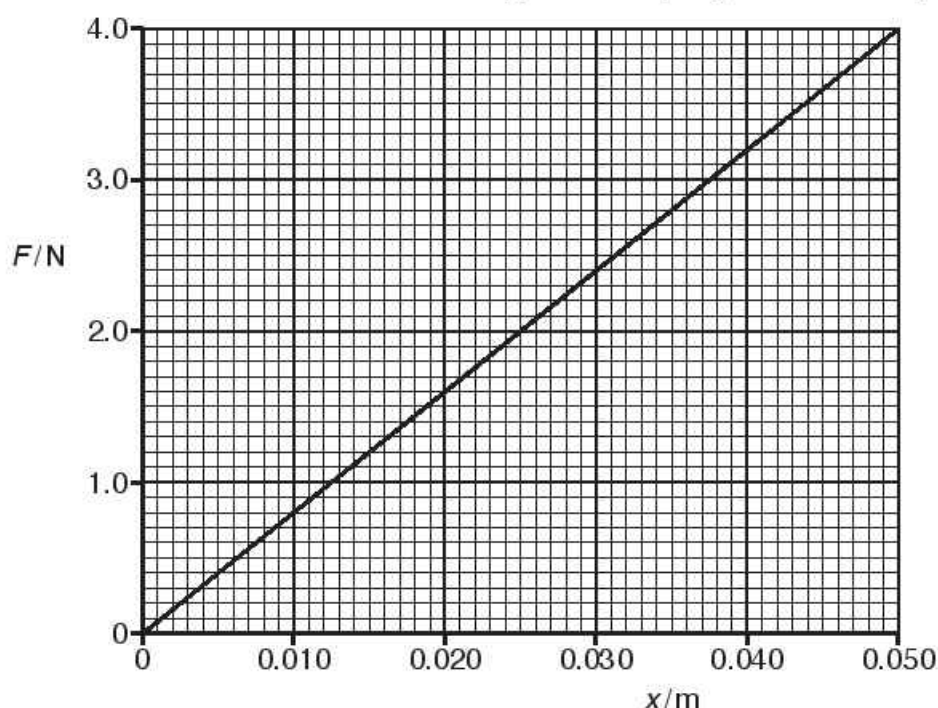


Fig. 4.1

The spring has an unstretched length of 0.080 m and is suspended vertically from a fixed point, as shown in Fig. 4.2.

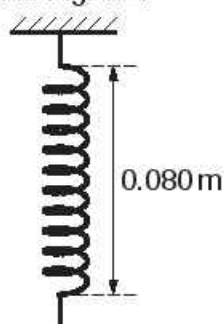


Fig. 4.2

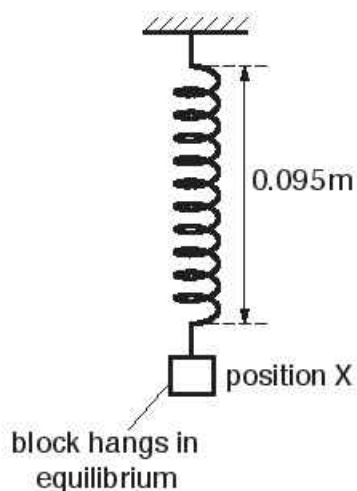


Fig. 4.3

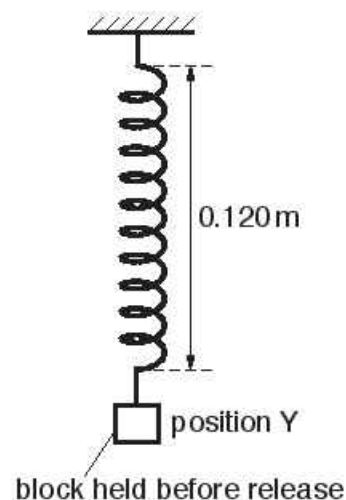


Fig. 4.4

A block is attached to the lower end of the spring. The block hangs in equilibrium at position X when the length of the spring is 0.095 m, as shown in Fig. 4.3.

The block is then pulled vertically downwards and held at position Y so that the length of the spring is 0.120 m, as shown in Fig. 4.4. The block is then released and moves vertically upwards from position Y back towards position X.

- (a) Use Fig. 4.1 to determine the spring constant of the spring.

spring constant = N m^{-1} [2]

- (b) Use Fig. 4.1 to show that the decrease in elastic potential energy of the spring is 0.055 J when the block moves from position Y to position X.

[2]

- (c) The block has a mass of 0.122 kg. Calculate the increase in gravitational potential energy of the block for its movement from position Y to position X.

increase in gravitational potential energy = J [2]

- (d) Use the decrease in elastic potential energy stated in (b) and your answer in (c) to determine, for the block, as it moves through position X:

- (i) its kinetic energy

kinetic energy = J [1]

- (ii) its speed.

speed = ms^{-1} [2]

- 32 (a) State Newton's third law of motion.

.....

.....

..... [2]

- (b) A block X of mass m_X slides in a straight line along a horizontal frictionless surface, as shown in Fig. 3.1.

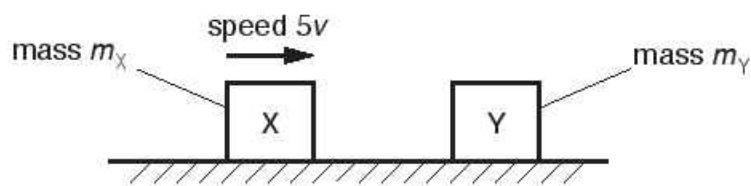


Fig. 3.1

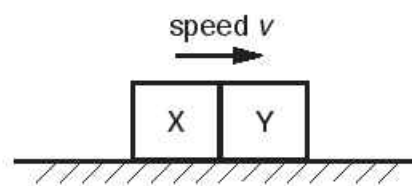


Fig. 3.2

The block X, moving with speed $5v$, collides head-on with a stationary block Y of mass m_Y . The two blocks stick together and then move with common speed v , as shown in Fig. 3.2.

- (i) Use conservation of momentum to show that the ratio $\frac{m_Y}{m_X}$ is equal to 4.

[2]

- (ii) Calculate the ratio

$$\frac{\text{total kinetic energy of X and Y after collision}}{\text{total kinetic energy of X and Y before collision}}.$$

ratio = [3]

(iii) State the value of the ratio in (ii) for a perfectly elastic collision.

ratio = [1]

(c) The variation with time t of the momentum of block X in (b) is shown in Fig. 3.3.

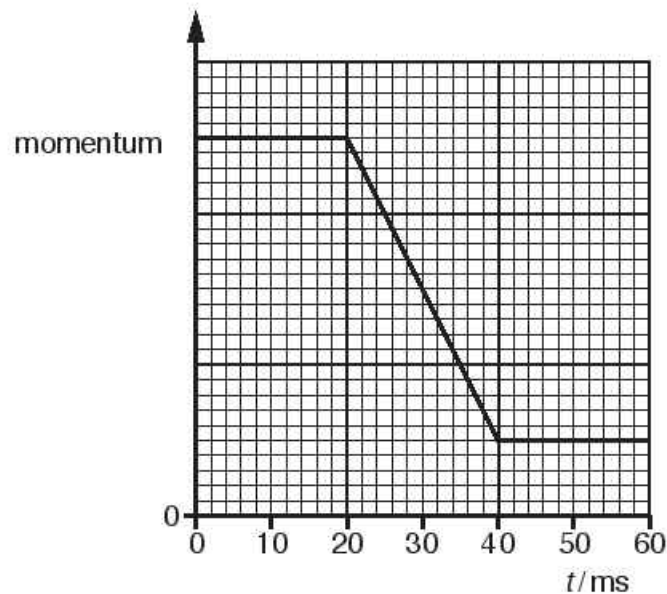


Fig. 3.3

Block X makes contact with block Y at time $t = 20$ ms.

(i) Describe, qualitatively, the magnitude and direction of the resultant force, if any, acting on block X in the time interval:

1. $t = 0$ to $t = 20$ ms

.....

2. $t = 20$ ms to $t = 40$ ms.

.....

.....

[3]

(ii) On Fig. 3.3, sketch the variation of the momentum of block Y with time t from $t = 0$ to $t = 60$ ms. [3]

33 (a) State **two** conditions for an object to be in equilibrium.

1.

.....

2.

.....

[2]

(b) A sphere of weight 2.4 N is suspended by a wire from a fixed point P. A horizontal string is used to hold the sphere in equilibrium with the wire at an angle of 53° to the horizontal, as shown in Fig. 3.1.

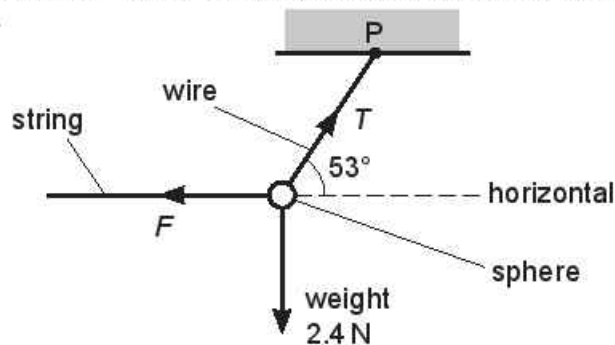


Fig. 3.1 (not to scale)

(i) Calculate:

1. the tension T in the wire

$$T = \dots\dots\dots \text{N}$$

2. the force F exerted by the string on the sphere.

$$F = \dots\dots\dots \text{N}$$

[2]

(ii) The wire has a circular cross-section of diameter 0.50 mm. Determine the stress σ in the wire.

$$\sigma = \dots\dots\dots \text{Pa} \quad [3]$$

- (c) The string is disconnected from the sphere in (b). The sphere then swings from its initial rest position A, as illustrated in Fig. 3.2.

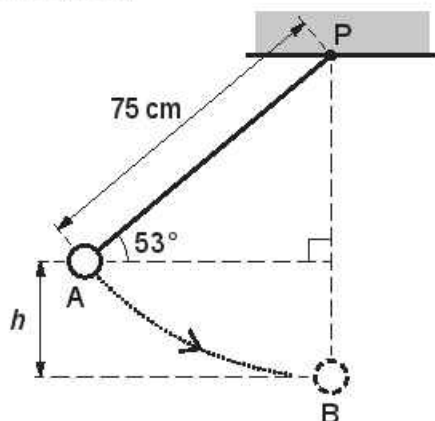


Fig. 3.2 (not to scale)

The sphere reaches maximum speed when it is at the bottom of the swing at position B. The distance between P and the centre of the sphere is 75 cm.

Air resistance is negligible and energy losses at P are negligible.

- (i) Show that the vertical distance h between A and B is 15 cm.

[1]

- (ii) Calculate the change in gravitational potential energy of the sphere as it moves from A to B.

change in gravitational potential energy = J [2]

- (iii) Use your answer in (c)(ii) to determine the speed of the sphere at B.
Show your working.

speed = m s^{-1} [3]

34 (a) Explain what is meant by *work done*.

.....
 [1]

(b) A ball of mass 0.42 kg is dropped from the top of a building. The ball falls from rest through a vertical distance of 78 m to the ground. Air resistance is significant so that the ball reaches constant (terminal) velocity before hitting the ground. The ball hits the ground with a speed of 23 m s^{-1} .

(i) Calculate, for the ball falling from the top of the building to the ground:

1. the decrease in gravitational potential energy

decrease in gravitational potential energy = J [2]

2. the increase in kinetic energy.

increase in kinetic energy = J [2]

(ii) Use your answers in (b)(i) to determine the average resistive force acting on the ball as it falls from the top of the building to the ground.

average resistive force = N [2]

- (c) The ball in (b) is dropped at time $t = 0$ and hits the ground at time $t = T$. The acceleration of free fall is g .

On Fig. 3.1, sketch a line to show the variation of the acceleration a of the ball with time t from time $t = 0$ to $t = T$.

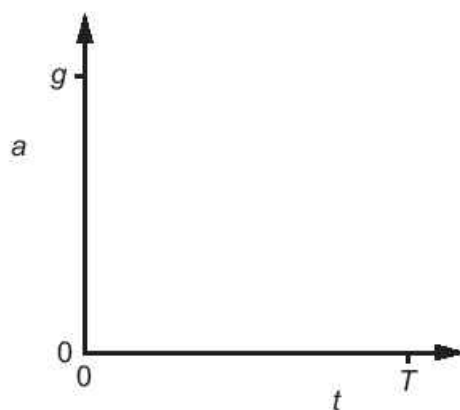


Fig. 3.1

[2]

35. 9702/22/F/M/21/Q2

- (a)** State what is meant by *work done*.

.....
..... [1]

- (b)** A beach ball is released from a balcony at the top of a tall building. The ball falls vertically from rest and reaches a constant (terminal) velocity. The gravitational potential energy of the ball decreases by 60 J as it falls from the balcony to the ground. The ball hits the ground with speed 16 m s^{-1} and kinetic energy 23 J.

- (i)** Show that the mass of the ball is 0.18 kg.

[2]

- (ii)** Calculate the height of the balcony above the ground.

height = m [2]

- (iii)** Determine the average resistive force acting on the ball as it falls from the balcony to the ground.

average resistive force = N [2]

- (c) State and explain the variation, if any, in the magnitude of the acceleration of the ball in (b) during the time interval when the ball is moving downwards **before** it reaches constant (terminal) velocity.

.....

.....

.....

.....

.....

..... [3]

[Total: 10]

36. 9702/21/M/J/21/Q2

A person uses a trolley to move suitcases at an airport. The total mass of the trolley and suitcases is 72 kg.

- (a) The person pushes the trolley and suitcases along a horizontal surface with a constant speed of 1.4 m s^{-1} and then releases the trolley. The released trolley moves in a straight line and comes to rest. Assume that a constant total resistive force of 18 N opposes the motion of the trolley and suitcases.

- (i) Calculate the power required to overcome the total resistive force on the trolley and suitcases when they move with a constant speed of 1.4 m s^{-1} .

power = W [2]

- (ii) Calculate the time taken for the trolley to come to rest after it is released.

time = s [3]

- (b) At another place in the airport, the trolley and suitcases are on a slope, as shown in Fig. 2.1.

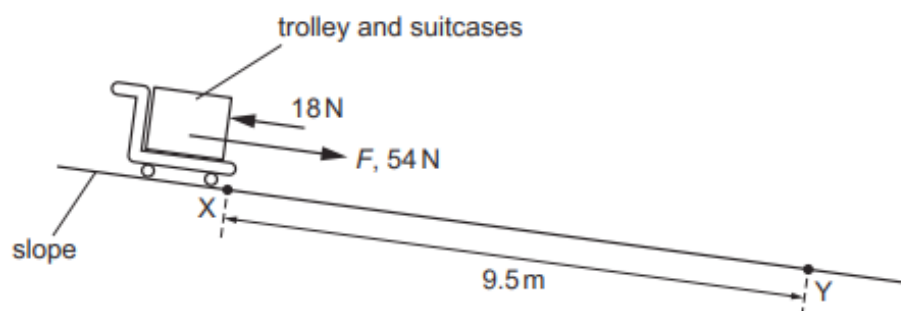


Fig. 2.1 (not to scale)

The person releases the trolley from rest at point X. The trolley moves down the slope in a straight line towards point Y. The distance along the slope between points X and Y is 9.5 m.

The component F of the weight of the trolley and suitcases that acts along the slope is 54 N. Assume that a constant total resistive force of 18 N opposes the motion of the trolley and suitcases.

- (i) Calculate the speed of the trolley at point Y.

speed = ms^{-1} [3]

- (ii) Calculate the work done by F for the movement of the trolley from X to Y.

work done = J [1]

- (iii) The trolley is released at point X at time $t = 0$.

On Fig. 2.2, sketch a graph to show the variation with time t of the work done by F for the movement of the trolley from X to Y.
Numerical values of the work done and t are not required.

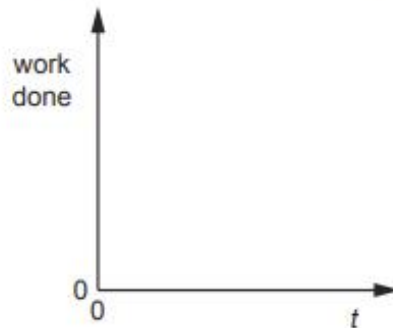


Fig. 2.2

[2]

- (c) The angle of the slope in (b) is constant. The frictional forces acting on the wheels of the moving trolley are also constant.

Explain why, in practice, it is incorrect to assume that the total resistive force opposing the motion of the trolley and suitcases is constant as the trolley moves between X and Y.

.....
..... [1]

[Total: 12]

37. 9702/22/M/J/21/Q3

A child of weight 330 N is at point X at the top of a slide. The slide is at the edge of a swimming pool, as shown in Fig. 3.1.

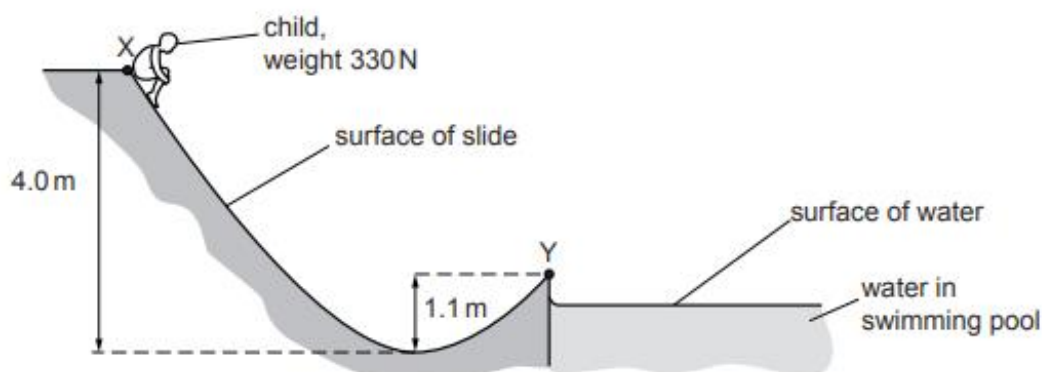


Fig. 3.1 (not to scale)

The child moves from rest to the lowest point of the slide that is a vertical distance of 4.0 m below X. The child continues moving towards point Y which is at the end of the slide and a vertical distance of 1.1 m above the lowest point. The kinetic energy of the child at Y is 540 J.

- (a)** Calculate the difference in the gravitational potential energy of the child at points X and Y.

difference in gravitational potential energy = J [2]

- (b)** An average frictional force of 52 N acts on the child when moving from X to Y.

By considering changes of energy, determine the distance moved by the child from X to Y.

distance moved = m [2]

- (c) The child leaves the slide at point Y with a velocity that is at an angle of 41° to the horizontal. The path of the child through the air is shown in Fig. 3.2.

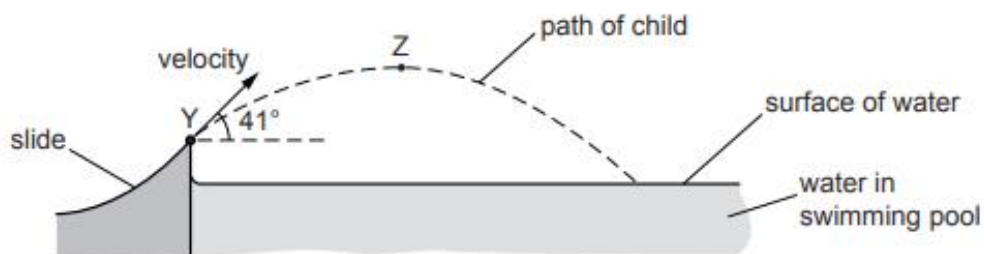


Fig. 3.2 (not to scale)

Point Z is the highest point on the path of the child through the air. Assume that air resistance is negligible.

Calculate the speed of the child at:

- (i) point Y

speed = ms^{-1} [2]

- (ii) point Z.

speed = ms^{-1} [2]

[Total: 8]

38. 9702/22/O/N/21/Q3

(a) Define *power*.

.....
..... [1]

(b) A car of mass 1700 kg moves in a straight line along a slope that is at an angle θ to the horizontal, as shown in Fig. 3.1.

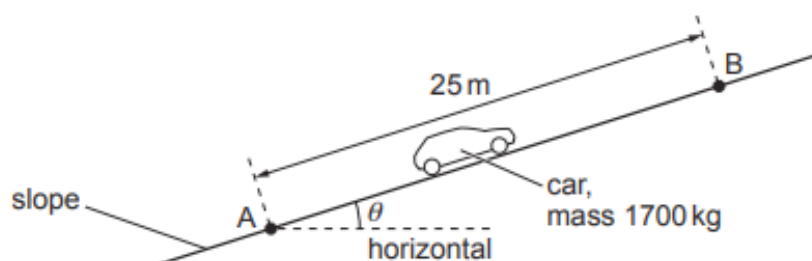


Fig. 3.1 (not to scale)

The car moves at constant velocity for a distance of 25 m from point A to point B.
Air resistance and friction provide a total resistive force of 440 N that opposes the motion of the car.

For the movement of the car from A to B:

(i) state the change in the kinetic energy

change in kinetic energy = J [1]

(ii) calculate the work done against the total resistive force.

work done = J [1]

- (c) The movement of the car in (b) from A to B causes its gravitational potential energy to increase by $4.8 \times 10^4 \text{ J}$.

Calculate:

- (i) the increase in vertical height h of the car for its movement from A to B

$$h = \dots\dots\dots \text{ m [2]}$$

- (ii) angle θ .

$$\theta = \dots\dots\dots^\circ \text{ [1]}$$

- (d) The engine of the car in (b) produces an output power of $1.7 \times 10^4 \text{ W}$ to move the car along the slope.

Calculate the time taken for the car to move from A to B.

$$\text{time} = \dots\dots\dots \text{ s [2]}$$

[Total: 8]

39. 9702/22/F/M/22/Q4

A child moves down a long slide, as shown in Fig. 4.1.

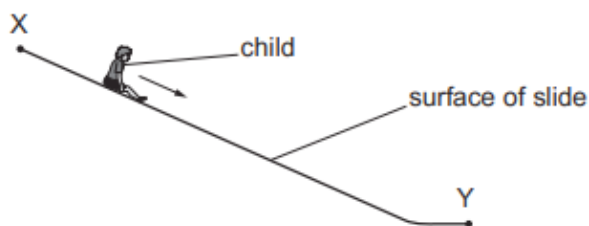


Fig. 4.1 (not to scale)

The child moves from rest at the top end X of the slide. An average resistive force of 76 N opposes the motion of the child as they move to the lower end Y of the slide. The kinetic energy of the child at Y is 300 J. The decrease in gravitational potential energy of the child as it moves from X to Y is 3200 J.

(a) Determine the ratio

$$\frac{\text{kinetic energy of the child at Y when the resistive force is 76 N}}{\text{kinetic energy of the child at Y if there is no resistive force}}.$$

ratio = [1]

(b) Use the answer in **(a)** to calculate the ratio

$$\frac{\text{speed of the child at Y when the resistive force is 76 N}}{\text{speed of the child at Y if there is no resistive force}}.$$

ratio = [2]

(c) Calculate the length of the slide from X to Y.

length = m [2]

- (d) At end Y of the slide, the child is brought to rest by a board, as shown in Fig. 4.2.

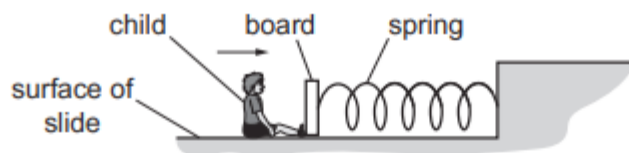


Fig. 4.2 (not to scale)

A spring connects the board to a fixed point. The spring obeys Hooke's law and has a spring constant of 63 N m^{-1} . The child hits the board so that it moves to the right and compresses the spring. The speed of the child becomes zero when the elastic potential energy of the spring has increased to its maximum value of 140 J .

- (i) Calculate the maximum compression of the spring.

maximum compression = m [2]

- (ii) Calculate the percentage efficiency of the transfer of the kinetic energy of the child to the elastic potential energy of the spring.

percentage efficiency = % [1]

- (iii) The maximum compression of the spring is x_0 . On Fig. 4.3, sketch a graph to show the variation of the elastic potential energy of the spring with its compression x from $x = 0$ to $x = x_0$. Numerical values are not required.

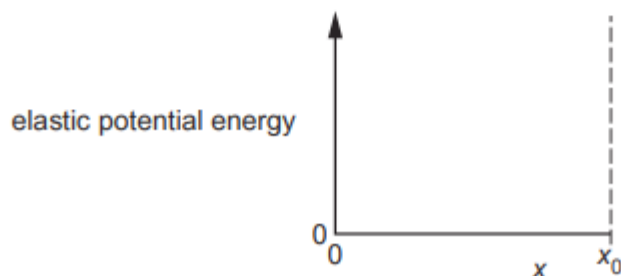


Fig. 4.3

[2]

40. 9702/21/M/J/22/Q3

(a) (i) Define power.

.....
 [1]

(ii) A force F takes time t to move an object through a displacement x at constant velocity v in the direction of the force. The work done by the force is W .

Use the definition of power to show that the power P transferred by the force is given by

$$P = Fv.$$

[2]

(b) A block is pulled up a slope by a wire attached to a motor, as shown in Fig. 3.1.

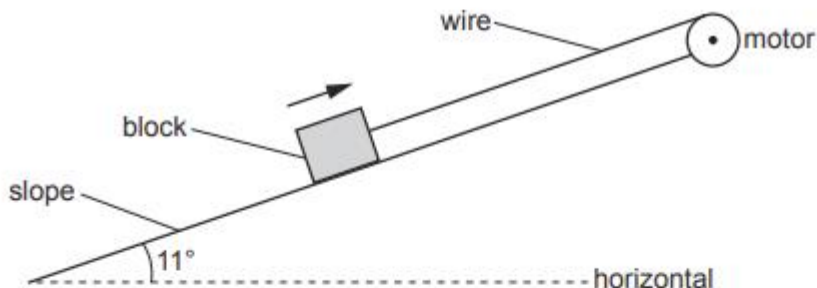


Fig. 3.1 (not to scale)

The useful power output of the motor is 56 W. The block has a weight of 430 N and travels with constant velocity along the slope at an angle of 11° to the horizontal.

Assume that there are no resistive forces opposing the motion of the block.

(i) Calculate the tension T in the wire.

$T = \dots\dots\dots$ N [2]

- (ii) Calculate the speed of the block.

speed = ms^{-1} [1]

- (iii) The rate of increase of gravitational potential energy of the block is equal to the useful power output of the motor. One of the reasons for this is that there is no work done against resistive forces.

By considering the motion of the block, state another reason for this.

.....
..... [1]

- (iv) The motor has an efficiency of 80%.

Calculate the time taken for an input energy of 1.2 kJ to be supplied to the motor.

time taken = s [2]

[Total: 9]

41. 9702/23/M/J/22/Q4

- (a) A mass m moves a vertical distance Δh in a uniform gravitational field and gains gravitational potential energy ΔE_p . The acceleration of free fall is g .

Use the concept of work done to show that

$$\Delta E_p = mg\Delta h.$$

[2]

- (b) A 0.60 kg mass is attached to a string which is wrapped around the wheel of a generator, as shown in Fig. 4.1.

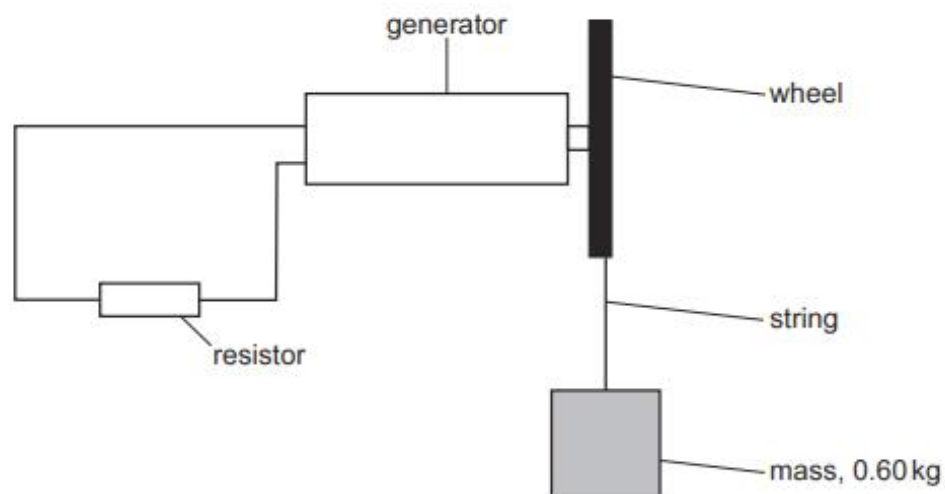


Fig. 4.1

The mass is held stationary above the floor. When released, the mass initially accelerates and then falls at a steady speed and spins the wheel. The generator causes a current in a resistor. Air resistance is negligible.

State the main energy change when the mass is falling at a steady speed.

..... energy to energy.

[1]

- (c) When falling at a steady speed, the mass in (b) falls through a vertical distance of 1.4 m in a time of 4.0 s. This causes a current of 90 mA in the resistor. The resistance of the resistor is $47\ \Omega$.

Calculate:

- (i) the rate of work done by the falling mass

rate of work done = W [2]

- (ii) the power dissipated in the resistor

power = W [2]

- (iii) the efficiency of the generator.

efficiency = [2]

[Total: 9]

42. 9702/21/O/N/22/Q3

(a) (i) Define power.

.....
..... [1]

(ii) Mechanical power P can be calculated using the formula $P = Fv$.

Use the concept of work and the definition of power to show how this formula is derived.

[2]

(b) The engine of a lorry provides 130 kW of power to the lorry's wheels when it is travelling at a constant speed of 25 m s^{-1} along a straight horizontal road.

Show that the resistive force opposing the forward motion of the lorry is 5200 N.

[11]

- (c) The lorry in (b) travels up a straight section of road that is inclined at an angle θ to the horizontal, as shown in Fig. 3.1.

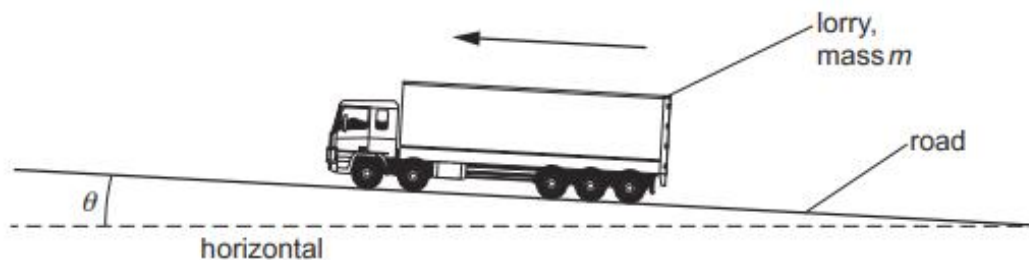


Fig. 3.1 (not to scale)

The lorry has mass m and the acceleration of free fall is g .

- (i) Determine an expression, in terms of m , g and θ , for the component of the weight of the lorry that acts parallel to the surface of the road.

[1]

- (ii) The total resistive force remains unchanged at 5200 N and the engine now provides greater power to maintain the speed of 25 m s^{-1} . The total mass m of the lorry is 36 000 kg. The angle θ is 1.4° .

Determine the power, in kW, now provided by the engine.

power = kW [3]

[Total: 8]

43. 9702/22/F/M/23/Q2

A motor uses a wire to raise a block, as illustrated in Fig. 2.1.

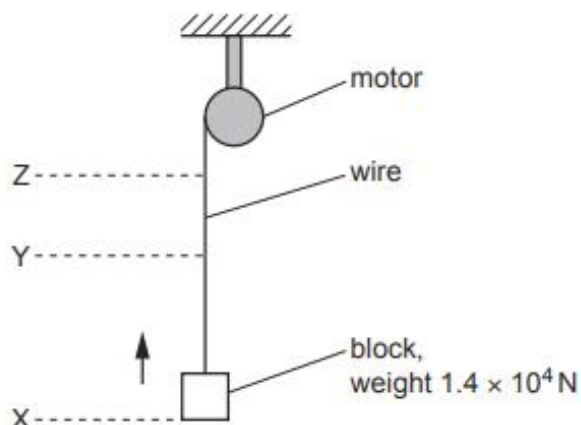


Fig. 2.1 (not to scale)

The base of the block takes a time of 0.49 s to move vertically upwards from level X to level Y at a constant speed of 0.64 m s^{-1} . During this time the wire has a strain of 0.0012 . The wire is made of metal of Young modulus $2.2 \times 10^{11} \text{ Pa}$ and has a uniform cross-section.

The block has a weight of $1.4 \times 10^4 \text{ N}$. Assume that the weight of the wire is negligible.

(a) Calculate:

(i) the cross-sectional area A of the wire

$$A = \dots\dots\dots \text{ m}^2 \quad [2]$$

(ii) the increase in the gravitational potential energy of the block for the movement of its base from X to Y.

$$\text{increase in gravitational potential energy} = \dots\dots\dots \text{ J} \quad [3]$$

- (b) The motor has an efficiency of 56%.

Calculate the input power to the motor as the base of the block moves from X to Y.

input power = W [3]

- (c) The base of the block now has a uniform deceleration of magnitude 1.3 m s^{-2} from level Y until the base of the block stops at level Z.

Calculate the tension T in the wire as the base of the block moves from Y to Z.

$T = \dots\dots\dots$ N [3]

- (d) The base of the block is at levels X, Y and Z at times t_X , t_Y and t_Z respectively.

On Fig. 2.2, sketch a graph to show the variation with time t of the distance d of the base of the block from level X. Numerical values of d and t are not required.

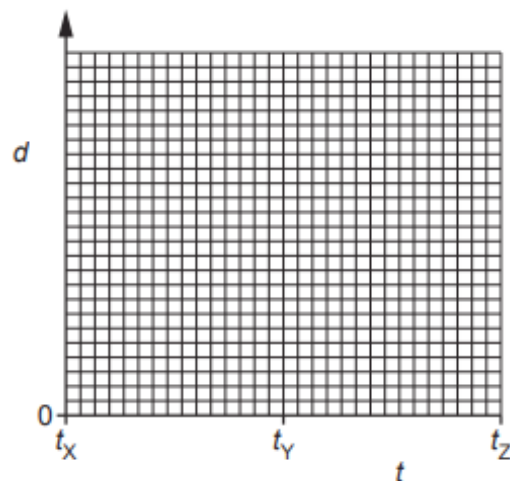


Fig. 2.2

[2]

[Total: 13]

44. 9702/22/M/J/23/Q2

- (a) State what is meant by the centre of gravity of an object.

.....
 [1]

- (b) Two blocks are on a horizontal beam that is pivoted at its centre of gravity, as shown in Fig. 2.1.

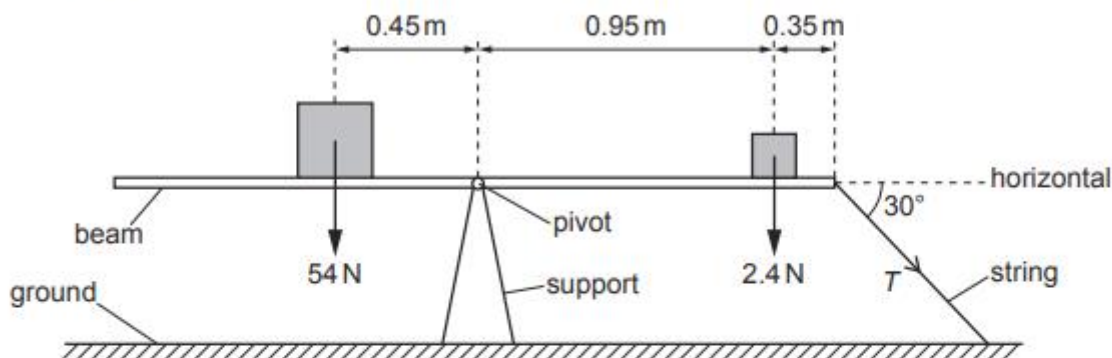


Fig. 2.1 (not to scale)

A large block of weight 54 N is a distance of 0.45 m from the pivot. A small block of weight 2.4 N is a distance of 0.95 m from the pivot and a distance of 0.35 m from the right-hand end of the beam.

The right-hand end of the beam is connected to the ground by a string that is at an angle of 30° to the horizontal. The beam is in equilibrium.

- (i) By taking moments about the pivot, calculate the tension T in the string.

$T = \dots\dots\dots$ N [3]

- (ii) The string is cut so that the beam is no longer in equilibrium.

Calculate the magnitude of the resultant moment about the pivot acting on the beam immediately after the string is cut.

resultant moment = $\dots\dots\dots$ Nm [1]

- (c) The beam in (b) rotates when the string is cut and the small block of weight 2.4 N is projected through the air. Fig. 2.2 shows the last part of the path of the block before it hits the ground at point Y.

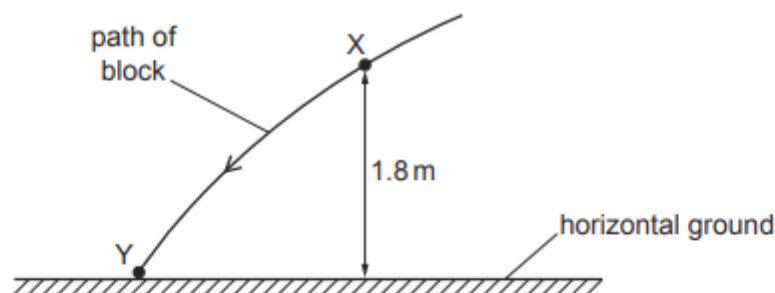


Fig. 2.2 (not to scale)

At point X on the path, the block has a speed of 3.4 m s^{-1} and is at a height of 1.8 m above the horizontal ground. Air resistance is negligible.

- (i) Calculate the decrease in the gravitational potential energy of the block for its movement from X to Y.

decrease in gravitational potential energy = J [2]

- (ii) Use your answer to (c)(i) and conservation of energy to determine the kinetic energy of the block at Y.

kinetic energy = J [3]

- (iii) State the variation, if any, in the direction of the acceleration of the block as it moves from X to Y.

..... [1]

- (iv) The block passes point X at time t_X and arrives at point Y at time t_Y .

On Fig. 2.3, sketch a graph to show the variation of the magnitude of the horizontal component of the velocity of the block with time from t_X to t_Y . Numerical values are not required.

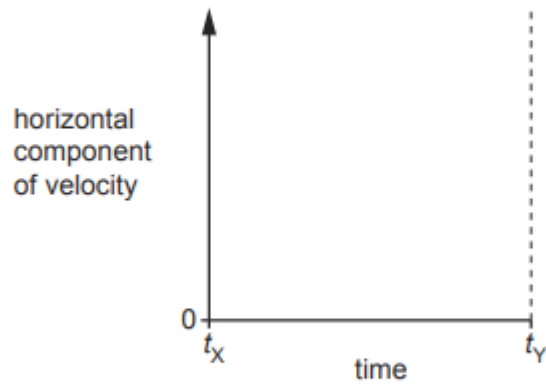


Fig. 2.3

[1]

[Total: 12]

ANSWERS

- 1 (a) *either* energy (stored)/work done represented by area under graph
or energy = average force $\times$ extension B1
 energy = $\frac{1}{2} \times 180 \times 4.0 \times 10^{-2}$ C1
 = 3.6 J A1 [3]
- (b) (i) *either* momentum before release is zero M1
 so sum of momenta (of trolleys) after release is zero A1
or force = rate of change of momentum (M1)
 force on trolleys equal and opposite (A1)
or impulse = change in momentum (M1)
 impulse on each equal and opposite (A1) [2]
- (ii) 1 $M_1 V_1 = M_2 V_2$ B1 [1]
 2 $E = \frac{1}{2} M_1 V_1^2 + \frac{1}{2} M_2 V_2^2$ B1 [1]
- (iii) 1 $E_k = \frac{1}{2} m v^2$ and $p = m v$ combined to give M1
 $E_k = p^2 / 2m$ A0 [1]
 2 m smaller, E_k is larger because p is the same/constant M1
 so trolley B A0 [1]
-
- 2 (a) work done is the force $\times$ the distance moved / displacement in the direction of the force
or
 work is done when a force moves in the direction of the force B1 [1]
- (b) component of weight = $850 \times 9.81 \times \sin 7.5^\circ$ C1
 = 1090 N A1 [2]
(use of incorrect trigonometric function, 0/2)
- (c) (i) $\Sigma F = 4600 - 1090 = (3510)$ M1
 deceleration = $3510 / 850$ A1
 = 4.1 ms^{-2} A0 [2]
- (ii) $v^2 = u^2 + 2as$
 $0 = 25^2 + 2 \times -4.1 \times s$ C1
 $s = 625 / 8.2$
 = 76 m A1 [2]
(allow full credit for calculation of time (6.05 s) & then s)
- (iii) 1. kinetic energy = $\frac{1}{2} m v^2$ C1
 = $0.5 \times 850 \times 25^2$
 = $2.7 \times 10^5 \text{ J}$ A1 [2]
2. work done = 4600×75.7
 = $3.5 \times 10^5 \text{ J}$ A1 [1]
- (iv) difference is the loss in potential energy (owtte) B1 [1]
-

3 (a) (i)	work done equals force $\times$ distance moved / displacement in the direction of the force	B1	[1]
(ii)	power is the rate of doing work / work done per unit time	B1	[1]
(b) (i)	kinetic energy $= \frac{1}{2} mv^2$	C1	
	$= 0.5 \times 600 (9.5)^2$	C1	
	$= 27075 \text{ (J)} = 27 \text{ kJ}$	A1	[3]
(ii)	potential energy $= mgh$		
	$= 600 \times 9.81 \times 4.1$	M1	
	$= 24132 \text{ (J)}$	A1	
	$= 24 \text{ kJ}$	A0	[2]
(iii)	work done $= 27 - 24 = 3.0 \text{ kJ}$	A1	[1]
(iv)	resistive force $= 3000 / 8.2$ (distance along slope $= 4.1 / \sin 30^\circ$)	C1	
	$= 366 \text{ N}$	A1	[2]

4 (a)	electrical potential energy (stored) when charge moved and gravitational potential energy (stored) when mass moved due to work done in electric field and work done in gravitational field	B1	
		B1	[2]
(b)	work done $=$ force $\times$ distance moved (in direction of force) and force $= mg$ $mg \times h$ or $mg \times \Delta h$	M1	
		A1	[2]
(c) (i)	$0.1 \times mgh = \frac{1}{2} mv^2$	B1	
	$0.1 \times m \times 9.81 \times 120 = 0.5 \times m \times v^2$	B1	
	$v = 15.3 \text{ ms}^{-1}$	A0	[2]
(ii)	$P = 0.5 m v^2 / t$	C1	
	$m / t = 110 \times 10^3 / [0.25 \times 0.5 \times (15.3)^2]$	C1	
	$= 3740 \text{ kg s}^{-1}$	A1	[3]

5 (a) (i)	force is rate of change of momentum	B1	[1]
(ii)	work done is the product of the force and the distance <u>moved</u> in the direction of the force	B1	[1]
(b) (i)	$W = Fs$ or $W = mas$ or $W = m(v^2 - u^2) / 2$ or $W = \text{force} \times \text{distance } s$	A1	[1]
(ii)	$as = (v^2 - u^2) / 2$ any subject	M1	
	$W = mas$ hence $W = m(v^2 - u^2) / 2$	M1	
	RHS represents terms of energy or with $u = 0$ $KE = \frac{1}{2}mv^2$	A1	[3]
(c) (i)	work done $= \frac{1}{2} \times 1500 \times [(30)^2 - (15)^2] (=506250)$	C1	
	distance $= WD / F = 506250 / 3800 = 133 \text{ m}$	A1	[2]
	or $F = ma$ $a = 2.533 \text{ (ms}^{-2}\text{)}$	C1	
	$v^2 = u^2 + 2as$ $s = 133 \text{ m}$	A1	
(ii)	the change in kinetic energy is greater or the work done by the force has to be greater, hence distance is greater (for same force)	A1	[1]
	allow: same acceleration, same time, so greater average speed and greater distance		

6 (a)	$V = h \times A$	
	$m = V \times \rho$	B1
	$W = h \times A \times \rho \times g$	B1
	$P = F / A$	B1
	$P = h\rho g$	
	P is proportional to h if ρ is constant (and g)	B1 [4]
(b)	density changes with height	B1
	hence density is not constant with link to formula	B1 [2]
7 (a)	power is the rate of doing work or power = work done / time (taken) or power = energy transferred / time (taken)	B1 [1]
(b)	(i) as the speed increases drag / air resistance increases resultant force reduces hence acceleration is less constant speed when resultant force is zero (allow one mark for speed increases and acceleration decreases)	B1 B1 B1 [3]
	(ii) force from cyclist = drag force / resistive force	B1
	$P = 12 \times 48$	M1
	$P = 576 \text{ W}$	A0 [2]
	(iii) tangent drawn at speed = 8.0 m s^{-1} gradient values that show acceleration between 0.44 to 0.48 ms^{-2}	M1 A1 [2]
	(iv) $F - R = ma$	C1
	$600 / 8 - R = 80 \times 0.5$ [using $P = 576$] $576 / 8 - R = 80 \times 0.5$	C1
	$R = 75 - 40 = 35 \text{ N}$ $R = 72 - 40 = 32 \text{ N}$	A1 [3]
	(v) at 12 ms^{-1} drag is 48 N , at 8 ms^{-1} drag is 35 or 32 N R / v calculated as 4 and 4 or 4.4 and consistent response for whether R is proportional to v or not	B1 [1]
8 (a)	loss in potential energy due to decrease in height (as $P.E. = mgh$)	(B1)
	gain in kinetic energy due to increase in speed (as $K.E. = \frac{1}{2} mv^2$)	(B1)
	<i>special case 'as PE decreases KE increases' (1/2)</i>	
	increase in thermal energy due to work done against air resistance	(B1)
	loss in P.E. equals gain in K.E. and thermal energy	(B1)
		max. 3 [3]
(b)	(i) kinetic energy = $\frac{1}{2} mv^2$	C1
	$= \frac{1}{2} \times 0.150 \times (25)^2$	C1
	$= 46.875 = 47 \text{ J}$	A1 [3]
	(ii) 1. potential energy ($= mgh$) = $0.150 \times 9.81 \times 21$	C1
	loss = $KE - mgh = 46.875 - (30.9)$	C1
	$= 15.97 = 16 \text{ J}$	A1 [3]
	2. work done = 16 J	
	work done = force $\times$ distance	C1
	$F = 16 / 21 = 0.76 \text{ N}$	A1 [2]

9 (a) power = energy / time	C1
= (force × distance / time) = kg m ² s ⁻² / s	C1
= kg m ² s ⁻³	A1 [3]
(b) (i) units of L^2 : m ² and units of ρ : kg m ⁻³ and units of v^3 : m ³ s ⁻³ ($C = P / L^2 \rho v^3$) hence units of C : kg m ² s ⁻³ m ⁻² kg ⁻¹ m ³ m ⁻³ s ³	C1
or any correct statement of component units	M1
argument / discussion / cancelling leading to C having no units	A1 [3]
(ii) power available from wind = $3.5 \times 10^5 \times 100 / 55$ (= 6.36×10^5)	C1
$v^3 = 3.5 \times 10^5 \times 100 / (55 \times 0.931 \times (25)^2 \times 1.3)$	C1
$v = 9.4 \text{ m s}^{-1}$	A1 [3]
(iii) not all kinetic energy of wind converted to kinetic energy of blades generator / conversion to electrical energy not 100% efficient / heat produced in generator / bearings etc	B1
(there must be cause of loss and where located)	B1 [2]
10 (a) (work =) force × distance <u>moved</u> / displacement in the direction of the force OR when a force moves in the direction of the force work is done	B1 [1]
(b) kinetic energy = $\frac{1}{2} mv^2$ = $\frac{1}{2} 0.4 (2.5)^2 = 1.25 / 1.3 \text{ J}$	C1
(c) (i) area under graph is work done / work done = $\frac{1}{2} Fx$ 1.25 = (14 x) / 2	C1
$x = 0.18 (0.179) \text{ m}$ [allow $x = 0.19 \text{ m}$ using kinetic energy = 1.3 J]	C1
(ii) smooth curve from $v = 2.5$ at $x = 0$ to $v = 0$ at Q curve with increasing gradient	M1
	A1 [2]
11 (a) gravitational PE is energy of a <u>mass</u> due to its position in a <u>gravitational field</u> elastic PE energy <u>stored</u> (in an object) <u>due to</u> (a force) changing its shape / deformation / being compressed / stretched / strained	B1
	B1 [2]
(b) (i) 1. kinetic energy = $\frac{1}{2} mv^2$ = $\frac{1}{2} \times 0.065 \times 16^2 = 8.3(2) \text{ J}$	C1
	A1 [2]
2. $v^2 = 2gh$ OR $PE = mgh$	C1
$h = 16^2 / (2 \times 9.81) = 13(.05) \text{ m}$	A1 [2]
(ii) speed at $t = \frac{1}{2}$ total time = $8 \text{ (m s}^{-1}\text{)}$ or total $t = 1.63$ or $t_{1/2} = 0.815 \text{ s}$	C1
KE is $\frac{1}{4}$ or h at $t_{1/2} = 9.78 \text{ (m)}$	C1
and PE is $\frac{3}{4}$ of max ratio = 3 or ratio = $9.78 / 3.26 = 3$	A1 [3]
(iii) time is less because (average) acceleration is greater OR average force is greater	B1 [1]

12 (a) (i) work (done)/ time (taken)	B1 [1]
(ii) work = force × displacement (in direction of force)	B1
power = force × displacement/ time (taken) = force × velocity	B1 [2]
(b) (i) weight = mg	C1
$P = Fv = 2500 \times 9.81 \times \sin 9^\circ \times 8.5$ (or use $\cos 81^\circ$)	C1
= 33 (32.6) kW	A1 [3]
(ii) no gain or loss of KE	B1
no work (done) against air resistance	B1 [2]
13 (a) kinetic energy = $\frac{1}{2} mv^2$	C1
= $\frac{1}{2} \times 0.040 \times (2.8)^2 = 0.157 \text{ J}$ or 0.16 J	A1 [2]
(b) (i) $k = F/x$ or $F = kx$	C1
$X_B = 14/800$	
= 0.0175 m	A1 [2]
(ii) area under graph = elastic potential energy stored	C1
or $\frac{1}{2} kx^2$ or $\frac{1}{2} Fx$	
(energy stored =) 0.1225 J less than KE (of 0.16 J)	A1 [2]
14 (a) GPE: energy of a <u>mass</u> due to its position in a <u>gravitational field</u>	B1
KE: energy (a mass has) due to its motion/ speed/ velocity	B1 [2]
(b) (i) 1. $KE = \frac{1}{2} mv^2$	C1
= $\frac{1}{2} \times 0.4 \times (30)^2$	C1
= 180 J	A1 [3]
2. $s = 0 + \frac{1}{2} \times 9.81 \times (2.16)^2$ or $s = (30 \sin 45^\circ)^2 / (2 \times 9.81)$	C1
= 22.88 (22.9) m	
= 22.94 (22.9) m	A1 [2]
3. $GPE = mgh$	C1
= $0.4 \times 9.81 \times 22.88 = 89.8$ (90) J	A1 [2]
(ii) 1. $KE = \text{initial KE} - GPE = 180 - 90 = 90 \text{ J}$	A1 [1]
2. (horizontal) velocity is not zero / (object) is still moving / answer explained in terms of conservation of energy	B1 [1]

15 (a)	work done is the product of force and the distance <u>moved</u> in the direction of the force or product of force and displacement in the direction of the force	B1	[1]
(b) (i)	work done equals the <u>decrease</u> in GPE – <u>gain</u> in KE	B1	[1]
(ii) 1.	distance = area under line $= (7.4 \times 2.5) / 2 = 9.3 \text{ m (9.25 m)}$	C1	
	or acceleration from graph $a = 7.4 / 2.5 (= 2.96)$ and equation of motion $(7.4)^2 = 2 \times 2.96 \times s$ gives $s = 9.3 \text{ (9.25) m}$	M1	[2]
		(C1)	
		(A1)	
2.	kinetic energy $= \frac{1}{2} m v^2$ $= \frac{1}{2} \times 75 \times (7.4)^2$ $= 2100 \text{ J}$	C1	
		C1	
3.	potential energy $= mgh$ $h = 9.3 \sin 30^\circ$ $\text{PE} = 75 \times 9.81 \times 9.3 \sin 30^\circ = 3400 \text{ J}$	A1	[3]
		C1	
4.	work done = energy loss $R = (3421 - 2054) / 9.3$ $= 150 \text{ (147) N}$	A1	[3]
16 (a) (i)	change in kinetic energy $= \frac{1}{2} m v^2$ $= 0.5 \times 25 \times (0.64)^2 = 5.1(2) \text{ J}$	C1	
		A1	[2]
(ii)	zero	A1	[1]
(iii)	(–) 5.1(2) J	A1	[1]
(b) (i)	PE $= mgh$ $= 350 \times 0.64 \times 25$ $= 5600 \text{ J}$	C1	
		C1	
		A1	[3]
	(If full length used allow 1/3)		
(ii)	$P = Fv$ or gain in PE / t , E_p / t or work done / t , W / t $= 350 \times 0.64$ or $5600 / 25$ $= 220 \text{ (224) W}$	C1	
		A1	[2]
17 (a)	(power =) work done / time (taken) or rate of work done	A1	[1]
(b) (i)	$F - R = ma$ $F = 1500 \times 0.82 + 1200$ $= 2400 \text{ (2430) N}$	C1	
		C1	
		A1	[3]
(ii)	$P = Fv$ $= (2430 \times 22) = 53\,000 \text{ (53\,500) W}$	C1	
		A1	[2]
(c)	(there is maximum power from car and) resistive force = force produced by car hence no acceleration or suggestion in terms of power produced by car and power wasted to overcome resistive force		
		B1	[1]

18 (a) (i)	(work =) force $\times$ distance <u>moved in the direction of the force.</u>	B1
	(ii) the energy <u>stored</u> (in an object) due to extension / compression / change of shape	B1
(b) (i)	$E_K = \frac{1}{2}mv^2$	C1
	$= 0.5 \times 0.40 \times 0.30^2$	
	$= 1.8 \times 10^{-2} \text{ (J)}$	A1
	(ii) (change in) kinetic energy = work done on spring / (change in) elastic potential energy	C1
	$1.8 \times 10^{-2} = \frac{1}{2} \times F \times 0.080$	C1
	$F_{\text{MAX}} = 0.45 \text{ (N)}$	A1
	(iii) $a = F/m = 0.45/0.40$	
	$= 1.1 \text{ (ms}^{-2}\text{)}$	A1
	(iv) 1. constant velocity / resultant force is zero, so in equilibrium	B1
	2. decelerating / resultant force is not zero, so not in equilibrium	B1
(c)	curved line from the origin	M1
	with decreasing gradient	A1

19 (a)	the energy (stored) in a body due to its extension/compression/deformation/ change in shape/size	B1	[1]
(b) (i)	two values of F/x are calculated which are the same		
	e.g. $10.4/40 = 0.26$ and $6.5/25 = 0.26$	B1	
	or		
	ratio of two forces and the ratio of the corresponding two extensions are calculated which are the same		
	e.g. $5.2/10.4 = 0.5$ and $20/40 = 0.5$	(B1)	
	or		
	gradient of graph line calculated and coordinates of one point on the line used with straight line equation $y = mx + c$ to show $c = 0$	(B1)	
	(so) force is proportional to extension (and so Hooke's law obeyed)	B1	[2]
(b) (ii) 1.	$k = F/x$ or $k = \text{gradient}$	C1	
	gradient or values from a single point used e.g. $k = 10.4 / (40 \times 10^{-2})$		
	$k = 26 \text{ Nm}^{-1}$	A1	[2]
	2. work done = area under graph		
	or $\frac{1}{2}Fx$ or $\frac{1}{2}(F_2 + F_1)(x_2 - x_1)$		
	or $\frac{1}{2}kx^2$ or $\frac{1}{2}k(x_2^2 - x_1^2)$	C1	
	$= \frac{1}{2} \times 10.4 \times 0.4 - \frac{1}{2} \times 5.2 \times 0.2$	C1	
	or $\frac{1}{2} \times (5.2 + 10.4) \times 20 \times 10^{-2}$		
	or $\frac{1}{2} \times 26 \times (0.4^2 - 0.2^2)$		
	$= 1.6 \text{ J}$	A1	[3]
(c)	remove the force and the spring goes back to its original length	B1	[1]

20 (a)	acceleration = change in velocity / time (taken) or rate of change of velocity	B1	[1]
(b) (i)	$v = 0 + at$ or $v = at$ ($a = 36/19 \Rightarrow 1.9$ (1.8947) m s^{-2})	C1 A1	[2]
(ii)	$s = \frac{1}{2}(u + v)t$ or $s = v^2/2a$ or $s = \frac{1}{2}at^2$ $= \frac{1}{2} \times 36 \times 19$ $= 36^2/(2 \times 1.89)$ $= \frac{1}{2} \times 1.89 \times 19^2$ $= 340 \text{ m}$ (342 m/343 m/341 m)	M1	[1]
(iii) 1.	$(\Delta KE =) \frac{1}{2} \times 95 \times (36)^2$ $= 62000$ (61 560) J	C1 A1	[2]
2.	$(\Delta PE =) 95 \times 9.81 \times 340 \sin 40^\circ$ or $95 \times 9.81 \times 218.5$ $= 200\,000 \text{ J}$	C1 A1	[2]
(iv)	work done (by frictional force) = $\Delta PE - \Delta KE$ or work done = $200\,000 - 62\,000$ (values from 1b(iii) 1. and 2.) (frictional force = $138\,000/340 \Rightarrow 410$ (406) N [420 N if full figures used])	C1 A1	[2]
(v)	$-ma = mg \sin 20^\circ - f$ or $ma = -mg \sin 20^\circ + f$ $-95 \times 3.0 = 95 \times 3.36 - f$ $f = 600$ (604) N	C1 A1	[2]

21 (a)	(gravitational potential energy is) the energy/ability to do work of a <u>mass</u> that it has or is stored due to its position/height in a gravitational field kinetic energy is energy/ability to do work a object/body/mass has due to its speed/velocity/motion/movement	B1 B1	[2]
(b) (i)	$s = [(u + v)t]/2$ or acceleration = $9.8/9.75$ (using gradient) $= [(7.8 + 3.9) \times 0.4]/2$ or $s = 3.9 \times 0.4 + \frac{1}{2} \times 9.75 \times (0.4)^2$ $s = 2.3(4) \text{ m}$	C1 C1 A1	[3]
(ii)	$a = (v - u)/t$ or gradient of line $= (7.8 - 3.9)/0.4 = 9.8$ (9.75) m s^{-2} (allow $\pm \frac{1}{2}$ small square in readings)	C1 A1	[2]
(iii)	$KE = \frac{1}{2}mv^2$ change in kinetic energy = $\frac{1}{2}mv^2 - \frac{1}{2}mu^2$ $= \frac{1}{2} \times 1.5 \times (7.8^2 - 3.9^2)$ $= 34$ (34.22) J	C1 C1 A1	[3]
(c)	work done = force $\times$ distance (moved) or Fd or Fx or mgh or mgd or mgx $= 1.5 \times 9.8 \times 2.3 = 34$ (33.8) J (equals the change in KE)	M1 A1	[2]

22 (a)	$\Delta E = mg\Delta h$ $= 0.030 \times 9.81 \times (-)0.31$ $= (-)0.091 \text{ J}$	C1 A1 [2]
(b)	$E = \frac{1}{2}mv^2$ (initial) $E = \frac{1}{2} \times 0.030 \times 1.3^2 (= 0.0254)$ $0.5 \times 0.030 \times v^2 = (0.5 \times 0.030 \times 1.3^2) + (0.030 \times 9.81 \times 0.31)$ so $v = 2.8 \text{ m s}^{-1}$ or $0.5 \times 0.030 \times v^2 = (0.0254) + (0.091)$ so $v = 2.8 \text{ m s}^{-1}$	C1 C1 A1 [3]
(c) (i)	$0.096 = 0.030(v + 2.8)$ $v = 0.40 \text{ m s}^{-1}$	C1 A1 [2]
(ii)	$F = \Delta p / (\Delta t)$ or $F = ma$ $= 0.096 / 20 \times 10^{-3}$ or $0.030 (0.40 + 2.8) / 20 \times 10^{-3}$ $= 4.8 \text{ N}$	C1 A1 [2]
(d)	<u>kinetic</u> energy (of ball and wall) decreases/changes/not conserved, so inelastic or (relative) speed of approach (of ball and wall) not equal to/greater than (relative) speed of separation, so inelastic.	B1 [1]
(e)	force = work done / distance moved $= (0.091 - 0.076) / 0.60$ $= 0.025 \text{ N}$	C1 A1 [2]
23 (a)	$v = u + at$ $v = 9.6 - (9.81 \times 0.37) = 6.0 \text{ m s}^{-1}$	A1
(b)	$s = \frac{1}{2} \times (9.6 + 6.0) \times 0.37$ or $6.0^2 = 9.6^2 - (2 \times 9.81 \times s)$ or $s = (9.6 \times 0.37) - (\frac{1}{2} \times 9.81 \times 0.37)$ or $s = (6.0 \times 0.37) + (\frac{1}{2} \times 9.81 \times 0.37^2)$ $s = 2.9 \text{ m}$	C1 A1
(c) (i)	$(\Delta)E = mg(\Delta)h$ $\Delta E = 0.056 \times 9.81 \times 2.9$ $= 1.6 \text{ J}$	C1 A1
(ii)	$E = \frac{1}{2}mv^2$ $\Delta E = \frac{1}{2} \times 0.056 \times (6.0^2 - 3.8^2)$ $= 0.60 \text{ J}$	C1 A1
(d)	force on ball (by ceiling) <u>equal</u> to force on ceiling (by ball) and opposite (in direction)	M1 A1
(e)	$(p =) mv$ or 0.056×6.0 or 0.056×3.8 change in momentum $= 0.056 \times (6.0 + 3.8)$ $= 0.55 \text{ N s}$	C1 A1
(f)	resultant force $= 0.55 / 0.085 (= 6.47 \text{ N})$ force by ceiling $= 6.47 - (0.056 \times 9.81)$ $= 5.9 \text{ N}$	C1 A1

24 (a) (i)	distance in a specified direction (from a point)	B1
(ii)	change in velocity/time (taken)	B1
(b) (i)	vertical component of velocity = $(5.5^2 - 4.6^2)^{1/2} = 3.0 \text{ (m s}^{-1}\text{)}$ or $5.5 \cos \theta = 4.6$ (so $\theta = 33.2^\circ$) and $5.5 \sin 33.2^\circ = 3.0 \text{ (m s}^{-1}\text{)}$	A1
(ii)	$s = ut + \frac{1}{2}at^2$ $0 = (3.0 \times t) - (\frac{1}{2} \times 9.81 \times t^2)$ or $v = u + at$ $-3.0 = 3.0 - 9.81t$ $t = 0.61 \text{ s}$	C1 A1
(iii)	$d = 4.6 \times 0.61$ $= 2.8 \text{ m}$	A1
(iv)	$E = \frac{1}{2}mv^2$ ratio = $(\frac{1}{2} \times m \times 4.6^2) / (\frac{1}{2} \times m \times 5.5^2)$ or ratio = $(\frac{1}{2} \times m \times 5.5^2 - m \times 9.81 \times 0.459) / (\frac{1}{2} \times m \times 5.5^2)$ ratio = 0.70	C1 C1 A1
(c)	straight line from positive value of v_y at $t = 0$ to negative value of v_y straight line ends at $t = T$ and final magnitude of v_y greater than initial magnitude of v_y	M1 A1

25 (a) (i)	work (done)/time (taken)	B1
(ii)	energy of a mass due to its position in a gravitational field	B1
(b) (i)	$P = Fv$ $= 2.0 \times 10^3 \times 45$ $= 9.0 \times 10^4 \text{ W}$	C1 A1
(ii)	1. $W = (2.0 \times 10^3) \times (45 \times 3.0 \times 60)$ or $W = 9.0 \times 10^4 \times 3.0 \times 60$ $W = 1.6 \times 10^7 \text{ J}$ 2. $(\Delta)E_p = mg(\Delta)h$ $= 1200 \times 9.81 \times 3.3 \times 3.0 \times 60$ $= 7.0 \times 10^6 \text{ J}$ 3. $W = 1.6 \times 10^7 - 7.0 \times 10^6$ $= 9.0 \times 10^6 \text{ J}$	C1 A1 C1 A1
(iii)	force = $(9.0 \times 10^6) / (45 \times 3.0 \times 60)$ $= 1.1 \times 10^3 \text{ N}$	A1
(iv)	constant velocity so no resultant force no resultant force so in equilibrium	B1 B1

- 26 (a)** $v_x = (6.0^2 - 4.8^2)^{1/2} = 3.6 \text{ (m s}^{-1}\text{)}$ **A1**
or
 $6.0 \sin \theta = 4.8 \text{ (so } \theta = 53.1^\circ \text{) and } v_x = 6.0 \cos 53.1^\circ = 3.6 \text{ (m s}^{-1}\text{)}$
- (b) (i)** straight line from (0, 4.8) to (0.49, 0) **M1**
 straight line continues with same slope to (0.98, -4.8) (labelled Y) **A1**
- (ii)** a horizontal line **M1**
 from (0, 3.6) to (0.98, 3.6) (labelled X) **A1**
- (c)** $s = ut + \frac{1}{2}at^2$ **C1**
 $= (4.8 \times 0.49) + (\frac{1}{2} \times -9.81 \times 0.49^2)$
or
 $s = \frac{1}{2}(u + v)t$ **or** area under graph
 $= \frac{1}{2} \times (4.8 + 0) \times 0.49$
or
 $s = vt - \frac{1}{2}at^2$
 $= \frac{1}{2} \times 9.81 \times 0.49^2$
or
 $v^2 = u^2 + 2as$
 $s = 4.8^2 / (2 \times 9.81)$
 $s = 1.2 \text{ m}$ **A1**
- (d)** $(\Delta)E = mg(\Delta)h$ **C1**
 $E = \frac{1}{2}mv^2$ **C1**
 $\text{ratio} = (\frac{1}{2} \times m \times 3.6^2) / (m \times 9.81 \times 1.2)$ **C1**
or
 $\text{ratio} = [(\frac{1}{2} \times m \times 6.0^2) - (m \times 9.81 \times 1.2)] / (m \times 9.81 \times 1.2)$
or
 $\text{ratio} = (\frac{1}{2} \times m \times 3.6^2) / (\frac{1}{2} \times m \times 4.8^2)$
 $\text{ratio} = 0.56$ **A1**
- (e)** (force due to) air resistance acts in opposite direction to the velocity **B1**
or
 (with air resistance, average) resultant force is larger (than weight)
-

27 (a)	energy (of a mass/body/object) due to motion/speed/velocity	B1
(b) (i)	$E = \frac{1}{2}mv^2$	C1
	$480 = \frac{1}{2} \times m \times 80^2$ so $m = 0.15 \text{ kg}$	A1
(ii) 1.	$E = mgh$ or $\Delta E = mg\Delta h$	C1
	$= 0.15 \times 9.81 \times 210$	A1
	$= 310 \text{ J}$	
	2. work done $= 480 - 310$	A1
	$= 170 \text{ J}$	
(iii)	work done $= Fs$	C1
	force $= 170/210$	A1
	$= 0.81 \text{ N}$	
(iv)	curved line from positive value on v-axis to (T, 0)	M1
	magnitude of gradient decreases	A1
(v)	as shell rises force decreases and as shell falls force increases	B1
	as shell rises force is downward and as shell falls force is upward	B1
	or	
	as shell rises the force decreases and is downward	(B1)
	as shell falls the force increases and is upward	(B1)
28 (a) (i)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$	C1
	$E = \frac{1}{2} \times 4.0 \times 0.32 = 0.64 \text{ J}$ or $E = \frac{1}{2} \times 12.5 \times (0.32)^2 = 0.64 \text{ J}$	A1
(ii)	$E = mgh$ or $E = Wh$	C1
	$= 2.5 \times 0.32$	
	$= 0.80 \text{ J}$	A1
(b) (i)	kinetic energy $= 0.80 - 0.64$	A1
	$= 0.16 \text{ J}$	
(ii)	$E = \frac{1}{2}mv^2$	C1
	$0.16 = \frac{1}{2} \times (2.5/9.81) \times v^2$	
	$v = 1.1 \text{ m s}^{-1}$	A1
29 (a)	the point where (all) the weight (of the body) is taken to act	B1
(b)(i)	vertical component $= 54 \sin 35^\circ$	A1
	$= 31 \text{ N}$	
(ii)	the (line of action of the) force (at B) passes through (point) A	B1
	or	
	the (line of action of the) force (at B) has zero (perpendicular) distance from (point) A	
(iii)	$54 \sin 35^\circ \times 0.68$ or $54 \cos 35^\circ \times 0.68$ or $W \times 0.34$	C1
	$54 \sin 35^\circ \times 0.68 + 54 \cos 35^\circ \times 0.68 = W \times 0.34$ so $W = 150 \text{ (N)}$	A1
(iv)	total vertical force $= 150 - 31$	A1
	$= 120 \text{ N}$	
(c)	$(\Delta)E = mg(\Delta)h$	C1
	$E = \frac{1}{2}mv^2$	C1
	ratio $= (m \times 9.81 \times 4.8) / (\frac{1}{2} \times m \times 9.2^2)$ or $(9.81 \times 4.8) / (\frac{1}{2} \times 9.2^2)$	C1
	$= 1.1$	A1

30 (a) (i)	1. $W = mas$	B1
	2. $s = (v^2 - u^2) / 2a$	B1
(ii)	W /work equals energy transferred/ <u>gain or change</u> in kinetic energy	B1
	$W (= mas) = ma(v^2 - u^2) / 2a$	B1
	leading to $W = m(v^2 - u^2) / 2$ (so $KE = \frac{1}{2}mv^2$)	
(b) (i)	1. solid curved line drawn from X to Y along path of ball and labelled D	B1
	2. solid straight line drawn from X to Y and labelled S	B1
(ii)	$(\Delta)E = mg(\Delta)h$	C1
	$4.5 = (0.040 \times 9.81 \times h) + (\frac{1}{2} \times 0.040 \times 9.5^2)$	C1
	$h = 6.9 \text{ m}$	A1
(iii)	line with a negative gradient starting from a non-zero value of kinetic energy when the vertical height is zero	M1
	<u>straight</u> line ends at a non-zero value of kinetic energy when the vertical height is h	A1
31 (a)	$k = F / x$ or $k = \text{gradient}$	
	e.g. $k = 4.0 / 0.050$	
	$k = 80 \text{ N m}^{-1}$	
(b)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$	
	$(\Delta)E = (\frac{1}{2} \times 3.2 \times 0.040) - (\frac{1}{2} \times 1.2 \times 0.015) = 0.055 \text{ J}$	
	or	
	$(\Delta)E = (\frac{1}{2} \times 80 \times 0.040^2) - (\frac{1}{2} \times 80 \times 0.015^2) = 0.055 \text{ J}$	
	or	
(c)	$(\Delta)E = mg(\Delta)h$	
	$= 0.122 \times 9.81 \times (0.120 - 0.095)$	
	$= 0.030 \text{ J}$	
	or	
	$(\Delta)E = W \times (\Delta)h$	
(d) (i)	$E = 0.055 - 0.030$	A1
	$= 0.025 \text{ J}$	
(ii)	$E = \frac{1}{2}mv^2$	C1
	$v = [(2 \times 0.025) / 0.122]^{0.5}$	A1
	$= 0.64 \text{ m s}^{-1}$	

32 (a)	force on body A (by body B) is equal (in magnitude) to force on body B (by body A)	B1
	force on body A (by body B) is opposite (in direction) to force on body B (by body A)	B1
(b) (i)	$m_X \times 5v$ or $(m_X + m_Y) \times v$	C1
	$m_X \times 5v = (m_X + m_Y) \times v$ (so) $m_Y / m_X = 4$	A1
(ii)	$(E =) \frac{1}{2}mv^2$	C1
	ratio = $[\frac{1}{2} \times (m_X + m_Y) \times v^2] / [\frac{1}{2} \times m_X \times (5v)^2]$	C1
	= 0.2	A1
(iii)	ratio = 1	A1
(c) (i)	1. (magnitude of resultant force is) zero	B1
	2. (magnitude of resultant force is) constant	B1
	(direction of resultant force is) opposite to the momentum	B1
(ii)	horizontal line from (0 ms, 0 squares) ending at (20 ms, 0 squares)	B1
	straight line from (20 ms, 0 squares) ending at (40 ms, 4.0 squares [= 4.0 cm vertically])	B1
	horizontal line from (40 ms, 4.0 squares) ending at (60 ms, 4.0 squares)	B1

33 (a)	resultant force (in any direction) is zero	B1
	resultant torque/moment (about any point) is zero	B1
(b) (i)	1. $T \sin 53^\circ = 2.4$	A1
	$T = 3.0 \text{ N}$	
	2. $F = T \cos 53^\circ$ or $F^2 = T^2 - 2.4^2$	A1
	$F = 1.8 \text{ N}$	
(ii)	$\sigma = T/A$ or $\sigma = F/A$	C1
	$A = \pi d^2 / 4$ or $A = \pi r^2$	C1
	$\sigma = 3.0 \times 4 / [\pi \times (0.50 \times 10^{-3})^2]$	A1
	= $1.5 \times 10^7 \text{ Pa}$	
(c) (i)	$h = 75 - 75 \sin 53^\circ = 15 \text{ cm}$	A1
(ii)	$(\Delta)E = mg(\Delta)h$ or $(\Delta)E = W(\Delta)h$	C1
	$(\Delta)E = 2.4 \times 15 \times 10^{-2}$	A1
	= 0.36 J	
(iii)	$E = \frac{1}{2}mv^2$	B1
	$0.36 = \frac{1}{2} \times (2.4 / 9.81) \times v^2$	C1
	$v = 1.7 \text{ m s}^{-1}$	A1

- 34 (a) (work done =) force $\times$ displacement in direction of the force **B1**
- (b) (i) 1. $(\Delta)E = mg(\Delta)h$ **C1**
 $= 0.42 \times 9.81 \times 78$ **A1**
 $= 320 \text{ J}$
2. $E = \frac{1}{2}mv^2$ **C1**
 $(\Delta)E = \frac{1}{2} \times 0.42 \times 23^2$ **A1**
 $= 110 \text{ J}$
- (ii) work done = $320 - 110$ (= 210 N) **C1**
average resistive force = $210 / 78$ **A1**
 $= 2.7 \text{ N}$
- (c) downward sloping line from (0, g) to a non-zero value on the time axis **M1**
line is curved with a gradient that becomes less negative **and** the line meets **A1**
 t -axis at time $t < T$
-

35.

2(a)	force $\times$ displacement in the direction of the force
(b)(i)	$E = \frac{1}{2}mv^2$
	$(m =) 23 \times 2 / 16^2 = 0.18 \text{ (kg)}$
(b)(ii)	$(\Delta)E = mg(\Delta)h$
	$60 = 0.18 \times 9.81 \times h$
	$h = 34 \text{ m}$
(b)(iii)	(work done $=$) $60 - 23$ $= 37 \text{ (J)}$
	average resistive force $= 37 / 34$ $= 1.1 \text{ N}$
2(c)	air resistance (acting on ball) increases
	or resultant force (on ball) decreases weight constant and air resistance increases
	acceleration decreases

36.

(a)(i)	$P = Fv$
	$= 18 \times 1.4$
	$= 25 \text{ W}$
(a)(ii)	$a = F / m$
	$a = 18 / 72 = 0.25 \text{ (ms}^{-2}\text{)}$
	$t = 1.4 / 0.25$
	$= 5.6 \text{ s}$
(b)(i)	$a = (54 - 18) / 72$ or $36 / 72 (= 0.50 \text{ ms}^{-2})$
	$v^2 = 2 \times 0.50 \times 9.5$
	$v = 3.1 \text{ m s}^{-1}$
(b)(ii)	$W = 54 \times 9.5$
	$= 510 \text{ J}$
b)(iii)	curved line from the origin
	gradient of line increases
2(c)	(force due to) air resistance increases/changes/not constant or air resistance increases with speed

37.

3(a)	$(\Delta)E = mg(\Delta)h \text{ or } W(\Delta)h$ $= 330 \times (4.0 - 1.1)$ $= 960 \text{ J}$
3(b)	$(\text{work}) = 960 - 540$ $(= 420 \text{ J})$ $\text{distance moved} = (960 - 540) / 52$ $= 8.1 \text{ m}$
(c)(i)	$E = \frac{1}{2}mv^2$ $540 = \frac{1}{2} \times (330 / 9.81) \times v^2$ $v = 5.7 \text{ m s}^{-1}$
(c)(ii)	$\text{speed} = \text{horizontal component of velocity}$ $= 5.7 \times \cos 41^\circ$ $= 4.3 \text{ m s}^{-1}$

38.

3(a)	work (done) / time (taken)
(b)(i)	zero / 0 J
(b)(ii)	$\text{work done} = 440 \times 25$ $= 1.1 \times 10^4 \text{ J}$
(c)(i)	$(\Delta)E_{(P)} = mg(\Delta)h$ $h = 4.8 \times 10^4 / (1700 \times 9.81)$ $= 2.9 \text{ m}$
(c)(ii)	$\theta = \sin^{-1} (2.9 / 25)$ $= 6.7^\circ$
3(d)	$\text{work done} = 4.8 \times 10^4 + 1.1 \times 10^4 (= 5.9 \times 10^4 \text{ J})$ $\text{time} = 5.9 \times 10^4 / 1.7 \times 10^4$ $= 3.5 \text{ s}$

39.

4(a)	ratio = $300 / 3200$ = 0.094
4(b)	$E = \frac{1}{2}mv^2$ or $E \propto v^2$ ratio = $(0.094)^{0.5}$ = 0.31
4(c)	work (done against frictional force) = $3200 - 300$ (=2900) length = $2900 / 76$ = 38 m
(d)(i)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$ $140 = \frac{1}{2} \times 63 \times x^2$ or $140 = \frac{1}{2}Fx$ and $F = 63x$ $x = 2.1$ m
(d)(ii)	percentage efficiency = $(140 / 300) \times 100$ = 47%
d)(iii)	curved line from the origin gradient of line increases

40.

(a)(i)	work (done) / time (taken)
(a)(ii)	$W = Fx$ or $E = Fx$ $(P =) Fx / t = Fv$ or $P = Fvt / t = Fv$
(b)(i)	$T = 430 \sin 11^\circ$ or $430 \cos 79^\circ$ = 82 N
(b)(ii)	speed = $56 / 82$ = 0.68 m s^{-1}
(b)(iii)	no change in kinetic energy (of block)
(b)(iv)	(input power) = $56 / 0.80$ (= 70 W) time taken = $1.2 \times 10^3 / 70$ = 17 s or (useful energy) = $1200 \times 80 / 100$ (= 960 J) time taken = $960 / 56$ = 17 s

41.

4(a)	work (done) = force $\times$ displacement
	(force = mg and distance = Δh)
	(so) work (done) = $mg\Delta h$ <u>and</u> work = $\Delta E_{(P)}$ (so $\Delta E_{(P)} = mg\Delta h$)
4(b)	gravitational potential (energy) to heat/thermal (energy)
(c)(i)	$P = mg(\Delta)h / (\Delta)t$ or Fv
	$P = (0.60 \times 9.81 \times 1.4) / 4.0$ or $0.60 \times 9.81 \times (1.4 / 4.0)$ = 2.1 W
(c)(ii)	$P = I^2 R$ or IV or V^2 / R
	= $0.09^2 \times 47$ or 0.09×4.23 or $4.23^2 / 47$ = 0.38 W
(c)(iii)	efficiency = $P_{out} / P_{in} (\times 100)$ or $E_{out} / E_{in} (\times 100)$
	= $0.38 / 2.1 (\times 100)$ or $0.38 \times 4.0 / 2.1 \times 4.0 (\times 100)$ = 0.18 or 18%

42.

(a)(i)	work done per unit time
(a)(ii)	$W = Fs$
	$P = Fs / t$ <u>and</u> (so) $P = Fv$
3(b)	($F =$) $130 \times 10^3 / 25 = 5200$ (N)
(c)(i)	(component of weight =) $mg \sin \theta$
(c)(ii)	F (along slope due to weight) = $36\,000 \times 9.81 \times \sin 1.4^\circ$ (= 8600 N)
	(total) $F = 5200 + 36\,000 \times 9.81 \times \sin 1.4^\circ$ (= 13 800 N)
	$P = 13\,800 \times 25$ = 350×10^3 (W) = 350 kW

43.

(a)(i)	$E = \sigma / \epsilon$ or $E = F / A\epsilon$
	$A = 1.4 \times 10^4 / (2.2 \times 10^{11} \times 0.0012)$
	$= 5.3 \times 10^{-5} \text{ m}^2$
(a)(ii)	$(\Delta)h = 0.64 \times 0.49 (= 0.3136)$
	$(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$
	$= 1.4 \times 10^4 \times 0.64 \times 0.49$
	$= 4.4 \times 10^3 \text{ J}$
2(b)	$P = Fv$ or W/t
	$= (1.4 \times 10^4 \times 0.64) / 0.56$ or $(4.4 \times 10^3 / 0.49) / 0.56$
	$= 1.6 \times 10^4 \text{ W}$
2(c)	$m = 1.4 \times 10^4 / 9.81$
	$(= 1427 \text{ kg})$
	$(\text{resultant}) F = (1.4 \times 10^4 / 9.81) \times 1.3$
	$(= 1855 \text{ N})$
	$T = 1.4 \times 10^4 - 1855$ or $(1.4 \times 10^4 / 9.81) \times (9.81 - 1.3)$
	$= 1.2 \times 10^4 \text{ N}$
2(d)	upward sloping straight line from $(t_x, 0)$ to t_y
	from t_y to t_z : an upward sloping curve with decreasing magnitude of gradient (that is horizontal at t_z)

44.

2(a)	the point where (all) the weight (of the object) is taken to act
(b)(i)	(54×0.45) or (2.4×0.95) or $(T \sin 30^\circ \times 1.3)$
	$(54 \times 0.45) = (2.4 \times 0.95) + (T \sin 30^\circ \times 1.3)$
	$T = 34 \text{ N}$
(b)(ii)	resultant moment $= (54 \times 0.45) - (2.4 \times 0.95)$ or $(34 \sin 30^\circ \times 1.3)$
	$= 22 \text{ N m}$
(c)(i)	$(\Delta)E = mg(\Delta)h$ or $W(\Delta)h$
	$= 2.4 \times 1.8$
	$= 4.3 \text{ J}$

(c)(ii)	$E = \frac{1}{2}mv^2$
	$= \frac{1}{2} \times (2.4 / 9.81) \times 3.4^2$
	$= 1.4 \text{ J (at X)}$
	kinetic energy at Y $= 4.3 + 1.4$ $= 5.7 \text{ J}$
	or
	$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mg(\Delta)h$
	$v^2 = 3.4^2 + 2 \times 9.81 \times 1.8$ $v^2 = 46.9 \text{ so } v = 6.85 \text{ (ms}^{-1}\text{)}$ $\text{KE} = \frac{1}{2} \times (2.4 / 9.81) \times 6.85^2$
	$= 5.7 \text{ J}$
(c)(iii)	no variation or acceleration is (always) vertically downwards
(c)(iv)	horizontal straight line at a non-zero value of velocity